

Gert Design Document

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Chapter 1

Overview

1.1 Problem Statement

Operational procedures are the connective tissue of production engineering. When a service degrades, an incident fires, or a change must be applied under pressure, engineers rely on runbooks—step-by-step guides that encode institutional knowledge about how to respond. The research literature is unambiguous: *unstructured runbooks fail*. Static documents drift from reality, skip governance checks, and produce no audit evidence. Teams that operate at scale cannot afford a response that depends entirely on whether the on-call engineer remembers to follow the wiki [5, 17].

The core operational problem is threefold. First, runbooks must be **executable**: a document that cannot be run is a suggestion, not a procedure. Second, runbooks must be **governed**: every action that touches infrastructure must be bounded by policy—command allowlists, environment variable controls, output redaction, and role-based approval gates. Third, runbooks must be **traceable**: every execution must produce an immutable, auditable record that satisfies SOC 2, ISO 27001, and HIPAA audit requirements [4, 7, 28] without manual effort. Without all three properties, organizations face audit failures, runbook drift, and uncontrolled blast radius during incidents.

Gert is designed to solve exactly this problem: a governed, executable, debuggable runbook engine that produces full traceability and evidence by construction.

1.2 Design Rationale

Gert is designed around three hard constraints: governance must be built in (not added on), traceability must be complete (every action leaves an immutable audit trail), and the component model must be stable (engine, schema, executors, and extensions evolve

independently under a documented contract). These constraints rule out ad-hoc scripting, general-purpose workflow engines, and cloud-execution services as the right foundation. Gert occupies a specific niche: a local-first, governed runbook execution engine with a stable component boundary and complete observability.

1.3 Scope

Gert is: a local-first, governed runbook execution engine with a stable component architecture, versioned extension contracts, a first-class governance and policy layer, full traceability by construction, and defined adapter interfaces for TUI, VS Code, and web clients. It ships as a single Go binary with an optional out-of-process extension runtime.

Gert is not: a managed cloud execution service, a general-purpose workflow orchestrator, a CI/CD pipeline system, a SOAR platform, or a replacement for systems like Temporal, Argo, or PagerDuty Runbook Automation [12, 23, 27]. It does not provide a central policy server, a shared execution registry, or multi-tenant execution isolation.

1.4 Design Priorities

- **A core runtime isolated from presentation adapters.** The execution engine—parser, planner, state machine, governance enforcer, trace writer—must have no dependency on any presentation layer. TUI, VS Code, and web clients attach via a stable event subscription API; they observe execution but never own it. This isolation is the prerequisite for every other architectural goal.
- **Explicit and versioned contracts for schema, tools, and events.** Every interface that crosses a component boundary must be named, versioned, and documented. The runbook schema carries an `apiVersion` field with formal compatibility guarantees. Tool definitions carry a version. The runtime event stream has a defined schema with ordered delivery guarantees. Breaking a contract requires a version increment, not a silent edit.
- **A secure extension runtime using out-of-process plugins.** Extensions run in isolated processes and are granted capabilities explicitly. No extension receives ambient authority. The capability model—what an extension may register, observe, or invoke—is a named, versioned list that the host enforces at admission time. This pattern follows the principle of least privilege endorsed by NIST SP 800-53 [1] and

applied in production by systems such as Kubernetes admission controllers [16] and OPA Gatekeeper [22].

1.5 Document Roadmap

This document is organized as follows. §2 defines measurable goals and explicit non-goals, establishing scope boundaries for implementors. §3 describes the five primary components and the dependency rules between them. §4 specifies the runbook schema, including versioning strategy and field inventory. §5 defines the extension runtime: capability taxonomy, handshake protocol, and trust model. §6 specifies the tool runtime: discovery, invocation, transport contracts, and governance integration. §7 defines the runtime event model: event schema, ordering guarantees, delivery semantics, and replay contract. §8 specifies the security and trust model: threat model, policy enforcement, supply chain controls, and audit requirements. §9 defines the testing and acceptance strategy: contract test categories, coverage requirements, and acceptance criteria. §10 documents open architectural questions with options and resolution criteria.

1.6 Success Definition

Gert is complete when: (1) a runbook executes end-to-end with governance checks, input resolution, tool invocation, and trace emission all functioning correctly; (2) the TUI, VS Code extension, and a third-party adapter can all be built against the published event and API contracts without importing internal packages; (3) the governance layer—allowlists, denylists, env blocking, redaction, approval gates—is enforced for every execution mode including extension-invoked runs; (4) every execution produces an append-only, crash-safe JSONL trace that is structurally valid against the trace schema; and (5) `gert test` passes a scenario suite covering all step types, governance rules, and evidence capture cases.

Chapter 2

Goals and Non-Goals

2.1 Goals

The following goals are the binding commitments of the design. Where possible, each goal is stated as a verifiable criterion rather than an aspiration.

- G1. Preserve and formalize the governance layer.** The governance primitives—command allowlists and denylists, environment variable blocking, output redaction with regex patterns, and per-step role-based approval gates—must be present with first-class semantics. Governance is gert’s core differentiator [8, 23]: approval gates integrated directly into step definitions. The runtime must additionally support RBAC scoping on runbook execution: which identities may run which runbooks, and under what conditions.
- G2. Preserve the append-only, crash-safe, tamper-evident trace.** The execution trace must be an append-only JSONL file. Each line is a self-describing JSON object with a unique event ID (UUID), ISO 8601 UTC timestamp [19], actor identity, step ID, and structured payload. The file must be crash-safe (partial writes must not corrupt prior entries) and must support optional cryptographic signing to satisfy SOC 2 and ISO 27001 (A.12.4) audit requirements [4, 7]. This is a launch blocker: no trace, no compliance.
- G3. Provide evidence capture, deterministic replay, TUI, VS Code extension, and gert test.** The runtime must provide: evidence capture with text, checklists, and SHA256-hashed attachments; scenario-based deterministic replay (`-mode replay`); the Bubble Tea terminal UI [14]; the VS Code extension powered by the JSON-RPC

server [18, 20]; and the `gert test` scenario runner with its assertion engine. These features represent validated operational value.

- G4. Establish stable, versioned extension contracts.** Tool definitions (`.tool.yaml`), input providers (`.provider.yaml`), and out-of-process extensions must bind to named, versioned contracts. A host version bump that breaks an extension contract requires a major version increment and a migration path. The capability taxonomy—what an extension may register, observe, and invoke—must be a published, enumerated list, not an implicit set of permissions inferred from code. This pattern is established by VS Code’s extension API [20] and Kubernetes admission controller design [16].
- G5. Introduce saga and compensation for multi-step rollback.** The runtime must support explicit compensation registration: when a forward step completes, the runbook may declare a compensating action to execute in reverse order on failure. This is the industry-standard saga pattern, implemented in production by Temporal [27] and AWS Step Functions [9]. Without compensation, partially-executed runbooks that modify infrastructure leave systems in undefined intermediate states.
- G6. Integrate OpenTelemetry for distributed tracing.** Every runbook execution must emit OpenTelemetry spans [6]: one root span per run, one child span per step, with W3C Trace Context [2] (`traceparent` / `tracestate`) propagated to tool invocations. Correlation IDs must appear in both the JSONL trace and the OTel span attributes, so that run traces can be joined with infrastructure observability data in Jaeger [15], Zipkin [30], or any OTLP-compatible backend. This is the current industry standard for operational visibility.
- G7. Decouple adapters from core via a typed event API.** The TUI, VS Code extension, and any future web adapter must attach to the core exclusively via a typed event subscription interface. No adapter may import internal runtime packages or call runtime methods directly. The event schema must be versioned: adapters declare which schema version they consume, and the host rejects incompatible connections. This constraint is necessary for `gert` to support concurrent adapters (e.g., TUI and VS Code running simultaneously against a single run).

2.2 Non-Goals

The following are explicitly outside the scope of gert. These boundaries are feature-scoped, not process-scoped, and are intended to prevent scope creep during implementation.

- NG1. Gert does not provide a managed cloud execution service.** There is no gert-as-a-service, no central execution registry, and no SaaS offering. Gert is a local-first binary. Cloud scheduling, multi-tenant isolation, and remote execution orchestration are concerns for a hypothetical v3 or a separate product.
- NG2. Gert does not implement a general-purpose workflow orchestrator.** Gert is not Temporal, Argo, or Prefect [12, 25, 27]. It does not support distributed workers, fan-out parallelism across remote nodes, or arbitrary DAG execution. Parallel execution within a single local run is a stretch goal; cross-host parallelism is explicitly out of scope.
- NG3. Gert does not ship a built-in policy server.** This design defines governance semantics and may integrate policy-as-code hooks [21], it does not bundle or operate an OPA server. Policy evaluation happens at the host process level. Central policy distribution, policy bundles over HTTP, and multi-host policy synchronization are out of scope.
- NG4. Gert does not define an extension marketplace or signing authority.** The runtime will define an extension signing *mechanism* (a verifiable public key model), but it will not operate a registry, curate extensions, or act as a certificate authority. Extension distribution is the responsibility of the extension author.
- NG5. Gert does not guarantee binary stability for internal packages.** The internal Go package structure, private interfaces, and runtime internals are not subject to backward compatibility guarantees. Only the documented public contracts—the runbook schema, the tool schema, the JSON-RPC API surface, the extension protocol, and the JSONL trace schema—carry compatibility commitments.

Chapter 3

Architecture

This chapter defines the system architecture: the five primary components, their interfaces and invariants, the complete data flow from runbook invocation to trace archive, the execution lifecycle, the concurrency model, and how the `gert serve` JSON-RPC layer integrates as a first-class adapter.

The single most important architectural rule is: **dependencies flow inward toward the Core Domain**. The Runtime Core knows nothing about TUI, VS Code, or JSON-RPC. Adapters know nothing about execution semantics. Every component boundary is enforced by a Go interface; no component imports a package that is “above” it in the dependency graph.

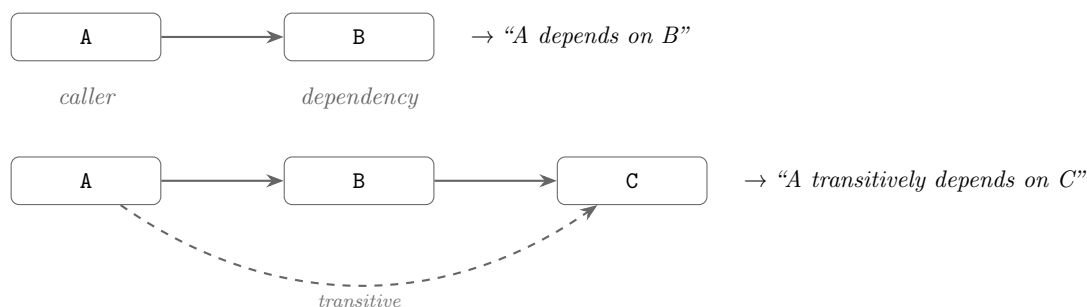


FIGURE 3.2: Dependency direction convention: an arrow from A to B means *A depends on B* (A imports/uses B). Arrows flow *toward* the depended-upon component.

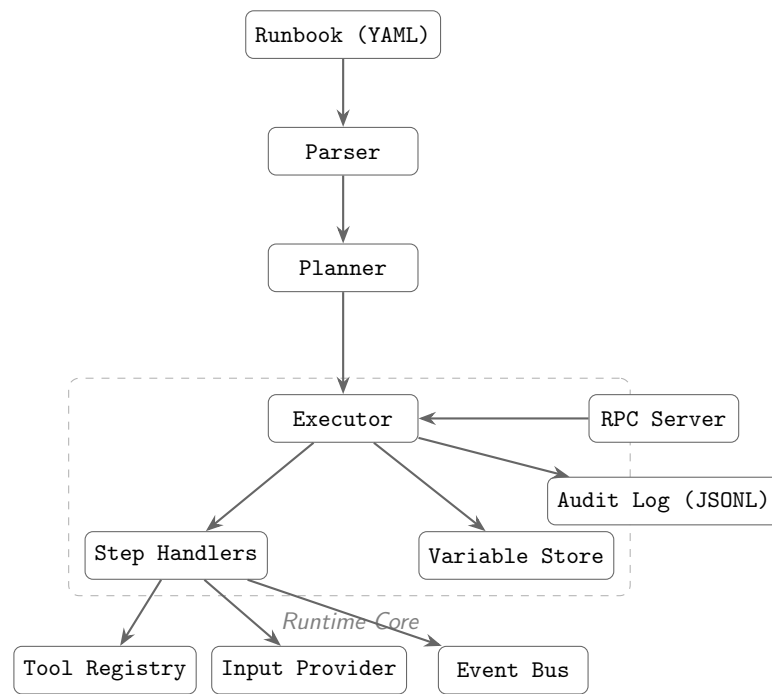


FIGURE 3.1: gert architecture — component dependency overview.

3.1 Primary Components

3.1.1 Parser

Responsibility. The Parser owns the transformation of a raw YAML document into a validated, semantically-resolved `ParsedRunbook` value. It does *not* own execution, planning, tool discovery, or input resolution. It validates schema structure and surface-level semantic rules (duplicate step IDs, unknown step types, malformed Go template expressions) but intentionally defers deep semantic validation (e.g., whether a `from:` binding can actually be resolved) to later pipeline stages.

Interface.

```
// Parser transforms raw YAML bytes into a validated ParsedRunbook.
type Parser interface {
    // Parse reads YAML from r, validates against the JSON Schema,
    // applies structural semantic checks, and returns a ParsedRunbook.
    // Errors carry structured diagnostics (file, line, column, message).
    Parse(ctx context.Context, r io.Reader, opts ParseOptions) (*ParsedRunbook,
        ↪ error)
}

type ParseOptions struct {
    // SchemaVersion overrides the apiVersion field for compatibility shims.
    SchemaVersion string
    // StrictMode causes deprecation warnings to be surfaced as errors.
    StrictMode bool
}

// ParsedRunbook is an immutable, fully-validated representation of a runbook.
// All fields have been structurally validated. Templates are not yet evaluated.
type ParsedRunbook struct {
    Meta          RunbookMeta
    Governance    GovernancePolicy // nil if no governance block
    Steps         []ParsedStep
    Imports       map[string]string // alias -> path (not yet resolved)
    Tools         []string          // tool names (not yet discovered)
    Schema        SchemaRef          // version and $schema URL
}
```

```
| }
```

Dependencies. The Parser depends only on the JSON Schema artifact (a static embedded file) and the expression evaluator (for template syntax validation). It has no runtime dependencies.

Key Invariants.

- A `ParsedRunbook` returned without error is structurally valid against the JSON Schema.
- All step IDs are unique within a `ParsedRunbook`.
- No I/O occurs during parsing (file imports are recorded as paths, not resolved).
- Parsing is pure and deterministic: the same input always produces the same output.

3.1.2 Planner

Responsibility. The Planner transforms a `ParsedRunbook` into an `ExecutionPlan`: a resolved, ordered, dependency-annotated representation of every step the runtime will execute. The Planner resolves imports (parsing and inlining included runbooks whose effective expansion mode is **eager**; recording **lazy** includes as opaque flow nodes for runtime materialization, see §3.1.2), discovers and validates tool definitions, resolves input provider bindings to their declared sources, and produces the static step ordering. It does *not* evaluate dynamic conditions (branch predicates, iterate count expressions) — those are runtime concerns.

What is an `ExecutionPlan`? An `ExecutionPlan` is a *flat, ordered list* of resolved steps with pre-computed metadata, not a DAG. Branch logic and iteration are left as runtime decisions because they depend on captured variables evaluated during execution. The plan encodes *which steps exist* and *what preconditions they carry*, not which steps will actually run. This design keeps the planner pure and the runtime authoritative over control flow.

```
| type Planner interface {  
|     // Plan resolves imports, discovers tools, validates bindings,  
|     // and returns a fully-resolved ExecutionPlan ready for the runtime.
```



```

Plan(ctx context.Context, rb *ParsedRunbook, opts PlanOptions)
↳ (*ExecutionPlan, error)
}

type PlanOptions struct {
  BaseDir      string           // working directory for import resolution
  ToolDirs     []string          // directories to search for .tool.yaml files
  ProviderDir  string           // directory for .provider.yaml files
  Vars         map[string]string // CLI-supplied variable overrides
}

// ExecutionPlan is an immutable, fully-resolved plan for a single run.
type ExecutionPlan struct {
  RunID        string
  RunbookPath  string
  Steps        []ResolvedStep // ordered; includes inlined include steps
  Tools        map[string]*ToolDef // name -> resolved tool definition
  Providers     map[string]*ProviderDef // binding name -> provider
  Governance   *GovernancePolicy
  Metadata     PlanMetadata
}

// ResolvedStep is a single step in the plan, fully resolved.
type ResolvedStep struct {
  ID          string
  Kind        StepKind // cli, choice, decision, collector, tool, include,
  ↳ branch, iterate
  Spec        StepSpec // kind-specific parameters (resolved, not templated)
  Contract    *contract.Contract //
  ↳ effects/reads/writes/idempotent/deterministic
  Depth       int // chain depth (0 = root runbook)
  Origin      string // source runbook path (for inlined include steps)
}

```

Dependencies. The Planner depends on the Parser (for import resolution), the Tool Registry (for `.tool.yaml` discovery), and the Provider Registry (for `.provider.yaml` discovery). It must not depend on the Runtime Core, Extension Host, or any adapter.

Key Invariants.

- An `ExecutionPlan` returned without error contains only resolvable steps.
- All tool names referenced by steps appear in `ExecutionPlan.Tools`.
- Import graphs are acyclic; the Planner detects and rejects cycles.
- All `from:` bindings that declare a provider appear in `ExecutionPlan.Providers`.
- The Planner does not execute or side-effect; it is safe to plan without running.

Include expansion modes. The Planner consults the active expansion policy (see §4.4.9) at every `include` site. When the effective mode is `eager`, the referenced runbook is loaded, planned, and inlined into the current plan; when it is `lazy`, the include appears in the plan as an opaque flow node whose only payload is the absolute path of the child runbook. Plan-time validation (cycle detection, max-depth, transitive contracts) walks only the eagerly expanded sub-tree; lazy edges are validated when they are entered at run time. The Adapter supplies the Planner with a *lazy loader* that propagates lazy mode recursively into any sub-include reached through a deferred edge, so a single eager include in a lazy sub-tree does not silently fall back to eager for the entire chain.

3.1.3 Runtime Core

Responsibility. The Runtime Core owns the execution loop: advancing through an `ExecutionPlan` step by step, evaluating templates and branch predicates, dispatching tool invocations, collecting evidence, writing the append-only trace, enforcing governance pre-flight checks, emitting structured events, and managing the mutable run state. It does *not* own user presentation, serialization formats, or network transport.

Interface.

```
type Runtime interface {
    // Start initializes a run from a plan and returns a RunHandle.
    // The run does not advance until Next is called.
    Start(ctx context.Context, plan *ExecutionPlan, opts RunOptions) (RunHandle,
        ↪ error)
```

```

    // Resume restores a previously checkpointed run from the run store.
    Resume(ctx context.Context, runID string, opts RunOptions) (RunHandle,
        ↪ error)
}

type RunHandle interface {
    // Next advances the run by one step. Blocks until the step completes
    // (or is paused waiting for evidence/approval).
    // Returns io.EOF when the run has no more steps.
    Next(ctx context.Context) (*StepResult, error)

    // Approve records approval for the current step's approval gate.
    Approve(ctx context.Context, decision ApprovalDecision) error

    // SubmitEvidence records evidence for interactive steps (choice, decision,
    ↪ collector).
    // For choice: stores selected option. For decision: stores route selection.
    // For collector: stores multi-field form data and artifact attachments.
    SubmitEvidence(ctx context.Context, stepID string, ev
        ↪ map[string]*EvidenceValue) error

    // Cancel requests a graceful stop of the run.
    Cancel(ctx context.Context, reason string) error

    // State returns the current mutable run state (read-only snapshot).
    State() RunState

    // Events returns a channel of structured events emitted during execution.
    Events() <-chan Event
}

type RunOptions struct {
    Mode      RunMode      // real, dry-run, replay
    Actor      string       // --as identity
    Vars       map[string]string // variable overrides
    ScenarioDir string       // replay scenario directory
    Store      RunStore     // durable store for checkpointing
}

```

```

    OnEvent      func(Event)      // synchronous event hook (nil = use Events
    ↪ channel)
}

```

Dependencies. The Runtime Core depends on the Governance Engine, Tool Runtime (for tool invocation), Expression Evaluator (for templates and branch predicates), Evidence store, and Trace Writer. It does not depend on the Extension Host, API/Adapter Layer, or any adapter.

Key Invariants.

- The trace is append-only: no event is ever deleted or modified after being written.
- Governance pre-flight is enforced before every `cli` and `tool` step dispatch.
- Redaction is applied to all captured output before it is written to the trace.
- The run state is checkpointed after every step completion.
- Events are emitted in a total order that matches step execution order.
- The runtime is re-entrant per run: multiple callers may call `Next` sequentially (but not concurrently) on the same `RunHandle`.

3.1.4 Extension Host

Responsibility. The Extension Host manages the lifecycle of out-of-process extensions: discovery, launch, capability negotiation, RPC dispatch, sandboxing, and graceful shutdown. It exposes extension-contributed tools and providers to the Tool Runtime and Provider Registry. It does *not* execute runbook steps directly and does not own the extension communication protocol (that belongs to the extension RPC layer, defined in §04).

Interface.

```

type ExtensionHost interface {
    // Load discovers and launches all extensions declared in the project
    ↪ manifest.
    Load(ctx context.Context, manifest *ProjectManifest) error
}

```

```

    // Shutdown gracefully stops all running extensions.
    Shutdown(ctx context.Context) error

    // ContributedTools returns all tools contributed by loaded extensions.
    ContributedTools() []*ToolDef

    // ContributedProviders returns all providers contributed by loaded
    ↪ extensions.
    ContributedProviders() []*ProviderDef

    // ContributedPolicyRules returns governance rules contributed by
    ↪ extensions.
    // These compose with host policy; host policy takes precedence on conflict.
    ContributedPolicyRules() []PolicyRule
}

```

Dependencies. The Extension Host depends on the extension manifest schema and the extension RPC transport (stdio JSON-RPC, as specified in §04). It does not depend on the Runtime Core or any adapter.

Key Invariants.

- No extension operates without an explicit capability grant from the host.
- Extension crashes do not crash the host process.
- The Extension Host enforces timeout and cancellation on all RPC calls to extensions.
- Extension-contributed policy rules are additive; they can only *restrict*, not *relax*, host-defined policy.

3.1.5 API / Adapter Layer

Responsibility. The API/Adapter Layer is the boundary between the Runtime Core and all external consumers (TUI, VS Code extension, web clients, CI pipelines). It translates the runtime’s event stream and `RunHandle` contract into transport-specific protocols. Each adapter is a thin, stateless renderer: it subscribes to the event stream, presents state to the user, and forwards user input (approvals, evidence, cancellation) back to the `RunHandle`. Adapters own *no* execution logic.

Interface.

```
// Adapter is the interface every consumer of the Runtime Core must implement.
type Adapter interface {
    // Attach binds the adapter to a RunHandle before the first Next call.
    Attach(handle RunHandle) error

    // Run blocks until the run completes or is cancelled. It is responsible
    // for draining Events(), rendering output, and forwarding user input.
    Run(ctx context.Context) error
}

// The gert serve HTTP layer implements Adapter for the VS~Code extension
// and the embedded web preview, exposing JSON-RPC at POST /rpc plus REST
// endpoints (/runs, /runs/{id}/state) and SSE event streams (/events,
// /runs/{id}/state, /runs/{id}/interactions).
// The TUI (Bubble Tea) implements Adapter over a terminal.
// A WebSocket transport (/ws) is also exposed for byte-stream consumers.
```

Dependencies. Adapters depend on the `RunHandle` and `Event` types from the Runtime Core. They must not import any package from the Parser, Planner, or Extension Host.

Key Invariants.

- An adapter may not call `Next` more than once concurrently.
- An adapter may not modify run state directly; all state changes go through `RunHandle`.
- Adapters must handle the case where the event channel is closed (run ended) without blocking.
- An adapter failure must not corrupt the run trace or run state.

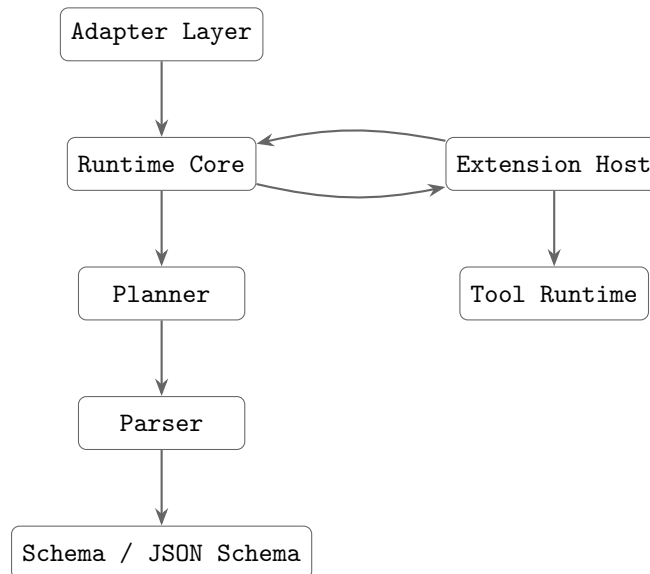


FIGURE 3.3: Component dependency direction. Arrows point from dependent to dependency. No upward arrows exist.

Dependency Direction Summary. No upward arrow exists. The Parser knows nothing about the Runtime. The Runtime knows nothing about adapters. The Extension Host is a peer of the Runtime Core, contributing tools and policy rules but not owning execution.

3.2 Data Flow

This section traces the complete path from runbook invocation to trace archive, identifying exactly where each responsibility is exercised.

3.2.1 Phase 1: Parse

The user invokes `gert exec runbook.yaml -as eng@corp.com`. The CLI layer reads the YAML file and calls `Parser.Parse()`. The Parser validates the document against the embedded JSON Schema, checks structural semantic rules, and returns a `ParsedRunbook`. If validation fails, structured diagnostics are returned and execution does not proceed.

3.2.2 Phase 2: Plan

The `ParsedRunbook` is passed to `Planner.Plan()`. The Planner:

1. Resolves all `imports`: recursively (parse $\rightarrow$ plan each imported runbook, inline its steps with incremented `Depth`).
2. Discovers `.tool.yaml` files for each named tool.
3. Resolves `from`: input bindings to their declared providers.
4. Validates that all step references (branch targets, iterate bodies) are valid step IDs.
5. Constructs and returns an `ExecutionPlan`.

Governance policy from the runbook's `meta.governance` block is extracted and embedded in the `ExecutionPlan` for runtime enforcement.

3.2.3 Phase 3: Run Initialization

`Runtime.Start(plan, opts)` is called. The runtime:

1. Generates a `RunID` (format: `YYYYMMDDTHHmmss-xxxxxxx`).
2. Creates the run directory: `.runbook/runs/<runID>/`.
3. Opens the append-only trace file: `.runbook/runs/<runID>/trace.jsonl`.
4. Writes the `run/started` trace event (runbook path, actor, mode, plan hash).
5. Initializes the `GovernanceEngine` from the plan's governance policy.
6. Compiles redaction rules from the governance policy.
7. Returns a `RunHandle`; the run is now paused at step 0.

3.2.4 Phase 4: Step Execution Loop

The adapter calls `RunHandle.Next(ctx)` in a loop until `io.EOF`. For each step, the runtime executes the following sequence:

Step Pre-flight (all step types)

1. Write `step/started` trace event.
2. Emit `StepStarted` runtime event to the event channel.
3. Evaluate template expressions in the step spec against current run variables.

Governance Pre-flight (cli and tool steps)

1. **Command check:** `GovernanceEngine.CheckCommand(argv[0])` evaluates the denylist (deny takes precedence) and then the allowlist. A violation writes a `governance/commandBlocked` trace event and returns an error; the step fails.
2. **Env var check:** `GovernanceEngine.FilterEnvVars()` removes variables matching `deny_env_vars` patterns. Blocked variable names are written to the `governance/envVarBlocked` trace event.
3. **Contract check:** The step's `Contract.Risk()` is evaluated. Steps with `RiskCritical` that do not have an explicit approval gate declaration in the plan emit a `governance/contractWarning` event (not a hard failure in ; upgraded to error in strict mode).
4. **Approval gate:** If the step declares `approvals.min > 0`, the runtime pauses and emits an `ApprovalRequired` event. Execution blocks at this step until `RunHandle.Approve()` is called with sufficient approvals. See §11.3 (Governance chapter) for the full approval protocol.

Dispatch (step-type-specific)

- **cli:** Fork a subprocess. Capture stdout/stderr. Apply redaction rules to captured output before storing in run variables or writing to trace.
- **tool:** Call `ToolRuntime.Invoke(toolName, input)`. The tool runtime dispatches to the appropriate transport (stdio, JSON-RPC, MCP). Redaction applied to returned output.
- **choice:** Emit `ChoiceRequired` event with option list. Block until `SubmitEvidence()` is called with selected option. Store result as named variable. No artifacts involved.

- **decision:** Emit `DecisionRequired` event with route map. Block until `SubmitEvidence()` is called with route selection. Alter execution flow by jumping to the selected runbook or label. Optionally store selection in variable for audit trail.
- **collector:** Emit `CollectorRequired` event with field schema. Block until `SubmitEvidence()` is called with multi-field form data. Validate each submitted value against its declared field type and constraints (see §4.4.5). SHA256-hash each `file` or `image` attachment. Store typed values as run variables: `number` and `integer` as JSON numbers, `boolean` as JSON boolean, multi-select `select` and `choice` as JSON arrays of strings, single-select `select` as a JSON string, `date` and `datetime` as ISO 8601 strings, `text` and `url` as JSON strings. Store `file` and `image` artifacts in the evidence store (not in run variables).
- **include:** Resolve the referenced runbook. Validate the include graph for cycles (load-time DFS — fail fast if cyclic). Evaluate `include.when` if present; skip all steps if false. Apply `include.with` overrides to the current variable scope. Inline-expand the referenced runbook's steps into the current execution queue at `Depth+1`. The include step itself does not appear in the run trace; expanded steps run as if defined inline, each carrying an `Origin` breadcrumb: [`included from: <path>`]. Includes may be materialized eagerly (loaded and expanded at plan time) or lazily (loaded, planned, and inlined when the step is actually entered) according to the runbook's `expand` field, the site's `include.expand`, or the runtime default; see §4.4.9. Lazy includes carry only an absolute path through the plan; at execution time the include executor delegates to a sub-engine that reuses the parent run identifier so deferred prompts surface on the parent's interaction stream.
- **branch:** Evaluate branch predicates (expression evaluator) against current variables. Select the matching branch arm. Write `event/branchResolved` trace event. Advance step cursor to the first step of the resolved arm.
- **iterate:** Evaluate the iteration expression. Write `event/iteratePassStart`. Execute body steps. Write `event/iteratePassEnd`. Repeat until condition is false.
- **wait_for_event:** Serialise run state to the run store. Register an event listener with the Event Dispatcher (source, event ID, filter predicate, and timeout). Set run state to `WAITING` and return to caller — execution is suspended. When the matching event arrives, the Event Dispatcher rehydrates the run, validates the payload against `event.payload_schema`, applies `event.filter` conditions, writes captured variables

via `capture` mappings, and resumes at the next step. If `timeout` elapses before an event arrives the `on_timeout` policy governs the outcome (see §3.3.4). This dispatch path is only available in `gert serve` mode; `gert run` MUST reject the step with error: "step type 'wait_for_event' requires gert serve mode".

Step Type Classification

All step types fall into one of three categories based on runtime behavior:

Category	Step Types	Runtime Behavior
Execution	cli, tool	Execute external commands or tools. Require governance pre-flight (allowlist, denylist, redaction). Capture output and exit codes.
Interactive	choice, decision, collector	Block execution, emit event requiring user input. choice stores selected option as variable. decision alters control flow by routing to a runbook or label. collector captures multi-field form data and artifacts (files, images). All three write evidence to trace.
Control Flow	branch, iterate, include	Alter execution path or loop body. branch evaluates predicates at runtime. iterate repeats a body until a condition is false. include resolves and inline-expands a referenced runbook's steps into the current execution queue; the include step itself is not emitted to the trace.
Synchronisation	wait_for_event	Suspend the run until an external event arrives via webhook, message queue, OS signal, or named in-process channel. Persists full run state to durable storage; resumes when the Event Dispatcher delivers a matching, validated event. Valid only in gert serve mode — rejected at runtime in gert run (local CLI) mode.
Approval Gate	approve	Pause execution pending sign-off from one or more authorised identities. Supports all/any/quorum modes with an explicit approver pool. Optionally enforces a business-day timeout (GAP-1) and M-of-N quorum tracking (GAP-2). See §3.3.5.
Presentation	display	Render template content to the operator without requiring input. content is evaluated via Go text/template against the current variable scope. format controls rendering hint (text or markdown). Fire-and-continue; no state mutation.
Utility	noop	No-op placeholder. Evaluates capture expressions. Zero side effects.

3.2.5 CLI Step Executor Contract

A **type**: `cli` step executes a shell command or script on the host. Two execution forms are supported:

- **Exec form** (`command/args`) — bypasses the shell entirely. The runtime forks the binary directly, which is both faster and injection-safe.
- **Shell-string form** (`run: "<string>"`) — passes a script to a shell interpreter. Supports multi-line scripts and shell built-ins at the cost of increased injection risk.
- **Multi-shell map form** (`run: {bash: "...", cmd: "...", ...}`) — provides per-shell scripts; the executor selects the first available shell at run time. Designed for cross-platform runbooks.

The fields `command` and `run` are mutually exclusive. If both are present the step fails validation with an error (see validation table below).

Shell Resolution — String Form

When `run` is a string the executor resolves the shell before forking.

Automatic detection (`shell: auto, the default`). The executor inspects the OS at plan-time and selects the platform-native shell:

- **Linux / macOS**: `/bin/sh` — POSIX-guaranteed to exist.
- **Windows**: `pwsh` if `exec.LookPath("pwsh")` succeeds; otherwise `cmd.exe`.

Explicit shell (`shell: bash | sh | pwsh | cmd`). The executor calls `exec.LookPath(shell)` at *plan time*. If the binary is not found the plan fails immediately with:

Shell 'bash' not found on this host (exec.LookPath returned empty)

This is a plan-time error, not a run-time error — the runbook is rejected before execution begins.

Invocation patterns. Once the shell is resolved the script is passed using the shell's standard non-interactive flag:

Shell	Invocation
bash	["bash", "-c", script]
sh	["sh", "-c", script]
pwsh	["pwsh", "-NonInteractive", "-Command", script]
cmd	["cmd.exe", "/c", script]

Template expansion. Template expansion (Sprig/Go template) is applied to the `run` string *before* the result is handed to the shell. The shell receives the already-expanded string. Shell interpretation therefore operates on the final expanded text, not on the template source.

Security note — shell injection. Shell-string form is **not** immune to shell injection. If a template variable contains shell metacharacters (`$`, `'`, `;`, `|`, `&`, parentheses, etc.) they will be interpreted by the shell after template expansion. *Authors are responsible for sanitizing template values used in shell-string steps.* For untrusted input, prefer the `exec` form (`command/args`), which bypasses the shell entirely.

Shell Resolution — Multi-Shell Map Form

When `run` is a YAML mapping (e.g. `run: {bash: "...", pwsh: "..."}`) the executor performs a two-phase resolution.

Phase 1 — Plan-time key validation. The executor reads all map keys and verifies each against the set of recognized shell names: `bash`, `sh`, `pwsh`, `cmd`. Any unrecognized key is a load-time validation error:

Unknown shell 'zsh' in run map. Valid values: bash, sh, pwsh, cmd

The plan is rejected before execution begins.

Phase 2 — Run-time shell selection. At the moment the step is dispatched the executor walks the map in *insertion order* and calls `exec.LookPath(key)` for each entry:

1. The **first key whose shell resolves successfully** is selected.

2. **Platform-family fallback:** if the `bash` key is present but `bash` is not found, the executor tries `sh` as a fallback before moving on to the next map key.
3. If **no key resolves** the step fails with:

```
CLI step '<id>': no shell from map {bash, pwsh} available on this host.  
Install one of: bash, pwsh.
```

The `shell` field is ignored when `run` is a map. Setting it produces a warning (see validation table).

Audit record. The JSONL trace records which map entry was selected. A `shell_selected` field is written alongside the step's `step/started` event:

```
{ "kind": "step/started", "...", "payload": {  
  "step_id": "deploy-linux",  
  "shell_selected": "bash",  
  ...  
}
```

This field appears in the `step/started` payload whenever a shell-string or map-form CLI step begins. For exec-form steps the field is omitted.

Template expansion. The same expansion-before-invocation rule applies: template expansion is applied to the selected map value before it is passed to the shell.

Platform Variable Injection

The runtime injects a `gert.platform` namespace into the variable scope at run start. These variables are available to all template expressions throughout the run, including `branch` predicates.

Variable	Type	Values
<code>gert.platform.os</code>	string	"linux" "darwin" "windows"
<code>gert.platform.arch</code>	string	"amd64" "arm64" "arm"
<code>gert.platform.shell</code>	string	Shell name selected for the current step (e.g. "bash"). Set after shell resolution; available in <code>capture</code> output and downstream steps. Not usable inside the <code>run</code> script of the same step — resolution happens before template expansion of that field is complete.

`gert.platform.os` is particularly useful for complementing the map form with `branch` steps in cases requiring more complex platform-conditional logic than the map form alone can express.

Validation Rules

Rule	Severity	Message
<code>command</code> and <code>run</code> both present	Error	Fields <code>`command`</code> and <code>`run`</code> are mutually exclusive on type: <code>cli</code>
<code>run</code> is a map and <code>shell</code> is set	Warning	<code>`shell`</code> is ignored when <code>`run`</code> is a map - shell is determined by map key resolution
<code>run</code> map has unknown key	Error	Unknown shell <code>'{key}'</code> in run map. Valid values: <code>bash</code> , <code>sh</code> , <code>pwsh</code> , <code>cmd</code>
<code>shell</code> is explicit but not found on host	Plan-time Error	Shell <code>'{shell}'</code> not found on this host (<code>exec.LookPath</code> returned empty)
<code>run</code> map, no key resolves at run time	Run-time Error	No shell from map <code>{keys}</code> available on this host

3.2.6 Include Step Executor Contract

The **include** step is the inline composition primitive. It expands a referenced runbook's steps directly into the caller's execution queue — no new run context, no isolated variable scope, no separate audit entry.

Executor Algorithm

When the executor reaches a **type: include** step it performs the following in order:

1. **Resolve** the referenced runbook by relative path or registry ID.
2. **Cycle check** — validate the include graph (see below). Fail with a hard error if any cycle is detected. This check must have already passed at load time; the executor re-asserts it as a defence-in-depth guard.
3. **Evaluate include.when** — if the condition is present and evaluates to **false**, skip: emit no steps and advance past the include step.
4. **Apply include.with overrides** to the current variable scope. These bindings are visible to all steps that follow (included or caller-defined).
5. **Inline expand** — replace the include step in the execution queue with the referenced runbook's steps, each at **Depth+1** with their **Origin** field set to the referenced runbook path.
6. **Continue execution** — the expanded steps execute as if they had been defined inline in the caller runbook.

Key Architectural Properties

- **No new run context.** The included steps execute within the caller's run; **RunID**, variable scope, and audit log are all shared.
- **No separate audit entry for the include step itself.** Expanded steps appear in the caller's audit log each carrying an **Origin** breadcrumb of the form: **[included from: ./shared/db-preflight.runbook.yaml]**.
- **Shared variable scope.** Variable mutations made by included steps are visible to all subsequent steps in the caller, and vice versa.

- **Include step invisible in trace.** Only the expanded steps appear in the run trace. The include step itself is not emitted as a trace event.

Cycle Detection

Load-time validation. When a runbook is loaded, the runtime builds a directed include graph over the full transitive closure of all `type: include` references. A depth-first search (DFS) with a visited set detects back-edges. A cycle found at any depth is a hard error:

```
cyclic include: A.yaml -> B.yaml -> A.yaml
```

Cyclic include graphs are rejected before execution begins. This check covers both direct and transitive includes.

Runtime note. Because cycles are validated at load time, the executor never needs to guard against infinite include loops at execution time. The executor-level re-assertion in step 2 above is a defence-in-depth check, not the primary enforcement point.

Future: Sub-Procedure Call. `type: include` is the the composition primitive: inline expansion into a shared variable scope. A future `type: call` step type would provide isolated scope with explicit input/output mapping, analogous to a function call. This pattern is deferred future; `type: include` covers the primary composition use case for the release.

Step Post-flight (all step types)

1. Apply outcome evaluation: test `when:` expressions; record `OutcomeRecord`.
2. Write captured variables to run state.
3. Checkpoint run state to `.runbook/runs/<runID>/snapshots/<stepID>.json`.
4. Write `step/completed` trace event (outcome, captures, elapsed time).
5. Emit `StepCompleted` runtime event.

3.2.7 Phase 5: Run Completion

When `Next` returns `io.EOF`:

1. Write `run/completed` trace event (final outcome, step counts, elapsed time).
2. Flush and sync the trace file.
3. Compute SHA256 of the trace file; write `run/evidenceHash` record.
4. Optionally archive the run directory (compress to `.runbook/archive/`).
5. Close the event channel.

3.3 Lifecycle Model

3.3.1 Run Start

A run starts in one of two ways:

1. **Direct invocation:** The CLI or test harness calls `Runtime.Start()` directly. This is the path for `gert exec`, `gert tui`, `gert debug`, and `gert test`.
2. **RPC invocation:** A client sends `exec/start` to the JSON-RPC server (`gert serve`). The server calls `Runtime.Start()` on behalf of the client and returns the `RunID` in the response. Subsequent step advancement is driven by `exec/next` RPC calls.

3.3.2 Step Advancement

Step advancement is *always client-driven*. The runtime does not auto-advance. After `Start`, the client calls `Next` to execute each step in sequence. This applies to all adapters, including headless CI mode. The only exception is the `-auto` flag, which causes the CLI adapter to call `Next` in a tight loop, producing the appearance of automatic advancement while retaining the single-step contract.

3.3.3 Pause and Resume

A run is *paused* when `Next` returns a non-terminal result that requires external input: interactive steps (`choice`, `decision`, `collector`) or approval gates. The run remains

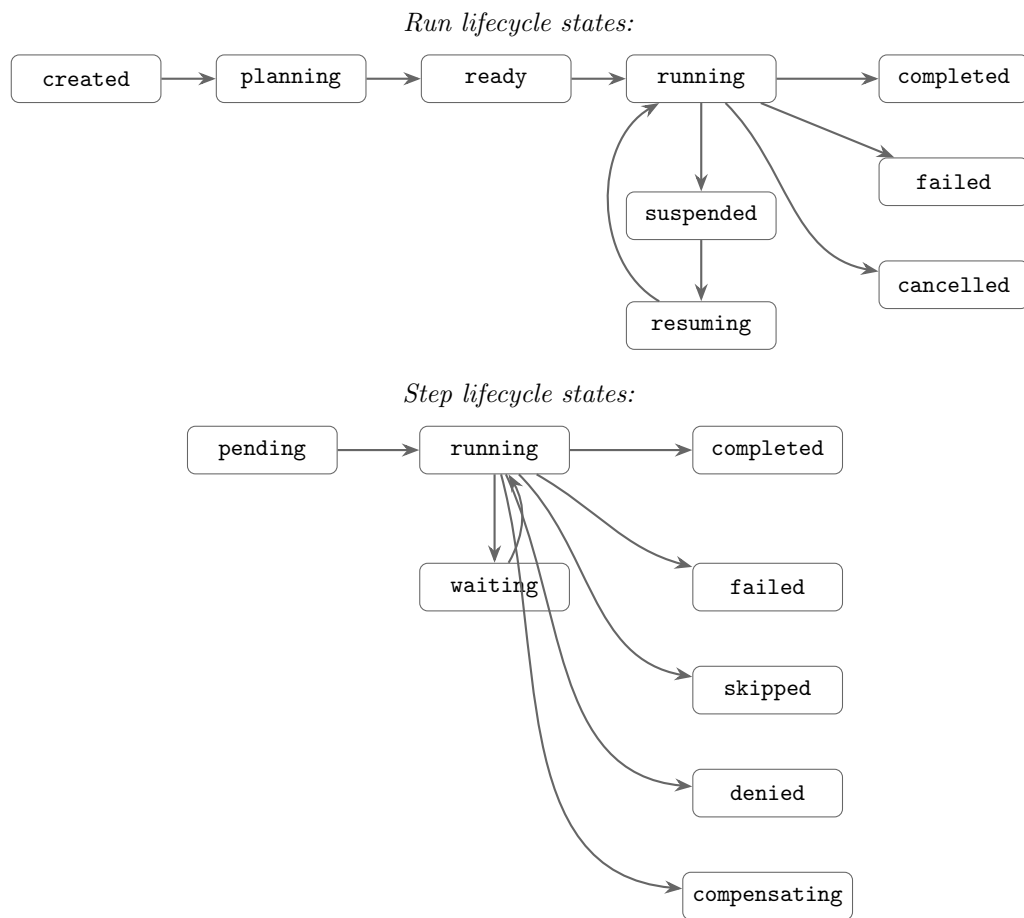


FIGURE 3.4: Execution lifecycle — run states (top) and step states (bottom).

paused until the required input is provided via `SubmitEvidence` or `Approve`. The adapter is responsible for presenting the pause to the user.

The checkpoint written after each completed step is sufficient to resume a run after a process restart. The checkpoint records: completed step IDs, current variable bindings, current evidence values, and the trace sequence number. Resumption is initiated by calling `Runtime.Resume(runID)` from any adapter.

3.3.4 Wait-for-Event Executor Contract

WAITING Run State

The `wait_for_event` step introduces a new **WAITING** run state that is distinct from the existing user-initiated **PAUSED** state:

State	Trigger	Exit Condition
RUNNING	<code>exec/start</code> or resume	Active step execution loop.
PAUSED	User-initiated pause; interactive / approval step	<code>SubmitEvidence</code> or <code>Approve</code> RPC call from adapter.
WAITING	<code>wait_for_event</code> step reached	Event Dispatcher delivers a matching event (WAITING → RUNNING); or timeout expires with <code>on_timeout: fail</code> (WAITING → FAILED) or timeout expires with <code>on_timeout branch:<id></code> (WAITING → RUNNING , jump to branch target).
COMPLETED	All steps finished	Terminal.
FAILED	Unhandled error or <code>on_timeout: fail</code>	Terminal (unless compensated in a future release).

WAITING is a *system-driven* suspension: the run is blocked on an external signal, not on user input. Clients **MUST** differentiate the two states in UX (e.g. “awaiting webhook” vs. “awaiting approval”) and **MUST NOT** send `exec/next` while the run is in **WAITING** state.

Pause / Resume Protocol

When the step execution loop reaches a `wait_for_event` step it **MUST**:

1. **Serialise run state** — write the full run snapshot (step index, all variable bindings, call stack, active listener registration) to the run store (embedded BoltDB / SQLite, same store used for checkpoint snapshots).
2. **Register event listener** — call `EventDispatcher.Register(runID, eventID, source, filter, payloadSchema, timeout)`. The Dispatcher returns a registration token.
3. **Suspend** — set `run.state = WAITING`, write a `step/waiting` trace event (includes `event.source`, `event.id`, timeout deadline), and return from the execution goroutine. The run goroutine is now idle; no `Next` call is outstanding.

When the Event Dispatcher delivers a matching event to a `WAITING` run it **MUST**:

1. Load the serialised run state from the run store via `store.Load(runID)`.
2. Validate the payload against `step.event.payload_schema` (JSON Schema Draft 2020-12). Validation failure → `FAILED` with schema violation detail.
3. Apply `step.event.filter` key-value predicates against the payload. Mismatch is treated as a non-matching event (listener remains registered; the event is discarded).
4. Apply `step.capture` mappings: extract `json.<path>` fields from the payload and write them into `run.variables`.
5. Write a `step/eventReceived` trace event (event source, sanitised payload summary, captured variable names).
6. Set `run.state = RUNNING` and call `ExecuteNextStep(run)`.

Executor Pseudocode

```
// Suspend path
ExecuteStep(step: WaitForEvent, run: Run):
    store.Persist(run)                // 1. durable checkpoint
    token = dispatcher.Register(      // 2. register listener
        run.id, step.event.id, step.event.source,
        step.event.filter, step.event.payload_schema,
        step.timeout)
    run.variables["gert.event."+step.event.id+".token"] = token // HMAC token
```

```

run.state = WAITING
emit(step/waiting, {event_id: step.event.id, deadline: now()+step.timeout})
return (suspended) // 3. goroutine exits

// Resume path
OnEventArrived(run_id, event_id, payload):
    run = store.Load(run_id) // 1. rehydrate
    step = run.currentStep() // (WaitForEvent step)
    ValidatePayload(payload, step.event.payload_schema) // 2. schema check
    if not MatchFilter(payload, step.event.filter): // 3. filter
        return // discard; listener stays registered
    ApplyCapture(payload, step.capture, run.variables) // 4. capture
    emit(step/eventReceived, {run_id, event_id}) // 5. trace
    run.state = RUNNING
    ExecuteNextStep(run) // 6. advance

// Timeout path
OnTimeout(run_id, event_id):
    run = store.Load(run_id)
    step = run.currentStep()
    switch step.on_timeout:
        case "fail":
            run.state = FAILED
            emit(step/failed, {reason: "wait_for_event timeout"})
        case "continue":
            run.state = RUNNING
            ExecuteNextStep(run)
        case "branch:<step-id>":
            run.state = RUNNING
            run.stepCursor = resolve(step-id)
            ExecuteNextStep(run)

```

Run Persistence Requirement

Runs in WAITING state MUST survive a `gert serve` process restart. The serialised run snapshot written in step 1 above MUST include:

- Current step index and depth.

- All variable bindings (including `gert.event.*` tokens).
- Call stack (for `include`-inlined steps).
- The active event listener registration (run ID, event ID, source, filter, timeout deadline as an absolute timestamp).

On `gert serve` restart, the server **MUST** re-register all persisted `WAITING` listener records with the Event Dispatcher before accepting new connections. Listeners whose timeout deadline has already passed are immediately moved to the timeout resolution path.

Serve-Only Constraint

`wait_for_event` is only valid when running under `gert serve`. If a runbook containing a `wait_for_event` step is executed via `gert run` (local CLI, non-server mode), the Runtime Core **MUST** fail at dispatch time with:

```
step type 'wait_for_event' requires gert serve mode
```

The Planner **MAY** emit a validation warning for `wait_for_event` steps in non-serve context (aiding the user before execution begins), but the enforcement point is the Runtime Core dispatch.

Security: HMAC Token for Webhook Transport

When `event.source: webhook`, each `wait_for_event` registration generates a cryptographically random one-time HMAC signing secret. The Event Dispatcher:

1. Generates 32 random bytes via `crypto/rand`.
2. Stores the secret alongside the listener registration in the run store.
3. Exposes the HTTP endpoint `POST /events/{run-id}/{event-id}` and validates incoming requests using `HMAC-SHA256` over the raw request body (header: `X-Gert-Signature: sha256=<hex>`).
4. Injects the secret into the run variable `gert.event.{event-id}.token` so the runbook author can communicate it to the external system (e.g. via a preceding `cli` step that calls `curl -data-urlencode`).

5. Rejects requests with invalid or missing signatures with HTTP 401.
6. Invalidates the secret after the first accepted delivery (one-time use).

The token is intentionally injected as a run variable (not hard-coded in the runbook YAML) so it is unique per run and never committed to source control.

3.3.5 Approve Step Executor Contract

The **approve** step is a first-class step type that pauses the run until one or more authorised identities submit a sign-off decision via the Input Provider interface. Two new capabilities — **business-day timeouts** (GAP-1) and **M-of-N quorum tracking** (GAP-2) — extend the base approval gate without changing its fundamental PAUSED-state semantics.

GAP-1: Business Calendar Engine

Motivation. Enterprise approval runbooks express deadlines in *business days* (“respond within 3 business days”), not wall-clock durations. The existing `timeout` field is wall-clock only and cannot satisfy this requirement.

Component responsibilities. The **Business Calendar Engine** is a stateless utility component within the Runtime Core (no external process, no network calls for built-in calendars). Its responsibilities are:

- Determine whether a given instant is a business moment (Monday–Friday, non-holiday in the specified calendar and timezone).
- Calculate the wall-clock deadline that is exactly N business days from a given start instant.
- Enumerate elapsed business days between two instants (for reporting and audit).
- Load and cache named business calendars; built-in calendars are compiled into the gert binary. Custom calendar definitions are deferred to a future release.

Go interface.

```
type BusinessCalendar interface {
    // IsBusinessDay reports whether t falls on a business day (Mon-Fri,
    ↪ non-holiday)
    // in the given timezone.
    IsBusinessDay(t time.Time, tz *time.Location) bool

    // AddBusinessDays returns the instant that is exactly n business days after
    ↪ t
    // in the given timezone. n must be >= 0.
    AddBusinessDays(t time.Time, days int, tz *time.Location) time.Time

    // ElapsedBusinessDays counts completed business days between start and end
    // in the given timezone. Returns 0 if end <= start.
    ElapsedBusinessDays(start, end time.Time, tz *time.Location) int
}
```

Built-in calendars for .

Calendar ID	Definition
default	Monday–Friday, no holidays. Appropriate for organisations that define their own holiday schedule externally.
us-federal	Monday–Friday, US federal public holidays (New Year’s Day, Martin Luther King Jr. Day, Presidents’ Day, Memorial Day, Juneteenth, Independence Day, Labor Day, Columbus Day, Veterans Day, Thanksgiving, Christmas).
uk-banking	Monday–Friday, UK bank holidays for England and Wales (New Year’s Day, Good Friday, Easter Monday, Early May Bank Holiday, Spring Bank Holiday, Summer Bank Holiday, Christmas, Boxing Day).

Custom calendar definitions (YAML-declared holiday lists referenced by a named ID) are **deferred to a future release** as a future-work item.

Timeout calculation at activation. The key architectural decision for business-day timeouts is a **one-time conversion at step activation**: when the runtime dispatches an **approve** step that carries `timeout_business_days`, it immediately calls:

```
deadline = calendar.AddBusinessDays(now, step.timeout_business_days, tz)
```

The resulting `deadline` is a plain `time.Time` value stored in the run state snapshot alongside the step. The existing timeout enforcement mechanism (the Event Dispatcher’s min-heap scheduler or the polling goroutine) then checks `now >= deadline` at regular intervals — *no business-calendar logic is needed at check time*. This design means the Business Calendar Engine is invoked exactly once per activate step and is entirely absent from the hot-polling path.

- **timezone** is resolved from the step’s `timezone` field (IANA timezone string, e.g. `America/New_York`). If absent, the operator’s local timezone is used; runbooks that require determinism SHOULD specify an explicit timezone.
- **calendar** is resolved from the step’s `business_calendar` field. Defaults to `default` if absent.
- If `timeout_business_days` and `timeout` (wall-clock) are both present on the same step, the *earlier* of the two computed deadlines applies. A Planner warning is emitted to flag the redundancy.

GAP-2: Quorum Approval Tracker

Motivation. Financial and regulatory runbooks require “3 of 5 department heads must approve” semantics. The previous `approvals.min` field was a bare integer with no explicit approver pool, making identity validation and audit reconstruction impossible.

Approval state per step instance. The runtime maintains an `ApprovalRecord` for each active approve step instance:

```
ApprovalRecord {
  step_id:    string
  run_id:     string
  mode:       all | any | quorum
  pool:       []string // eligible approver role names
  required:   int       // 0 = all (for mode: all / any)
  approvals:  []Approval // received approvals
  rejections: []Approval // received rejections
}
```

```
Approval {
  approver_id: string    // verified identity of the submitter
  role:        string    // which pool role the approver satisfied
  timestamp:   time.Time
  notes:       string    // optional free-text from the approver
}
```

The `ApprovalRecord` is persisted to the run store at each update (same durable store as run snapshots) so it survives a process restart.

Progression rules.

Mode	Proceed when
all	<code>len(approvals) == len(pool)</code> — every pool member has approved.
any	<code>len(approvals) >= 1</code> — at least one pool member has approved.
quorum	<code>len(approvals) >= required</code> — the required count is reached.

Deduplication: each pool member may submit **exactly one** approval. If the same `approver_id` submits a second approval for the same step instance, the runtime rejects the submission with `ErrDuplicateApproval` and returns an error to the caller — it is not silently ignored.

Rejections are recorded in the audit trail and do *not* block progression by themselves, *unless* quorum becomes mathematically impossible (see Deadlock Detection below).

Deadlock detection. After *every* approval or rejection event, the runtime evaluates:

```
remaining_possible = len(pool) - len(rejections) - len(approvals)
if (len(approvals) + remaining_possible) < required:
    // quorum is impossible - fail immediately
    step.state = FAILED
    emit(step/failed, {reason: "QuorumImpossible", ...})
```

For mode: `all`, `required` is treated as `len(pool)` for this check. For mode: `any`, the check is skipped (one approval always suffices unless the pool is empty, which is a

schema validation error). If the check fires, the step fails immediately with error code `QuorumImpossible`; the `on_timeout` policy does *not* apply.

Audit trail. Every approval and rejection is appended to the run's append-only trace as a `approve/decision` trace event containing: `approver_id`, `role`, `decision` (`approve` or `reject`), `timestamp` (RFC 3339), and `notes`.

This satisfies electronic-signature audit requirements including SOC 2 Type II and FDA 21 CFR Part 11.

Input Provider integration. Approval decisions are delivered via the Input Provider JSON-RPC interface. The `approve` step executor registers a pending input request of type `input/submitApproval` with the fields:

```
{
  "run_id":  string,           // identifies the run
  "step_id": string,           // identifies the approve step
  "decision": "approve"|"reject",
  "notes":   string            // optional
}
```

Before recording the decision, the runtime:

1. Resolves the caller's `approver_id` from the authenticated session (or the `-as` flag in CLI mode).
2. Verifies that `approver_id` maps to at least one role in `approvals.pool`. Unknown identities receive `ErrNotAuthorised`.
3. Checks for duplicate submission (`ErrDuplicateApproval` if already recorded).
4. Appends the `Approval` or rejection record to the `ApprovalRecord`.
5. Re-evaluates the progression rule and the deadlock-detection invariant.
6. Persists the updated `ApprovalRecord` to the run store.

Composition: Business-Day Timeout with Quorum Approval

Both GAP-1 and GAP-2 **compose freely**. A step may declare `timeout_business_days`, `timezone`, `business_calendar`, and `approvals (mode: quorum)` simultaneously. At step activation:

1. The Business Calendar Engine computes the wall-clock **deadline**.
2. The Quorum Tracker initialises an empty `ApprovalRecord`.
3. The step enters PAUSED state; the timeout min-heap is armed with the deadline.

The two mechanisms then run **independently**: the Quorum Tracker processes each incoming approval/rejection without consulting the calendar, and the timeout scheduler checks `now >= deadline` without consulting the quorum state. Whichever condition fires first wins: quorum reached → step proceeds; deadline fired → `on_timeout` policy governs the outcome.

3.3.6 Display Step Executor Contract

A `type: display` step renders template content to the operator's terminal without blocking for input. It is the standard primitive for presenting collected data, reports, and status summaries mid-flow.

Schema

```
- step:
  id: show_health_report
  type: display
  title: "Service Health Report"      # step-list label
  display:
    content: "{{ .report }}"          # template body (required)
    format: text                      # text / markdown (default: text)
```

Fields

Field	Type	Required	Description
<code>content</code>	string	Yes	Template body rendered via <code>text/template</code> against the current variable scope. YAML <code> </code> and <code>></code> blocks are supported. Must not be empty.
<code>format</code>	enum	No	Render hint for the display surface. <code>text</code> (default): pre-formatted plaintext, whitespace preserved. <code>markdown</code> : render with Markdown formatting (bold, lists, headers, code blocks).

Execution Semantics

The display executor:

1. Evaluates `content` through the template engine (same evaluator used by `title`, `subtitle`, and `capture` expressions).
2. Writes the rendered string to the operator's output stream. In CLI mode this is stdout; TUI clients may render into a styled panel.
3. Returns immediately — the step never blocks for acknowledgment.
4. Records a `step/completed` trace event. The rendered content is included in the event payload subject to the run's redaction policy.
5. Never mutates the variable scope. `capture`: is disallowed on display steps (semantic validation rejects it).

Surface Behaviour

Surface	Behaviour	Notes
CLI (TTY)	<code>fmt.Fprintln(stdout, rendered)</code>	ANSI passthrough if format is markdown
CLI (pipe / CI)	Same as TTY — output is emitted to stdout	Markdown stripped to plain-text
TUI	Rendered in the step panel / output card	<code>format: markdown</code> enables styled rendering
Dry-run	Emits trace event; no stdout write	Content is logged as-is

Disallowed Common Fields

The following common **Step** fields are incompatible with `type: display` and are rejected by semantic validation:

- `capture`: — `display` produces no output values; nothing to bind.
- `retry`: — `display` cannot fail in a retrieable sense.
- `contract`: — `display` has no side effects; a contract declaration is misleading.

Runbook Example

```
flow:
  - iterate:
    id: collect_health_loop
    over: services
    as: svc
    steps:
      - step:
        id: check_svc
        type: include
        include:
          runbook: check-service.runbook.yaml
          with:
            service_host: "{{ .svc }}"
```

```

        capture:
            last_status: service_status

- step:
    id: accumulate
    type: noop
    capture:
        report: "{{ .report }}"{{ .svc }}: {{ .last_status }}\n"

- step:
    id: show_report
    type: display
    title: "Health Report"
    display:
        content: |
            Service Health Report
            =====

            {{ .report }}
    format: text

- step:
    id: done
    type: end
    outcome:
        category: resolved
        code: health_report_complete

```

The `type: display` step is the correct primitive for presenting accumulated data to the operator. Using `type: noop` with a multiline `title:` is explicitly an anti-pattern: `title` is a navigation label for the step list, not a display surface.

3.3.7 Collector Field Validation Contract

When the executor receives evidence from a `SubmitEvidence` call for a `collector` step, it validates each field value against the field's declared type and constraints *before* writing any value to run state or the trace.

Validation Algorithm

For each field in submission order:

1. **Required check** — if the field is `required: true` and the submitted value is absent or empty, record a validation error.
2. **Type coercion and format check** — attempt to parse or coerce the value to the declared type (see per-type rules below). A parse failure is a validation error.
3. **Constraint check** — evaluate `validation.min`, `validation.max`, `validation.step`, and option-list membership as applicable. A constraint violation is a validation error.
4. **Re-prompt** — if any field produced a validation error, the executor sends a new `CollectorRequired` event back to the input provider with an `errors` map populated (field name → human-readable error message). The provider **SHOULD** re-render the form highlighting the invalid fields. The re-prompt counts as one retry attempt.
5. **Max retries** — if validation still fails after **3** re-prompt attempts, the executor fails the step: `step.state = FAILED`, writes a `step/failed` trace event, and populates the `gert.error` reserved run variable with structured error detail (see below).

Per-Type Validation and Storage Rules

Type	Validation	Variable Storage
text	Required check; then optional length and pattern constraints: <code>validation.min_length</code> , <code>validation.max_length</code> (character count); <code>validation.pattern</code> (RE2 regex — see below); <code>validation.pattern_hint</code> (shown on pattern failure). <code>multiline: true</code> is a UI hint only; no effect on validation or storage. If <code>format: yaml</code> or <code>format: json</code> is declared, structural parse validation is applied <i>after</i> the pattern check (see below).	JSON string (raw; parsed form not stored)
number	Parse as IEEE 754 float64 (reject non-numeric input). Check <code>validation.min</code> and <code>validation.max</code> (both inclusive). Check <code>validation.step</code> : $(\text{value} - \text{min}) \% \text{step} < \text{epsilon}$.	JSON number (float64)
integer	Parse as int64; reject fractional input. Same <code>min</code> / <code>max</code> / <code>step</code> checks as <code>number</code> .	JSON number (int64)
date	Must match YYYY-MM-DD (RFC 3339 full-date). Validate <code>validation.min</code> / <code>max</code> as date strings (lexicographic comparison is correct for ISO dates). Wall clock at submission time; timezone UTC.	JSON string (YYYY-MM-DD)
datetime	Must be valid RFC 3339 (ISO 8601 with time offset). Validate <code>validation.min</code> / <code>max</code> as RFC 3339 strings; normalise to UTC before comparison. Wall clock at submission time.	JSON string (RFC 3339, e.g. 2026-04-18T14:00:00Z)
boolean	Truthy/falsy coercion: <code>true</code> , <code>"true"</code> , <code>1</code> , <code>"yes"</code> → <code>true</code> ; <code>false</code> , <code>"false"</code> , <code>0</code> , <code>"no"</code> → <code>false</code> . Any other value is a validation error. No <code>min</code> / <code>max</code> constraints.	JSON boolean
select (multiple: false)	Submitted value must exactly match one of the declared <code>options[*].value</code> strings. If <code>options_from</code> was used, validation is against the dynamically fetched list (§14.5.4).	JSON string
select (multiple: true)	All submitted values must be members of the	JSON array of strings

`multiline: true` **Hint**

The `multiline: true` flag on a `text` field is a **UI hint only**. The executor treats the submitted value as a plain string regardless of this flag. Input providers **SHOULD** render a multi-line textarea (or open `$EDITOR`) rather than a single-line input box when `multiline: true` is set.

text Field Pattern Validation

`text` fields support optional length, pattern, and structural-format constraints applied *after* the required check. The full sequence for a `text` field is:

1. **Required check** — same as all field types (step 1 of the general algorithm above).
2. **Min-length check** — if `validation.min_length` is declared, fail if `len(value) < min_length` (character count, not bytes).
3. **Max-length check** — if `validation.max_length` is declared, fail if `len(value) > max_length`.
4. **Pattern check** — if `validation.pattern` is declared, compile and test the full value against the RE2 regex. On failure, the re-prompt error message is `validation.pattern_hint` if present; otherwise the generic message “*Value does not match required format*” is used.
5. **Format check** — if `format: yaml` is declared, attempt `yaml.Unmarshal([]byte(value), &interface{})`. On error, re-prompt with “*Value must be valid YAML.*”. If `format: json` is declared, attempt `json.Unmarshal([]byte(value), &interface{})`. On error, re-prompt with “*Value must be valid JSON.*”. Both checks are **parse-only**: the parsed result is discarded; the value remains a raw string in the run variable. Downstream steps use the `fromYAML` and `fromJSON` template functions for traversal (see §4.3.3). If both `validation.pattern` and `format` are declared, both must pass; the pattern check (step 4) runs first.
6. **Re-prompt** — any failure at steps 2–5 triggers a re-prompt. The 3-attempt limit is a **combined total** across all validation steps: if the operator fails step 2 once, step 4 once, and step 5 once, the step fails on the third attempt regardless of which check failed each time.

RE2 vs ECMA 262. The gert schema specification uses the term *ECMA 262 regex* for compatibility with JSON Schema tooling and browser-based UI validators. At runtime, gert uses Go’s `regexp` package, which implements RE2 semantics — not a full ECMA 262 engine.

Patterns **must** be valid RE2 expressions. ECMA 262 features absent from RE2 (lookahead, lookbehind, backreferences) are **rejected at schema load time** with a validation error naming the offending field:

```
error: step "gather-details", field "ticket_ref":  
  validation.pattern uses lookahead (?=...) which is not supported in RE2;  
  replace with an anchored RE2 equivalent
```

Runbook authors targeting both the VS Code extension (which uses a JavaScript regex engine) and the gert runtime (RE2) should restrict patterns to the safe common subset: character classes, quantifiers, anchors, alternation, and non-capturing groups.

Conditional Field Evaluation Contract

`collector` fields MAY declare a **when** expression. The executor evaluates **when** before rendering each field, determining whether the field is shown, prompted, and bound to a variable.

Evaluation model. Fields are evaluated left-to-right in array order:

1. Before rendering field *i*, the executor evaluates its **when** expression against:
 - **Collected values so far** — the values returned by the input provider for fields $0 \dots i - 1$ in the current step (those fields whose **when** evaluated to **true** and were rendered).
 - **Global variable scope** — the run’s accumulated variable bindings from all previously completed steps.
2. If **when** evaluates to **false**: the field is **not rendered, not prompted, and its variable binding is skipped entirely**. The field’s `id` is NOT added to the step’s variable scope for this run.
3. If **when** evaluates to **true**, or if **when** is absent (unconditionally visible), the field is rendered and validated normally.

This is a **pull model**: the executor drives field-by-field evaluation in order; the input provider renders one field at a time, or a full form with some fields conditionally hidden — both are valid provider strategies.

Input provider integration. Two delivery strategies are both valid; providers declare their preference via capability flags (see §14.5.3):

- **Full-form delivery** (recommended for rich UI providers): the executor sends the complete `fields` array in a single `provider/collect` request. Each field carries a pre-evaluated `visible: bool` flag reflecting the current `when` result at dispatch time. Rich UI providers SHOULD implement dynamic client-side show/hide as the user fills in earlier fields, re-evaluating `when` locally against live form state so the form responds without a round-trip to the executor.
- **Streaming delivery** (recommended for CLI providers): the executor sends only the currently-visible fields, evaluating each `when` expression in real time as previous field values are received. This is the natural mode for the built-in `prompt` provider, which renders fields sequentially at the terminal. Fields where `visible: false` are simply never sent.

Variable binding on skip. When a field is skipped (`when = false`):

- Its `id` is NOT added to the step's variable bindings.
- Template expressions in subsequent steps that reference the skipped field's variable resolve to the *zero value*: `"` for string, `0` for number, `false` for boolean.
- If the skipped variable is referenced with `.require` in a downstream template expression, the executor produces a template error at that step.
- Runbook authors should guard downstream template expressions that may depend on conditionally-bound fields:

```
{{/* Safe: test for empty before using a conditionally-bound variable */}}
{{ if .ticket_ref }}JIRA: {{ .ticket_ref }}{{ else }}(no ticket){{ end }}
```

Load-time validation — forward reference check. At schema load time, the parser performs a forward-reference check for every field that declares **when**:

1. Parse the **when** expression and extract all variable references.
2. For each referenced variable, determine whether it is bound by a field declared *earlier* in the same **fields** array (by array index).
3. If any referenced variable is bound by a field declared *later* in the same **fields** array, **reject the runbook at load time** with a validation error:

```
error: step "gather-details", field "region" (index 2):
  when expression references "country" which is bound by field at index 4;
  forward references in when expressions are not allowed
```

Variables from the global scope (prior steps) and the run context are always valid references in **when** expressions; only same-step forward references are rejected.

Variable Storage — Downstream Expression Semantics

Typed storage ensures that template comparison operators in downstream steps work without manual coercion:

```
{{/* number/integer: .amount is float64, not a string */}}
{{ if gt .amount 50000.0 }}HIGH_VALUE{{ end }}

{{/* boolean: .confirmed is bool, not "yes"/"no" */}}
{{ if .confirmed }}PROCEED{{ end }}

{{/* select/multiple or choice/multiple: .systems is []string */}}
{{ if has .systems "payments-api" }}CRITICAL{{ end }}

{{/* date/datetime: .start_date is ISO 8601 string; parseable by time.Parse */}}
{{ if lt .start_date "2026-07-01" }}BEFORE_CUTOVER{{ end }}
```

Error Variable Schema

When a collector step exhausts its re-prompt attempts and fails, the executor populates the **gert.error** reserved run variable:

```
{
  "step_id": "gather-approval-details",
  "kind": "collector.validation_failed",
  "attempts": 3,
  "fields": {
    "amount": "value 49000 is below minimum 50000",
    "start_date": "2025-12-01 is before minimum 2026-01-01",
    "department": "\"legal\" is not a valid option"
  }
}
```

The `gert.error` variable is a reserved JSON object populated on any step failure. A subsequent `branch` step may inspect it to route to an error-handling path. Each new step failure overwrites the previous value.

Ephemeral Field Handling

A collector field may declare `ephemeral: true` to indicate that its value is sensitive and must never be written to any durable store.

Audit redaction. After a field value passes all validation checks, the executor writes the step's output to the JSONL trace file. For fields where `ephemeral: true`:

- The JSONL audit record stores the string literal "[REDACTED]" as the field value — not the actual value.
- The *actual* value IS placed in the in-memory `VariableScope` for the duration of the run and is available to downstream steps via normal template expressions.
- The actual value is **NEVER written to disk** in any log, trace, snapshot, or checkpoint file. This includes the run trace (`trace.jsonl`), the per-step checkpoint (`snapshots/<stepID>.json`), and the evidence hash record.

Go interface contract. The in-memory `VariableScope` type MUST expose the method:

```
IsEphemeral(key string) bool
```


The JSONL trace writer calls `IsEphemeral` for every variable before serialising a `step/completed` or `collector/submitted` event. If the method returns `true`, the writer substitutes "[REDACTED]" for the value in the emitted JSON. No other layer in the stack applies this substitution — the redaction responsibility belongs exclusively to the trace writer.

Tool step compatibility. Ephemeral values MAY be passed as arguments to `type: tool` steps. The tool receives the plaintext value. This is intentional: `ephemeral: true` is an *audit redaction* feature, not end-to-end encryption. Runbook authors are responsible for using ephemeral fields only with tools that handle the value securely.

Suspension/resume constraint. When a run is suspended (e.g., a `wait_for_event` or `approve` step puts the run into `WAITING` state), the in-memory `VariableScope` is not persisted. Because ephemeral values exist only in memory, they are **dropped** on suspension. On resume, any downstream step that requires an ephemeral variable will find it absent from the reconstructed scope.

Constraint: ephemeral variables and suspension

Steps that depend on ephemeral variables **MUST** appear before any `wait_for_event` or `approve` step that could suspend the run. The schema validator **SHOULD** emit a load-time warning (not a hard error) if an ephemeral variable is referenced in a step that follows a suspendable step in the same runbook's execution path.

3.3.8 Cancellation

`RunHandle.Cancel(ctx, reason)` initiates graceful cancellation:

1. The currently-executing subprocess (if any) receives `SIGTERM` followed by `SIGKILL` after a 5-second grace period.
2. The runtime writes a `run/cancelled` trace event with the reason and actor.
3. Subsequent `Next` calls return `ErrRunCancelled`.
4. The run state is checkpointed as `cancelled` (resumable).

The cancellation contract does not invoke compensation handlers. Saga/compensation (stepping backward through a compensation chain on failure) is deferred to a future release as specified in §09 (Open Questions).

3.3.9 Crash Recovery

If the gert process crashes mid-run, the trace and checkpoint files remain on disk in a consistent state due to the append-only, sync-after-write trace discipline. On restart, the user (or the VS Code extension) can call `Runtime.Resume(runID)` to continue from the last completed step. The runtime reads the checkpoint to reconstruct run state. Steps already recorded as completed in the trace are not re-executed.

3.4 Concurrency Model

3.4.1 Per-Process Run Isolation

The runtime is **single-run-per-process**. Each `gert exec` invocation is a separate OS process. `gert serve` is the exception: it is a long-running process that may host multiple sequential runs (one at a time) for a single VS Code extension session. Concurrent runs within a single `gert serve` instance are explicitly out of scope for the (see §09).

Rationale. Governance enforcement, file-level trace writes, and subprocess management are dramatically simpler with one run per process. Process isolation also provides a natural crash boundary: a failed run cannot corrupt the state of a concurrent run.

3.4.2 Goroutine Model

Within a single run, the concurrency model is:

- **One goroutine per run:** the step execution loop runs on a single goroutine. Steps execute sequentially; there is no goroutine-per-step parallelism.
- **Event fan-out goroutine:** a second goroutine reads from an internal event buffer and writes to the `Events()` channel. This decouples event emission (which must be synchronous with step execution) from event consumption (which may be slow, e.g., a WebSocket write).

- **Subprocess goroutine:** when a `cli` step spawns a subprocess, a goroutine is used to drain `stdout/stderr` concurrently, preventing deadlock on full pipe buffers.
- **Tool RPC goroutine:** tool invocations over JSON-RPC or MCP use a goroutine per call, managed by the Tool Runtime's connection pool.

3.4.3 Thread Safety at Component Boundaries

- `RunHandle` is **not safe for concurrent use**. The adapter must serialize all calls. `gert serve` enforces this with a per-session mutex.
- The `Events()` channel is safe for a single consumer goroutine.
- The `GovernanceEngine` is **read-only after construction** and therefore safe for concurrent read access (relevant if future versions support parallel steps).
- The Trace Writer is protected by an internal mutex; write calls are safe to issue from any goroutine within the run.

3.5 gert serve Integration

`gert serve` is the HTTP server that powers the VS Code extension, the embedded web preview, the MCP tools layer, and any future external client. It exposes a JSON-RPC 2.0 endpoint at `POST /rpc` alongside REST and Server-Sent Events (SSE) endpoints. In the The architecture it is classified as an **adapter in the API/Adapter Layer**.

3.5.1 Architectural Position

`gert serve` is a process that:

1. Starts a long-lived HTTP listener (default `:7778`) exposing JSON-RPC 2.0 at `POST /rpc` alongside REST endpoints (`POST /runs`, `GET /runs/{id}/state`, `DELETE /runs/{id}`, ...) and Server-Sent Events streams (`GET /events`, `GET /runs/{id}/state`, `GET /runs/{id}/interactions`).
2. On receiving `exec/start` (or `POST /runs`), calls `Runtime.Start()` and stores the resulting `RunHandle` in a session map (keyed by `RunID`).

3. On receiving `exec/next`, calls `RunHandle.Next()` and returns the `StepResult` as the RPC response.
4. Drains the `Events()` channel in a background goroutine and forwards each event to subscribed SSE clients as a named SSE event whose data payload is the event's JSON representation. JSON-RPC clients receive the same events as notifications on a long-lived RPC channel.
5. On receiving `exec/approve`, `exec/submitEvidence`, or `exec/cancel`, dispatches to the corresponding `RunHandle` method.

3.5.2 Event Forwarding

Every event emitted by the Runtime Core is forwarded to subscribed clients. Two transports are offered concurrently against the same in-process event bus:

- **SSE** (`text/event-stream`): the VS Code extension and the web preview attach an `EventSource` to `GET /events` (or scope-narrowed streams like `GET /runs/{id}/state` and `GET /runs/{id}/interactions`). Each event is emitted with an `id` header carrying the event sequence so clients can resume with `Last-Event-ID` after a transient disconnect. A 15 s comment-frame heartbeat (`: hb\n\n`) keeps idle connections alive through reverse proxies.
- **JSON-RPC notifications**: clients holding a long-lived RPC channel receive each event as a notification (no `id` field) with method `event/eventType`. The payload is identical to the SSE data field.

A WebSocket endpoint (`GET /ws`) is exposed for byte-stream consumers that prefer framed messages over SSE; it carries the same event schema.

3.5.3 Crash Recovery in `gert serve`

If the VS Code extension reconnects after a crash (e.g., the `gert` process was killed), `gert serve` exposes `exec/list` and `exec/resume` methods. The extension discovers the interrupted run, calls `exec/resume`, and receives a new `RunHandle` over the same session. Mid-run SSE disconnects are recovered by the browser/`EventSource` layer automatically: the client reconnects with the most recent `Last-Event-ID`, and `gert serve` replays buffered events past that sequence number. The trace and checkpoint on disk ensure no steps are lost.

3.5.4 Event Dispatcher

The **Event Dispatcher** is a new runtime component that lives exclusively inside the `gert serve` process. It is not instantiated by `gert run`.

Responsibilities

- **Listener lifecycle** — maintain an in-memory registry of active event listeners (indexed by `run_id + event_id`). Each entry records: source type, filter predicates, payload schema URL, timeout deadline, and HMAC secret (webhook source only). Persist the registry to the run store so it survives process restart.
- **Webhook transport** — expose `POST /events/{run-id}/{event-id}` authenticated with HMAC-SHA256 (see security note in §3.3.4). Parse the request body as JSON and dispatch to the matching listener. Return `HTTP 202 Accepted` on success; `404` if no listener is registered; `401` on signature failure; `410 Gone` if the run is no longer in `WAITING` state.
- **Channel transport** — maintain an in-process `map[string]chan Payload` keyed by `event.id`. Any component within the same `gert serve` process can post to a named channel via `EventDispatcher.Send(eventID, payload)`.
- **Message broker transport** — delegate to a pluggable `MessageBroker` interface (implementation-defined for ; a no-op stub ships by default). The interface follows the same pattern as the `ToolTransport` plug-in model: a registered factory function keyed by broker type string.
- **Signal transport** — register `os/signal` handlers scoped to a specific run. Signal handling is multiplexed: the same OS signal may be registered by multiple active listeners (e.g. `SIGUSR1` on two concurrent waiting runs). Each listener receives its own notification; signal handlers are deregistered on listener expiry or run completion.
- **Timeout enforcement** — maintain a min-heap of timeout deadlines. A background goroutine fires on the earliest deadline and invokes `OnTimeout(runID, eventID)`.

Go Interface

```
type EventDispatcher interface {
    // Register creates a listener for an inbound event on a waiting run.
```

```

// Returns a one-time HMAC token (non-empty for webhook source only).
Register(ctx context.Context, reg ListenerRegistration) (token string, err
↳ error)

// Deregister removes a listener (called on run completion or cancellation).
Deregister(runID, eventID string) error

// Send posts a payload to a named in-process channel listener.
Send(eventID string, payload json.RawMessage) error

// RestoreFromStore re-registers all WAITING listeners after process
↳ restart.
RestoreFromStore(store RunStore) error
}

type ListenerRegistration struct {
    RunID          string
    EventID        string
    Source         EventSource      // webhook / message / signal / channel
    Filter         map[string]string
    PayloadSchema  string           // JSON Schema URL, may be empty
    TimeoutAt      time.Time
    OnTimeout      TimeoutPolicy    // fail / continue / branch:<step-id>
}

```

Availability Constraint

The Event Dispatcher is instantiated by `gert serve` only. It is **not** part of the `Runtime` interface exposed to step executors directly; instead, the `Runtime Core` receives a `DispatcherHandle` at construction time (nil in `gert run` mode). The `wait_for_event` dispatch path checks for a nil handle and fails immediately with the `serve-mode` error before attempting any state serialisation.

3.5.5 RPC Method Summary

<code>exec/start</code>	→ Start a new run (returns <code>RunID</code> , first <code>StepResult</code>)
<code>exec/next</code>	→ Advance run by one step (returns <code>StepResult</code>)
<code>exec/approve</code>	→ Submit approval decision <code>for</code> current gate

[illegible]

Chapter 4

Schema

This chapter is the normative specification for all gert data structures: runbook documents, tool definitions, and provider definitions. It defines the contracts that Brian’s parser, Ken’s runtime, and the VS Code extension must implement.

The design philosophy is *minimal core, extensible periphery*: the core language fields that the runtime must understand are small, typed, and stable; everything else is declared as an extension with an explicit namespace. Gert draws a hard boundary and enforces it at parse time.

4.1 Schema Identity and Versioning

4.1.1 apiVersion Values

Every gert document carries a mandatory **apiVersion** field that identifies both the document kind and the schema version. The current values are:

Document kind	apiVersion
Runbook	runbook/v1
Tool definition	tool/v1
Provider definition	provider/v1

The format is `<kind>/<major>`. Minor and patch changes do not alter the **apiVersion** string; they are tracked by the schema artifact version (see §4.9).

4.1.2 Self-Description via \$schema

Following JSON Schema [29] best practice and the pattern used by GitHub Actions, Argo Workflows, and Kubernetes CRDs, every runbook document *should* carry a `$schema` field pointing to the canonical JSON Schema URL:

```
$schema: "https://schemas.gert.dev/runbook/v1.json"
apiVersion: runbook/v1
id: service-health
name: Service Health Check
...
```

The `$schema` field is optional at runtime (the runtime uses `apiVersion` for dispatch) but is *strongly recommended* because:

- VS Code and JetBrains IDEs provide schema-backed auto-completion and inline validation when `$schema` is present.
- The schema URL carries an implicit version contract: updating the URL updates the validation rules visible to tooling.
- It enables offline validation with any JSON Schema Draft 2020-12 validator (ajv, jsonschema, check-jsonschema).

Schema URLs for each document kind:

```
https://schemas.gert.dev/runbook/v1.json
https://schemas.gert.dev/tool/v1.json
https://schemas.gert.dev/provider/v1.json
```

4.1.3 Versioning Strategy

gert schema versioning follows a simplified semantic versioning model applied to the `apiVersion` major component:

- **Major increment** (e.g. `runbook/v1` → `runbook/v1`): A breaking change. The parser *must not* silently accept the old format. Automated migration is provided via `gert migrate`.

- **Minor/patch changes within a major:** Additive, backward-compatible. New optional fields may be added. Existing fields may not be removed or renamed. The JSON Schema artifact version tracks these with a semver suffix (e.g. `runbook/v1.json` at artifact version `2.1.0`).

What constitutes a breaking change A breaking change is any of:

- Renaming or removing a required field.
- Narrowing an existing enum (removing a valid value).
- Changing a field type (e.g. `string` $\rightarrow$ `object`).
- Changing the meaning of a field such that existing runbooks produce different runtime behavior.
- Removing a step type.

Adding new optional fields, adding new enum values, and loosening existing constraints are *not* breaking changes.

4.2 Core Runbook Fields

4.2.1 Top-Level Field Inventory

The runbook document has the following top-level fields:

Field	Type	Required	Description
<code>\$schema</code>	string	No	JSON Schema URL (recommended)
<code>apiVersion</code>	string	Yes	Must be <code>runbook/v1</code>
<code>id</code>	string	Yes	Unique machine identifier (slug)
<code>name</code>	string	Yes	Human-readable display name
<code>kind</code>	enum	No	<code>mitigation</code> <code>reference</code> <code>composable</code> <code>rca</code>
<code>description</code>	string	No	Short prose summary
<code>vars</code>	map	No	Default variable values
<code>inputs</code>	map	No	Input declarations (§4.3)
<code>outputs</code>	map	No	Output declarations (§4.3)
<code>toolRefs</code>	array	No	Tool references (§4.2.2)
<code>imports</code>	map	No	Named runbook imports
<code>defaults</code>	object	No	Default step settings
<code>governance</code>	object	No	Governance policy
<code>prose</code>	object	No	Human-readable TSG sections
<code>metadata</code>	map	No	Arbitrary key-value annotations
<code>expand</code>	enum	No	Default include-expansion mode for this runbook: <code>eager</code> <code>lazy</code> <code>auto</code> . Empty inherits the platform default (<code>lazy</code>). See §4.4.9.
<code>flow</code>	array	Yes	Execution tree (TreeNode array)

id field The `id` is a slug: lowercase alphanumeric with hyphens, matching `^[a-z][a-z0-9-]*[a-z0-9]$`. It uniquely identifies a runbook within a project and is used by the `include` step and the trace record. It must be stable across renames; changing it is a breaking change for any runbook that includes it by `id`.

kind values

- **procedure** — Workflow procedure; governance defaults to `require-approval` for destructive effects.
- **reference** — Documentation-first runbook; dry-run by default.

- **composable** — Sub-runbook intended for include from other runbooks; not directly executable from CLI.
- **rca** — Root-cause analysis template.

4.2.2 toolRefs

toolRefs is a typed array that allows aliasing and explicit path overrides:

```
toolRefs:
- name: curl
- name: nslookup
- name: k8s
  path: ./tools/k8s.tool.yaml    # override default discovery
  alias: kubect1-wrapper        # optional local alias
```

The default discovery path for **name: foo** is **tools/foo.tool.yaml** relative to the runbook file. The **path** field overrides this. The **alias** field lets the runbook reference the tool by a local name that differs from the tool's declared **meta.name**.

4.2.3 Minimal Valid Runbook

```
$schema: "https://schemas.gert.dev/runbook/v1.json"
apiVersion: runbook/v1
id: hello-world
name: Hello World

flow:
- step:
  id: greet
  type: cli
  title: Print greeting
  command: echo
  args: ["Hello, world!"]
  capture:
    output: stdout
- step:
  id: done
```

```
type: end
title: Complete
outcome:
  category: resolved
  code: success
```

4.2.4 Complete Annotated Example

```
$schema: "https://schemas.gert.dev/runbook/v1.json"
apiVersion: runbook/v1
id: service-health-check
name: Service Health Check
kind: mitigation
description: |
  Validate DNS resolution and HTTP endpoint reachability
  for a named service. Escalates on failure.

vars:
  default_timeout: "30s"

inputs:
  hostname:
    type: string
    required: true
    description: "Hostname to check"
    from: prompt
  request_id:
    type: string
    required: false
    description: "Request tracking ID"
    from: tracking.id      # provider-resolved

outputs:
  health_status:
    type: string
    description: "Final health assessment"
    from: captures.http_result
```

```
toolRefs:
  - name: curl
  - name: nslookup

governance:
  rules:
    - effects: ["network.probe"]
      action: allow
    - effects: ["service.restart"]
      action: require-approval
      min_approvers: 2
  redact:
    - pattern: "(?i)Authorization:\\s*\\S+"
      replace: "Authorization: <redacted>"

defaults:
  timeout: "60s"

prose:
  safety: |
    Run this runbook only when a request is active. Do not
    loop - each execution creates a trace record.
  prerequisites: "Requires curl >= 7.80 and nslookup on PATH."

metadata:
  team: platform-sre
  runbook-url: "https://wiki.example.com/runbooks/service-health"
  severity: P2

flow:
  - step:
      id: check_dns
      type: tool
      title: "Resolve DNS: {{ .hostname }}"
      tool:
        name: nslookup
        action: lookup
        args:
```

```
    host: "{{ .hostname }}"
capture:
  dns_result: stdout
contract:
  effects: ["network.probe"]
  reads: ["hostname"]
  writes: ["dns_result"]

- step:
  id: check_http
  type: tool
  title: "HTTP health check: {{ .hostname }}"
  tool:
    name: curl
    action: head
    args:
      url: "https://{{ .hostname }}/healthz"
  capture:
    http_result: stdout
  timeout: "{{ .default_timeout }}"

- step:
  id: evaluate
  type: branch
  title: Evaluate health results
  branches:
    - condition: 'str.contains(http_result, "200")'
      label: Healthy
      steps:
        - step:
            id: resolved
            type: end
            title: Service is healthy
            outcome:
              category: resolved
              code: http_ok
    - condition: '!str.contains(http_result, "200")'
      label: Unhealthy
```

```
steps:
  - step:
      id: escalate
      type: end
      title: Escalate to on-call
      outcome:
        category: escalated
        code: http_fail
```

4.3 Input and Output Declarations

4.3.1 Input Declarations

Inputs declare variables that must be resolved before execution begins. In the schema the `inputs` map is a top-level field (promoted from `meta.inputs`).

```
inputs:
  <name>:
    type: string | number | boolean | secret
    required: true | false           # default: true
    description: "..."/>

```

type field The `type` field declares the value's type. Supported values:

- **string** — Default. UTF-8 text.
- **number** — Integer or float, passed as string in templates.
- **boolean** — `true` or `false`.
- **secret** — String, redacted in traces and log output.

from: binding syntax The **from** field specifies how the value is resolved. Binding forms:

Binding	Behavior
prompt	Ask the operator interactively at runtime
env.<NAME>	Read from environment variable NAME
file.<path>	Read first line from a local file
<provider>.<field>	Delegate to a registered input provider (e.g. tracking.id)
var.<name>	Copy from another variable or capture

If **from** is omitted, the field must have a **default** value or the runtime will reject the runbook at semantic validation time.

4.3.2 Output Declarations

Outputs declare values that a composable runbook exposes to its callers. With **type: include**, the caller's variable space is shared directly; explicit output declarations are reserved for the future **type: call** step type (future scope, isolated scope with explicit I/O mapping).

```
outputs:
  <name>:
    type: string | number | boolean | secret
    description: "..."/>

```

The **from: captures.*** binding is the only supported form in the schema. The named capture must be written by at least one step in the runbook; the semantic validator checks this.

4.3.3 Expression Language

Gert uses **expr-lang/expr** [11] as the expression language for all conditional fields (**when**, **condition**, **until**), providing natural, Python-like infix expressions. The choice of **expr** is grounded in three facts:

1. Infix notation (**a == "x" && b > 1**) is significantly more readable and familiar to operators than prefix notation (**{{ and (eq .a "x") (gt .b 1) }}**).

2. `expr` provides a safe, sandboxed evaluation environment with no access to Go internals, improving security posture.
3. Expressions are localized to `when/condition` fields; shell command interpolation (`args`, `title`) continues to use Go templates.

Syntax overview Expressions operate on the *run variable map*: all `vars`, resolved `inputs`, and accumulated `captures` merged into a single flat map. Variables are referenced directly by name (`hostname`), without prefix notation.

Operators:

- Comparison: `==`, `!=`, `<`, `>`, `<=`, `>=`
- Logical: `&&`, `||`, `!`
- Grouping: `(expr)`

Built-in operators and predicates (expr-lang):

Expression	Example
<code>len(s)</code>	<code>len(output) > 0</code>
<code>matches(s, pattern)</code>	<code>matches(version, "^v[0-9]+")</code>

Examples:

```
# Equality check
when: severity == "critical"

# Logical AND
when: environment == "prod" && count > 1

# Logical OR with grouping
when: (severity == "medium" || severity == "critical") && customer_impact == true

# str.* helper (canonical gert syntax)
condition: str.contains(http_result, "200")

# Negation
condition: '!str.contains(http_result, "200")'
```

```
# Numeric comparison
until: operation_status == "Running"
```

String-operation helpers: `str.*` Gert defines a canonical string-operation vocabulary accessed through the `str` namespace. The `namespace.method()` pattern is engine-portable: any future gert runtime (Go, C#, etc.) implements the same vocabulary under the same names. The `str` object is injected into the `expr` evaluation environment as a helper struct; member-access syntax (`str.contains(...)`) avoids conflicts with `expr-lang` reserved keywords such as `contains` and `startsWith`, which are infix operators in the underlying library and cannot be called as top-level functions.

Function	Signature	Description
<code>str.contains</code>	<code>str.contains(s, substr)</code>	True if <code>s</code> contains <code>substr</code>
<code>str.startsWith</code>	<code>str.startsWith(s, prefix)</code>	True if <code>s</code> begins with <code>prefix</code>
<code>str.endsWith</code>	<code>str.endsWith(s, suffix)</code>	True if <code>s</code> ends with <code>suffix</code>
<code>str.toLower</code>	<code>str.toLower(s)</code>	Returns <code>s</code> converted to lowercase
<code>str.toUpper</code>	<code>str.toUpper(s)</code>	Returns <code>s</code> converted to uppercase
<code>str.trim</code>	<code>str.trim(s)</code>	Returns <code>s</code> with leading/trailing whitespace removed

YAML runbook examples:

```
# Check if DNS output contains an address record
- step:
  id: check_dns
  type: branch
  branches:
    - condition: 'str.contains(dns_output, "Address")'
      label: Resolved
    - condition: '!str.contains(dns_output, "Address")'
```

```

        label: Unresolved

# Environment-aware gate
- step:
  id: prod_check
  type: cli
  when: 'str.startsWith(hostname, "prod-") && str.endsWith(region, "-1")'
  title: Verify prod-region-1 host

# Case-insensitive method comparison
- step:
  id: route_request
  type: branch
  branches:
    - condition: 'str.toLower(http_method) == "get"'
      label: Read path
    - condition: 'str.toLower(http_method) == "post"'
      label: Write path

# Trim and validate captured input
- step:
  id: validate_input
  type: assert
  condition: 'str.trim(user_input) != ""'
  title: Input must not be blank

```

Future namespaces. The `namespace.method()` pattern is the standard for all gert helper vocabularies. Future additions will follow the same convention:

Namespace	Planned scope
<code>list.*</code>	Collection predicates (<code>list.contains</code> , <code>list.any</code> , <code>list.all</code>)
<code>regex.*</code>	Pattern matching (<code>regex.match</code> , <code>regex.replace</code>)
<code>math.*</code>	Numeric helpers (<code>math.round</code> , <code>math.abs</code>)
<code>json.*</code>	JSON traversal (<code>json.get</code> , <code>json.has</code>)

Note on infix syntax. The `expr-lang` library exposes `x contains "y"` as an infix operator. This form evaluates correctly at runtime but is *not* part of the gert condition vocabulary: it is engine-specific, non-portable, and excluded from gert's normative spec. Runbook authors **MUST** use `str.contains(x, "y")` in all condition fields.

String interpolation: still Go templates Shell command arguments (`args`), `title`, `instructions`, and other non-conditional fields *still use Go templates* for variable interpolation. This is intentional: the full power of Go template pipelines, custom function maps, and `fromJSON`/`fromYAML` parsing is necessary for constructing command lines and text output.

Access variables by dot notation: `{{ .hostname }}`.

fromYAML and fromJSON (template-only) `fromYAML` and `fromJSON` parse a raw string variable into a structured `interface{}` value. They are available in Go template contexts (`args`, `title`, etc.) but *not* in `expr` conditions. They are the primary mechanism for traversing values collected from `text` fields that declare `format: yaml` or `format: json` (see §4.4.5). The raw string is *always* what is stored in the run variable; these functions perform the parse on each template evaluation. A parse error in a template expression is a hard step failure.

```
{{/* Access a key from a JSON-typed text field */}}
{{ $cfg := fromJSON .service_config }}
{{ $cfg.region }}
```

```
{{/* Access a nested key from a YAML-typed text field */}}
{{ $spec := fromYAML .operation_spec }}
{{ $spec.replicas }}
```

Expression evaluation errors An evaluation error in a `condition`, `when`, or `until` field is a hard step failure (not a soft skip). The step transitions to `failed` and execution halts unless `continue_on_fail: true` is set on the step.

4.4 Step Types

Each step in flow is a `TreeNode`: either a `step` object (with optional `branches` and `iterate` siblings) or a standalone `iterate` block.

The defines fifteen step types:

Type	Purpose
<code>cli</code>	Execute a local command (§4.4.2)
<code>choice</code>	Operator selects from predefined options (§4.4.3)
<code>decision</code>	Operator selects an execution path (§4.4.4)
<code>collector</code>	Operator provides structured input across 9 field types (§4.4.5)
<code>approve</code>	Standalone approval gate with quorum and business-day timeout (§4.4.6)
<code>tool</code>	Invoke a declared tool action (§4.4.7)
<code>wait_for_event</code>	Pause until an inbound event arrives (§4.4.8)
<code>include</code>	Expand another runbook inline into the current run path (§4.4.9)
<code>branch</code>	Conditional fan-out to labelled step sequences (§4.4.10)
<code>iterate</code>	Loop over a collection (§4.4.11)
<code>parallel</code>	Fan-out concurrent branches with join semantics (§4.4.12)
<code>assert</code>	Evaluate a guard expression and halt on failure (§4.4.13)
<code>compensate</code>	Declare a saga compensation action (§4.4.14)
<code>noop</code>	No-operation step; applies delay and capture without side effects (§4.4.15)
<code>end</code>	Declare a terminal outcome (§4.4.16)

4.4.1 Common Step Fields

The following fields are available on every step type:

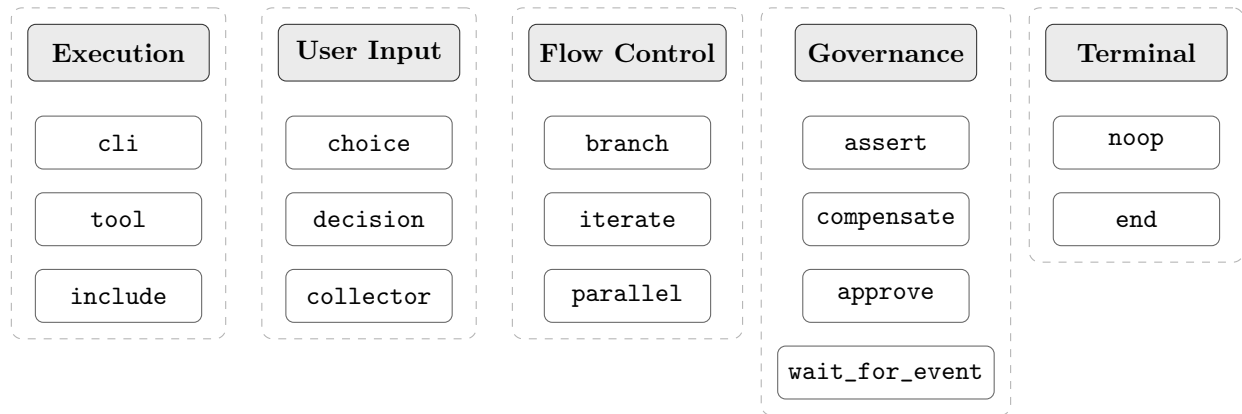


FIGURE 4.1: Step type taxonomy — 15 types organized by category.

Field	Type	Required	Description
id	string	Yes	Unique within the runbook
type	enum	Yes	Step type (see below)
title	string	No	Template; displayed in TUI/trace
subtitle	string	No	Secondary template annotation
when	string	No	Guard expression; skips step if false
timeout	string	No	Duration (e.g. 30s); overrides default
delay	string	No	Pre-execution pause (e.g. 5s)
retry	object	No	Retry config (§4.4.1)
continue_on_fail	bool	No	Continue execution on failure
on_error	string object	No	Error handling strategy (§4.4.1)
scope	string	No	Named scope for variable isolation
export	string array	No	Variables to export from scope
capture	map	No	Output capture bindings
required_evidence	array	No	Evidence requirements (§4.4.1)
contract	object	No	Behavioral contract for governance

Retry configuration

```

retry:
  max: 3          # maximum retry attempts (required)
  interval: "5s"  # initial wait between attempts
  backoff: linear  # linear / exponential (default: linear)

```

Step contract The `contract` field annotates a step with its behavioral properties for governance policy evaluation:

```

contract:
  effects:
    - network.probe
    - service.restart
  reads:  [hostname, port]
  writes: [http_result]
  deterministic: true
  idempotent: false
  secrets: [auth_token]

```

Effects are free-form strings matched against `governance.rules[].effects`. The `secrets` list names variables whose values must be redacted in traces.

Error routing The `on_error` field declares the error handling strategy for a step. Three modes are supported:

- `stop` — Halt execution immediately (default behavior)
- `continue` — Suppress the error and advance to the next step
- `goto:step_id` — Jump to a named step on error

```

# Suppress error and continue
- step:
  id: optional_check
  type: cli
  run: ./health_check.sh
  on_error: continue

# Explicit stop on error (default behavior, made explicit)

```



```
- step:
  id: critical_step
  type: cli
  run: ./deploy.sh
  on_error: stop

# Jump to a named step on error
- step:
  id: risky_operation
  type: cli
  run: ./risky.sh
  on_error: "goto:rollback_handler"

- step:
  id: rollback_handler
  type: cli
  run: |
    echo "Error: {{ __error_message }}"
    echo "Failed step: {{ __error_step_id }}"
    ./rollback.sh
```

When `on_error: goto` fires, the following variables are automatically set:

- `__error_message` — Error message text
- `__error_step_id` — ID of the step that failed
- `__error_code` — Error code (if available)
- `__error_timestamp` — ISO8601 timestamp of error

Target resolution: Goto targets must be top-level flow steps; jumping into branch arms or iterate bodies is not supported. Self-referential goto targets are rejected at parse time.

Precedence: When both `continue_on_fail` and `on_error` are present, `on_error` takes precedence and `continue_on_fail` is ignored.

Required evidence The `required_evidence` field declares evidence artifacts the operator must provide before the step can be marked complete. Three evidence kinds are supported:

- **text** — Free-form text input
- **checklist** — List of items the operator must acknowledge
- **attachment** — File upload with content-addressed storage

```
- step:
  id: deploy_service
  type: cli
  run: ./deploy.sh
  required_evidence:
    - kind: text
      name: deployment_notes
      label: "Deployment notes"
    - kind: checklist
      name: pre_deploy_check
      label: "Pre-deployment checklist"
      items:
        - "Change ticket approved"
        - "Rollback plan ready"
        - "Monitoring dashboards open"
    - kind: attachment
      name: approval_screenshot
      label: "Approval screenshot"
```

Enforcement: The TUI surfaces evidence requirements to the operator and blocks step completion until all evidence is provided. The engine records evidence artifacts in the trace.

Field-level evidence in collector steps: Collector fields can declare evidence metadata using the `evidence` subfield:

```
- step:
  id: collect_deployment_info
  type: collector
  prompt: "Provide deployment evidence"
  fields:
```

```
- name: notes
  type: text
  label: "Deployment notes"
  evidence:
    kind: text
    name: deployment_evidence
- name: approval_doc
  type: file
  label: "Approval document"
  evidence:
    kind: attachment
    name: approval_attachment
```

This allows collector steps to explicitly mark which fields serve as compliance evidence.

4.4.2 Step Type: cli

Executes a local command directly. `cli` is the canonical command step.

type: `cli` supports two mutually exclusive input modes:

- **Exec form** (`command + args`): the executor invokes the binary directly without a shell. No shell expansion occurs; arguments are passed verbatim (after template substitution).
- **Run form** (`run`): a shell-script string or a map of per-shell scripts is passed to a shell process. The `shell` field controls which shell is used when `run` is a string.

Using both `command/args` and `run` in the same step is a **schema validation error**.

Field	Type	Required	Description
<code>command</code>	string	See note	Executable name or full path. Mutually exclusive with <code>run</code> .
<code>args</code>	string array	No	Arguments (templates expanded). Only valid with <code>command</code> .
<code>run</code>	string map	See note	Shell-string or per-shell map. Mutually exclusive with <code>command</code> .
<code>shell</code>	string	No	Shell to use with <code>run</code> string form. Enum: <code>auto</code> <code>bash</code> <code>sh</code> <code>pwsh</code> <code>cmd</code> . Default: <code>auto</code> . Ignored when <code>run</code> is a map (validation warning if both are present).
<code>env</code>	map	No	Additional environment variables
<code>workdir</code>	string	No	Working directory
<code>stdin</code>	string	No	Template string piped to stdin
<code>capture</code>	map	No	<code>stdout</code> <code>stderr</code> <code>exit_code</code> → var

Note: Exactly one of `command` or `run` must be present.

Governance checks against the `allowed_commands` / `denied_commands` lists and any matching `governance.rules[].effects` are evaluated before the command executes. If the rule action is `require-approval`, the step pauses for human approval before proceeding.

Run form: string When `run` is a string, the executor spawns the shell specified by the `shell` field and passes the string as a single script argument (equivalent to `shell -c "script"`). When `shell: auto` (the default), the executor selects the platform-native shell: `bash` on Linux/macOS, `pwsh` on Windows (falling back to `cmd` if PowerShell is absent).

Run form: map When `run` is a map, the keys are shell names (`bash`, `sh`, `pwsh`, `cmd`) and the values are shell-script strings. The executor selects the entry that matches the shell available on the current host.

Map key resolution order

1. **Exact match:** the host's configured or detected shell matches a map key (e.g. host has `bash` → use `bash` entry).
2. **Platform-family fallback:** if the exact match fails, fall back according to the table below.
3. **No match:** the step **fails** with the error: *"No matching shell entry for current platform. Available entries: {keys}. Platform shell: {detected}."*

Requested shell	Falls back to
<code>bash</code>	<code>sh</code>
<code>sh</code>	(none — fail)
<code>pwsh</code>	<code>cmd</code> (Windows only)
<code>cmd</code>	(none — fail)

When `run` is a map and a `shell` field is also present, the `shell` field is **ignored** and the runtime emits a validation warning: *"shell field is ignored when run is a map. Remove shell or convert run to a string."*

Example: exec form (unchanged)

```
- step:
  id: tail_logs
  type: cli
  title: "Tail last 100 lines from {{ .log_file }}"
  command: tail
  args: ["-n", "100", "{{ .log_file }}"]
  capture:
    output: stdout
    exit: exit_code
  contract:
    effects: [filesystem.read]
    reads: [log_file]
    writes: [output]
```

Example: run string form with explicit shell

```
- step:
  id: run_script
  type: cli
  shell: bash
  run: |
    if [ "${ .env }}" = "prod" ]; then
      echo "Production deploy"
    fi
  capture:
    stdout: script_output
```

Example: run string form with auto shell

```
- step:
  id: echo_env
  type: cli
  run: echo "Environment is ${ .env }"
  capture:
    stdout: echo_result
```

Example: run map form (multi-shell)

```
- step:
  id: clear_cache
  type: cli
  run:
    bash: rm -rf /tmp/cache/${ .service }
    cmd: rd /s /q C:\tmp\cache\${ .service }
    pwsh: Remove-Item -Recurse -Force C:\tmp\cache\${ .service }
  capture:
    exit_code: clear_result
```

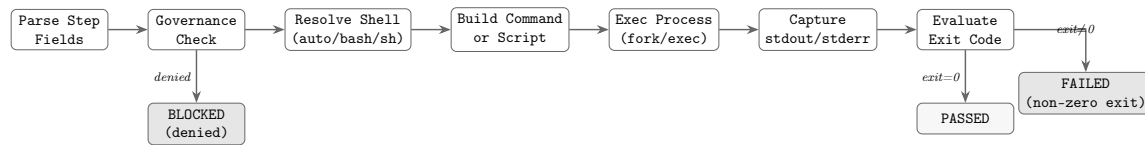


FIGURE 4.2: `type: cli` step execution flow: from field parsing through governance check, shell resolution, command building, process execution, output capture, and exit-code evaluation.

4.4.3 Step Type: choice

Prompts the user to select from a set of predefined options and stores the result in a variable. This is a value-capture primitive, analogous to a form field or radio-button group. The selected value is available to subsequent steps via variable substitution.

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Question or prompt displayed to user
<code>options</code>	array	Yes	List of option objects
<code>variable</code>	string	Yes	Variable name to store selected value
<code>default</code>	string	No	Default option value if user skips
<code>multiple</code>	bool	No	Allow selecting more than one option (default: false)
<code>min_selections</code>	integer	No	Minimum number of selections when <code>multiple: true</code>
<code>max_selections</code>	integer	No	Maximum number of selections when <code>multiple: true</code>

When `multiple: true` the stored variable becomes an array of selected values rather than a single string. The first selected value is used when a downstream template expression expects a scalar.

Option structure Each option is an object with:

Field	Type	Required	Description
label	string	Yes	Display text shown to user
value	string	Yes	Value stored in variable when selected
hint	string	No	Additional context or help text

Constraints

- At least 2 options must be defined.
- All option.value entries must be unique within a step.
- If default is provided, it must match one of the option values.
- The variable name follows standard identifier rules (alphanumeric + underscore).
- When multiple: true, min_selections and max_selections are optional but recommended to constrain the selection count.
- min_selections must be ≥ 1 ; max_selections must be $\leq$ the total number of options and $\geq$ min_selections.

Example: Environment selection

```
- step:
  id: select_environment
  type: choice
  title: Select operation target
  prompt: Which environment are you provisioning to?
  options:
    - label: Staging
      value: staging
      hint: Non-production environment for validation
    - label: Production
      value: production
      hint: Live customer-facing environment
    - label: Canary (10%)
      value: canary
      hint: Gradual rollout to 10% of production traffic
```



```
variable: operation_env
default: staging
```

The value `staging`, `production`, or `canary` is stored in `{{ .operation_env }}` and can be referenced by subsequent steps.

4.4.4 Step Type: decision

Presents the user with 2 or more execution paths and routes the flow to the selected one. This is a control-flow primitive: the user picks a branch, and execution forks to a different runbook or labeled section. Unlike `choice`, which merely stores a value, `decision` directs the execution path itself.

Also acceptable name: `router` (alias for `decision`).

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Question or scenario description
<code>routes</code>	array	Yes	List of route objects (minimum 2)
<code>variable</code>	string	No	Variable to store selected route label (audit)

Route structure Each route is an object with:

Field	Type	Required	Description
<code>label</code>	string	Yes	Display text shown to user
<code>runbook</code>	string	No	Runbook ID to invoke (mutually exclusive with <code>goto</code>)
<code>goto</code>	string	No	Step ID to jump to within current runbook
<code>hint</code>	string	No	Additional context or help text

Constraints

- At least 2 routes must be defined.
- Each route must specify exactly one of `runbook` or `goto`, not both.

- All `route.label` entries must be unique within a step.
- If `variable` is provided, it stores the selected route's `label` for audit trails.

Example: Workflow priority fork

```
- step:
  id: priority_decision
  type: decision
  title: Choose workflow path
  prompt: |
    Based on the impact level and stakeholder requirements, which
    path should we take?
  routes:
    - label: Standard procedure
      runbook: workflow-standard-procedure
      hint: Impact < 5%, no stakeholder escalations
    - label: Emergency cleanup
      runbook: workflow-emergency-cleanup
      hint: Impact > 10% or critical stakeholder concern
    - label: Escalate for review
      goto: escalate_for_review
      hint: Unclear priority; needs senior review
  variable: chosen_path
```

After the user selects a route, execution continues at the target runbook or step. The `chosen_path` variable contains the selected label ("Standard procedure", "Emergency cleanup", or "Escalate for review") for audit purposes.

4.4.5 Step Type: collector

Prompts the user to provide structured input across a rich set of field types: free-form text, numbers, dates, booleans, dropdowns, file attachments, and more. All fields in a single `collector` step are presented and submitted atomically — the operator fills them all in and confirms once. The collected values are stored as named variables or artifacts for downstream steps and evidence capture. This replaces the overly-generic `manual` step type from early drafts.

Field	Type	Required	Description
<code>prompt</code>	string	Yes	Markdown instructions displayed to user
<code>fields</code>	array	Yes	List of field objects to collect (minimum 1)
<code>approvals</code>	object	No	Approval gate config (§4.4.5)

Field structure Each field is an object with:

Field	Type	Required	Description
<code>name</code>	string	Yes	Variable name to store collected value
<code>type</code>	enum	Yes	Field type — see field type inventory below
<code>label</code>	string	Yes	Prompt or label text shown to user
<code>required</code>	bool	No	If true, user must provide a value (default: false)
<code>when</code>	template	No	Go template — field shown and prompted only when truthy; variable binding skipped when falsy
<code>hint</code>	string	No	Additional context or validation help
<code>default</code>	any	No	Pre-filled default value (must match field type)
<code>validation</code>	object	No	Type-specific validation constraints
<code>options</code>	array	No	Option list for <code>select</code> type
<code>options_from</code>	object	No	Dynamic option list provider for <code>select</code> and <code>autocomplete</code> types
<code>multiple</code>	bool	No	Allow multiple selections for <code>select</code> and <code>autocomplete</code> types (default: false)
<code>multiline</code>	bool	No	Enable multi-line textarea for <code>text</code> type (default: false)
<code>ephemeral</code>	bool	No	Value is never written to the audit log and never persisted; in-memory only for the duration of the run (default: false)

Ephemeral fields When a field carries `ephemeral: true`, gert applies the following constraints to the collected value:

- **Variable binding:** The value *is* bound into the runbook variable space and may be used in downstream template expressions (e.g. `{{ .deploy_token }}`) for the duration of the current run.
- **Audit log:** The audit record for this field writes "[REDACTED]" instead of the actual value. The field name, type, and timestamp are still recorded; only the value is suppressed.
- **Persistence:** The value is *never* written to any store — not the JSONL trace, not the evidence artifact store, not any variable snapshot. It exists only in the in-memory execution context.
- **Run suspension:** If the run is suspended after collection (e.g. after a `wait_for_event` step) and then resumed, ephemeral field values are gone. The `collector` step must be re-executed and the operator must re-supply the values.

`ephemeral: true` is valid on any field type, but is semantically meaningful on `text`, `select`, and `boolean` fields. Setting it on `file` or `image` fields is a validation *warning* (not an error) — file content is large and a [REDACTED] placeholder in the audit trail is still useful; however, the *file payload* will not be persisted to the artifact store.

Security note: Ephemeral fields reduce audit footprint for secrets. They do *not* prevent the value from being transmitted to tool steps or sub-processes. Authors are responsible for ensuring ephemeral values are only passed to secure execution contexts.

```
- id: collect_credentials
  type: collector
  label: "Deploy credentials"
  fields:
    - id: operation_user
      type: text
      label: "Service account username"
      required: true
    - id: operation_token
```

```
    type: text
    label: "Deploy token"
    required: true
    ephemeral: true      # never written to audit log or persisted
- id: target_env
  type: select
  label: "Target environment"
  required: true
  options:
    - { value: "staging",    label: "Staging" }
    - { value: "production", label: "Production" }
```

Dynamic Forms When one or more fields carry a **when** expression, the **collector** step renders as a *dynamic form*: fields are shown, hidden, and re-evaluated as the operator fills in earlier fields. **gert** enforces the following evaluation order guarantees:

1. Fields are evaluated in the order they appear in the **fields** array.
2. A field's **when** expression may reference only variables that are declared *earlier* in the same **fields** array, or variables already present in the global runbook variable space (variables set by prior steps, runbook inputs, or **gert.*** built-ins). A forward reference — referencing a field declared *later* in the same **fields** array — is a schema validation error detected at parse time.
3. When a field's **when** expression evaluates to falsy: the field is hidden in the UI, not prompted in a TUI, and its variable binding is *skipped*. The variable does not appear in the runbook variable space; it is not set to **null** or empty string.
4. In interactive UIs (VS Code, browser), if an earlier field value changes after later fields have been filled, all downstream **when** expressions are re-evaluated and dependent fields are shown or hidden accordingly.
5. A **required: true** field with a **when** expression is required only when its **when** condition is truthy. When the condition is falsy, the **required** constraint does not apply.

Field type inventory **collector** supports ten field types:

Type	Description	Notes
<code>text</code>	Single-line or multi-line text input	Set <code>multiline: true</code> for textarea; <code>format</code> for structured validation
<code>number</code>	Integer or float numeric input	Supports <code>validation.min</code> , <code>max</code> , <code>step</code>
<code>integer</code>	Integer-only numeric input	Shorthand for <code>number</code> with <code>step: 1</code>
<code>date</code>	ISO 8601 date (YYYY-MM-DD)	UI hint: date picker
<code>datetime</code>	RFC 3339 datetime with timezone	UI hint: datetime picker
<code>boolean</code>	True/false checkbox	Stored as boolean in variable space
<code>select</code>	Dropdown or option list	Set <code>multiple: true</code> for multi-select
<code>autocomplete</code>	Live search text field	Options fetched from provider as user types; degrades to <code>select</code> in CLI
<code>file</code>	File attachment	Stored as artifact with SHA256 hash
<code>image</code>	Image/screenshot attachment	Stored as artifact

Field type: `text`

Single-line text input by default. Set `multiline: true` to render a multi-line textarea — appropriate for descriptions, reproduction steps, or any field where newlines are meaningful.

Text field validation constraints The optional validation object on a `text` field supports five sub-fields:

Validation field	Type	Required	Description
pattern	string	No	ECMA 262 regular expression; field value must match
pattern_hint	string	No	Human-readable error shown when pattern validation fails
min_length	integer	No	Minimum character count (≥ 0)
max_length	integer	No	Maximum character count (> 0 ; must exceed <code>min_length</code> when both are present)
format	enum	No	Structural parse validation for <code>multiline: true</code> fields: <code>yaml</code> , <code>json</code> , or <code>toml</code>

```

# Basic multiline with hint
- name: description
  type: text
  label: Incident description
  multiline: true
  required: true
  hint: Describe what happened, when, and which services were affected

# Pattern validation - IPv4 address
- name: target_ip
  type: text
  label: Target IP address
  required: true
  validation:
    pattern:
      ↪ "^(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\\.\\.{3}(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)$"
    pattern_hint: "Must be a valid IPv4 address (e.g. 192.168.1.1)"

# Length constraints
- name: summary
  type: text
  label: Executive summary
  required: true

```



```
validation:
  min_length: 20
  max_length: 500
```

Structured multiline format validation When a text field has `multiline: true`, the optional `validation.format` field declares that the submitted value must be parseable as a specific structured data format:

- `format: yaml` — value must parse as valid YAML.
- `format: json` — value must parse as valid JSON.
- `format: toml` — value must parse as valid TOML . *Note: TOML format validation is deferred to a future release; declaring it is a validation warning.*
- No `format` (default) — plain text; no structural validation.

Parse validation occurs *after submission*. On failure, the operator is re-prompted with the message: "Value must be valid {FORMAT}. Please correct and resubmit." On success, the value is stored as a raw string — the original text, not a parsed object. Parsing is validation only.

Note: The stored value is always a string. Downstream template expressions that need to traverse the structure must use gert's `fromYAML`/`fromJSON` template functions.

```
# YAML-format multiline field
- id: config_patch
  type: text
  label: "Configuration patch (YAML)"
  multiline: true
  required: true
  validation:
    format: yaml
    min_length: 10

# JSON-format multiline field
- id: event_payload
```

```

type: text
label: "Event payload (JSON)"
multiline: true
required: true
validation:
  format: json

```

`validation.format` is silently ignored when `multiline` is `false` or absent — a validation warning is emitted at schema load time.

Field type: number

Integer or floating-point numeric input. The optional `validation` object constrains acceptable values:

Validation field	Type	Required	Description
<code>min</code>	number	No	Minimum acceptable value (inclusive)
<code>max</code>	number	No	Maximum acceptable value (inclusive)
<code>step</code>	number	No	Granularity; set to 1 to enforce integers

```

- name: amount_usd
  type: number
  label: Amount (USD)
  required: true
  validation:
    min: 0
    max: 10000000
- name: port_number
  type: number
  label: Port number
  required: true
  validation:
    min: 1
    max: 65535
    step: 1

```

```
- name: severity_score
  type: number
  label: Severity score (1.0 - 10.0)
  required: true
  validation:
    min: 1.0
    max: 10.0
    step: 0.1
```

Field type: integer

Convenience shorthand for `number` with `step: 1`. The stored variable value is always an integer. Accepts the same `validation.min` and `validation.max` fields as `number`.

```
- name: instance_count
  type: integer
  label: Number of instances
  required: true
  validation:
    min: 1
    max: 100
```

Field type: date

ISO 8601 date input (YYYY-MM-DD). The UI renders a date picker. The stored variable value is the ISO date string.

Validation field	Type	Required	Description
min	ISO date string	No	Earliest acceptable date (inclusive)
max	ISO date string	No	Latest acceptable date (inclusive)

```
- name: start_date
  type: date
  label: Start date
  required: true
  validation:
    min: "2024-01-01"
```

Field type: datetime

RFC 3339 datetime input with explicit timezone offset or Z for UTC (e.g. 2026-04-19T14:30:00-05:00). The UI renders a datetime picker.

Validation field	Type	Required	Description
min	RFC 3339 string	No	Earliest acceptable datetime (inclusive)
max	RFC 3339 string	No	Latest acceptable datetime (inclusive)

```
- name: event_detected_at
  type: datetime
  label: When was the event first detected?
  required: true
```

Field type: boolean

True/false checkbox. The stored variable value is a boolean (**true** or **false**), not a string. No additional validation fields are defined.

```
- name: background_check_completed
  type: boolean
  label: Background check completed?
  required: true
- name: no_disqualifying_issues
  type: boolean
  label: No disqualifying issues found?
  required: true
```

The value is available in templates as a boolean: `{{ if .background_check_completed }}`...`{{ end }}`.

Field type: select

Dropdown or option list. Options are defined inline via **options** (static list) or loaded at runtime via **options_from** (dynamic list from a provider). Exactly one of **options** or **options_from** must be present.

options — static list

Field	Type	Required	Description
value	string	Yes	Value stored in variable when selected
label	string	Yes	Display text shown to user

options_from — dynamic provider

Field	Type	Required	Description
provider	string	Yes	Name of the input provider to query
field	string	No	Provider response field to use as option list

`options` and `options_from` are mutually exclusive. When `multiple: true` is set, the operator may select more than one option and the stored variable becomes an array of selected values.

```
# Static options
- name: department
  type: select
  label: Department
  required: true
  options:
    - value: engineering
      label: Engineering
    - value: sales
      label: Sales
    - value: marketing
      label: Marketing
    - value: finance
      label: Finance

# Dynamic options from provider
- name: budget_line_item
  type: select
  label: Budget line item
```

```
required: true
options_from:
  provider: finance-system
  field: budget_line_items

# Multi-select
- name: affected_services
  type: select
  label: Affected services
  required: true
  multiple: true
  options:
    - value: api-gateway
      label: API Gateway
    - value: auth-service
      label: Auth Service
    - value: payments
      label: Payments
```

Field type: autocomplete

A text field with live search: as the operator types, the configured `options_from` provider is queried and matching options are returned in real time. Useful for searching service names, users, ticket IDs, configuration keys, and any other dataset too large to enumerate statically.

`autocomplete` behaves like `select` in its storage and validation semantics — the selected value is stored as a string (or an array of strings when `multiple: true`). It differs in that options are fetched dynamically on each keystroke rather than declared inline.

The `options_from` object for an `autocomplete` field extends the `select` variant with two additional search-control fields:

options_from field	Type	Required	Description
provider	string	Yes	Input provider ID that handles search queries
field	string	Yes	Provider input field that receives the search query string
min_chars	integer	No	Minimum characters typed before search fires (default: 2)
debounce_ms	integer	No	Debounce delay in milliseconds (default: 300)

CLI / non-interactive fallback: Providers that do not support live search *must* gracefully degrade to a **select** by calling the provider once with an empty query string to fetch all available options. The resulting list is then rendered as a standard dropdown.

```
# Live service search
- name: affected_service
  type: autocomplete
  label: "Affected service"
  required: true
  options_from:
    provider: service-registry
    field: query
    min_chars: 2
    debounce_ms: 300

# Multi-select user lookup
- name: reviewers
  type: autocomplete
  label: "Reviewers"
  multiple: true
  options_from:
    provider: hr-directory
    field: query
    min_chars: 2
```

Approval gate If the `approvals` field is present, the step requires explicit approval from designated roles before proceeding.

Field	Type	Required	Description
<code>min</code>	integer	Yes	Minimum approvals required
<code>roles</code>	array	Yes	List of role names authorized to approve
<code>timeout</code>	string	No	Duration before timeout (e.g. 4h)
<code>on_timeout</code>	enum	No	<code>escalate</code> , <code>fail</code> , <code>skip</code>
<code>escalate_to</code>	array	No	Roles to escalate to if timeout occurs

The `timeout` and `on_timeout` fields are new in this schema, addressing the SLA enforcement gap identified in the research brief.

Example: Incident evidence collection

```
- step:
  id: collect_evidence
  type: collector
  title: Collect workflow evidence
  prompt: |
    Gather all relevant evidence for the workflow record.
    Include screenshots, logs, and a summary of impact.
  fields:
    - name: event_summary
      type: text
      label: Event summary
      multiline: true
      required: true
      hint: Describe what happened, when, and what services were affected
    - name: error_log
      type: file
      label: Error log file
      required: true
```



```

    hint: Attach the relevant error log from the affected service
  - name: dashboard_screenshot
    type: image
    label: Monitoring dashboard screenshot
    required: false
    hint: Screenshot showing metrics during the event
  - name: related_ticket
    type: url
    label: Related ticket or alert URL
    required: false
  approvals:
    min: 1
    roles: [approver, responder]
    timeout: 30m
    on_timeout: fail

```

After the user provides the requested evidence and approval is granted, the collected values are available as `{{ .event_summary }}`, `{{ .error_log }}`, etc. File and image attachments are stored as artifacts in the execution trace with SHA256 hashes for tamper-evidence.

Questionnaire Pattern A `collector` step with multiple `fields` is the canonical way to present a questionnaire in gert. All fields are presented and submitted atomically: the operator fills in the entire form and confirms once, rather than answering questions in sequential individual steps. This pattern is preferred over a series of single-field `collector` steps when the questions are logically related and should be answered together.

Benefits of the questionnaire pattern:

- Single approval gate covers all related data.
- Atomic submission — partial forms cannot leave variables in inconsistent state.
- Cleaner audit trail: one evidence entry per logical form.
- Better UX: operator sees the full context before filling in any field.

Example: Request details form (4-field form with conditional logic using `when`)

```

- step:

```

```
id: collect_request_details
type: collector
title: Request Details
prompt: |
  Complete the request details form.
fields:
  - name: severity
    type: select
    label: Severity
    required: true
    options:
      - value: low
        label: Low
      - value: medium
        label: Medium
      - value: critical
        label: Critical
    # Shown only when severity = critical (forward-reference guard)
  - name: escalation_reason
    type: text
    label: Escalation reason
    multiline: true
    required: true
    when: "severity == 'critical'"
  - name: affected_systems
    type: text
    label: Affected systems
    required: true
    # Shown for medium and critical; hidden for low
  - name: customer_impact
    type: boolean
    label: Customer-facing impact?
    when: "severity == 'medium' || severity == 'critical'"
```

Validation rules for collector

- A collector step **MUST** have at least one field in `fields`. An empty `fields` array is a schema validation error. Structural validation enforces `minItems: 1` on the `fields`

array.

- Each field `name` must be a valid identifier (alphanumeric and underscore, must start with a letter). Field names must be unique within a step.
- For `select` fields, exactly one of `options` or `options_from` must be present.
- For `number` fields, if both `validation.min` and `validation.max` are present, `min` must be $\leq$ `max`.
- For `date` and `datetime` fields, if both `validation.min` and `validation.max` are present, `min` must be chronologically $\leq$ `max`.
- A field `when` expression **MUST NOT** reference a field declared later in the same `fields` array (no forward references). Forward references are detected at parse time and are a schema validation error.
- A `required: true` field with a `when` expression is only required when the `when` condition is truthy. When `when` evaluates to falsy, `required` does not apply and the variable binding is skipped entirely.
- For `text` fields, if both `validation.min_length` and `validation.max_length` are present, `max_length` **MUST** be strictly greater than `min_length`.
- For `text` fields, `validation.pattern` **MUST** be a valid ECMA 262 regular expression; it is validated at schema load time and will cause a structural validation error if invalid.
- For `text` fields, `validation.pattern_hint` is silently ignored when `validation.pattern` is absent (warning, not error).
- For `autocomplete` fields, `options_from` **MUST** be present and **MUST** include both `provider` and `field`. `options` (static list) is not valid on `autocomplete` fields and is a schema validation error.
- For `autocomplete` fields, `options_from.min_chars` **MUST** be ≥ 0 when specified (default 2). `options_from.debounce_ms` **MUST** be ≥ 0 (default 300).
- `validation.format` is only meaningful when `multiline: true` is also set. If `format` is declared on a non-multiline field, a validation warning is emitted and the constraint is ignored at runtime.

- `validation.format: toml` is a validation warning in (TOML parse validation is deferred); the runbook is accepted and the format constraint is not enforced at runtime.
- A field with `ephemeral: true` and `type: file` or `type: image` is a validation *warning* (not error). File payloads are large; writing `[REDACTED]` in the audit trail is acceptable, but authors should consider whether attachment evidence is appropriate alongside an ephemeral field.

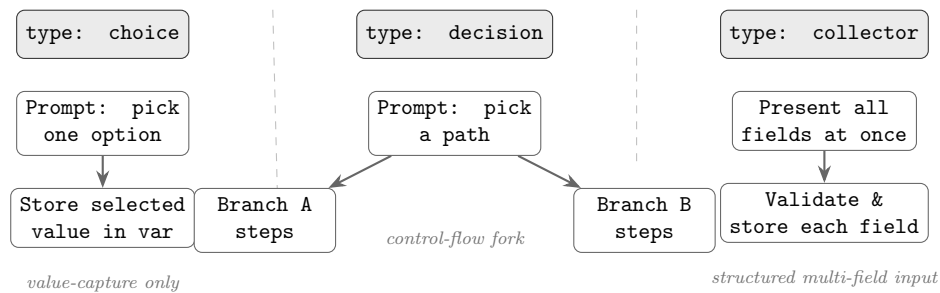


FIGURE 4.3: Human-input step triad: **choice** captures a single selection as a value; **decision** forks execution to the chosen branch; **collector** presents multiple typed fields atomically and stores all captured values.

4.4.6 Step Type: approve

An **approve** step is a dedicated, standalone approval gate that suspends execution until one or more authorized roles explicitly approve (or until the configured timeout fires). Unlike the inline **approvals** block available on **collector** and **choice** steps, **approve** carries no data-collection fields — its sole purpose is to capture an authoritative go/no-go decision and record it in the run audit trail.

Field	Type	Required	Description
<code>title</code>	string	No	Human-readable approval prompt
<code>approvals</code>	object	Yes	Approval gate config (see below)
<code>timeout</code>	string	No	Wall-clock duration (e.g. 48h); mutually exclusive with <code>timeout_business_days</code>
<code>timeout_business_days</code>	integer	No	Timeout in business days; mutually exclusive with <code>timeout</code> ; requires <code>timezone</code>
<code>timezone</code>	string	No	IANA timezone (e.g. <code>America/New_York</code>); required when <code>timeout_business_days</code> is set
<code>business_calendar</code>	string	No	Named calendar ID (e.g. <code>us-federal</code> , <code>uk-banking</code>); defaults to Mon–Fri with no holidays
<code>on_timeout</code>	string	No	<code>fail</code> , <code>continue</code> , or <code>branch:<id></code>
<code>contract</code>	object	No	Behavioral contract for governance

approvals block The `approvals` object controls who may approve and the quorum semantics:

Field	Type	Required	Description
<code>mode</code>	string	No	<code>all</code> (default) — all listed roles must approve; <code>any</code> — one approval suffices; <code>quorum</code> — M-of-N from <code>pool</code>
<code>roles</code>	string[]	No	Authorized role names; used with <code>mode: all</code> or <code>mode: any</code> ; mutually exclusive with <code>pool</code>
<code>pool</code>	string[]	No	Eligible role names for <code>quorum</code> ; used with <code>mode: quorum</code> ; mutually exclusive with <code>roles</code>
<code>required</code>	integer	No	Number of approvals required from <code>pool</code> ; used with <code>mode: quorum</code> ; must be ≥ 1 and $\leq \text{len}(\text{pool})$

Business-Day Timeout `timeout_business_days: 3` means the approval must be granted within 3 working days from the moment the `approve` step became active. The runtime counts elapsed business days using the following rules:

- The clock starts when the step enters the `waiting` state.
- Weekends (Saturday and Sunday) do not count toward the timeout.
- Holidays defined in the named `business_calendar` do not count.
- If `business_calendar` is omitted, the calendar defaults to Monday–Friday with no holidays.
- `timezone` determines the midnight boundary used for day counting; it is required whenever `timeout_business_days` is set.

If both `timeout` and `timeout_business_days` are present in the same step, schema validation **MUST** reject the runbook with a structural error before execution begins. The `on_timeout` semantics remain unchanged: `fail` terminates the run, `continue` advances to the next step, and `branch:<id>` jumps to the named branch.

```
- step:
  id: legal_review
  type: approve
  title: "Legal team approval"
  approvals:
    roles: [legal-counsel]
  timeout_business_days: 5
  timezone: "America/New_York"
  business_calendar: us-federal
  on_timeout: branch:escalate_legal
  contract:
    effects: [approval.legal]
    reads: []
    writes: []
```

Quorum Approval Setting mode: `quorum` together with `pool` and `required` enables M-of-N approval semantics:

- The runtime accepts approvals from any role listed in `pool`.
- Once the number of distinct approvals reaches `required`, the step proceeds immediately without waiting for remaining pool members.
- Each pool member may approve only once; their identity (role + principal) is recorded in the run audit trail.
- If a pool member explicitly rejects, the rejection is recorded in the audit trail. A rejection does not by itself block the step — the step is blocked only if so many members have rejected that the remaining approvers can no longer reach `required`. (For example, if `required: 2` and `pool` has 3 members and 2 reject, the step immediately fails because quorum is no longer reachable.)

```
- step:
  id: security_quorum
  type: approve
  title: "Security committee approval (2 of 5)"
  approvals:
    mode: quorum
```

```
pool: [security-lead, ciso, security-eng-1,
       security-eng-2, security-eng-3]
required: 2
timeout_business_days: 3
timezone: "Europe/London"
on_timeout: fail
contract:
  effects: [approval.security]
  reads: []
  writes: []
```

Validation Rules The following conditions are schema validation errors that **MUST** be reported before execution:

1. `timeout` and `timeout_business_days` are mutually exclusive; both present is an error.
2. `timeout_business_days` requires `timezone`; omitting `timezone` when `timeout_business_days` is set is an error.
3. `approvals.mode: quorum` requires both `approvals.pool` and `approvals.required`; either missing is an error.
4. `approvals.required` must satisfy $1 \leq \text{required} \leq \text{len}(\text{pool})$; violating this range is an error.
5. `approvals.roles` and `approvals.pool` are mutually exclusive; both present is an error.
6. `approvals.mode: all` or `mode: any` requires `approvals.roles` (not `pool`); using `pool` with these modes is an error.

4.4.7 Step Type: tool

Invokes a named tool action. The tool must be declared in `toolRefs`. The runtime dispatches to the tool's configured transport (stdio, jsonrpc, or mcp).

Field	Type	Required	Description
<code>tool.name</code>	string	Yes	Tool name (must match tool-Refs entry)
<code>tool.action</code>	string	Yes	Action declared in tool definition
<code>tool.args</code>	map	No	Argument key-value map (templates expanded)
<code>tool.version</code>	string	No	Optional version pin
<code>capture</code>	map	No	<code>stdout stderr json.<path> → var</code>

Output capture, adds a `json.<path>` form for JSON-structured tool outputs, eliminating the need for follow-up `cli` steps with `jq`:

```
capture:
  pod_name: json.items[0].metadata.name
  status:   json.items[0].status.phase

- step:
  id: list_pods
  type: tool
  title: "List pods in {{ .namespace }}"
  tool:
    name: kubectl
    action: get-pods
    args:
      namespace: "{{ .namespace }}"
    output: json
  capture:
    pod_count: json.items | length
    first_pod: json.items[0].metadata.name
  contract:
    effects: [k8s.read]
    reads: [namespace]
    writes: [pod_count, first_pod]
```

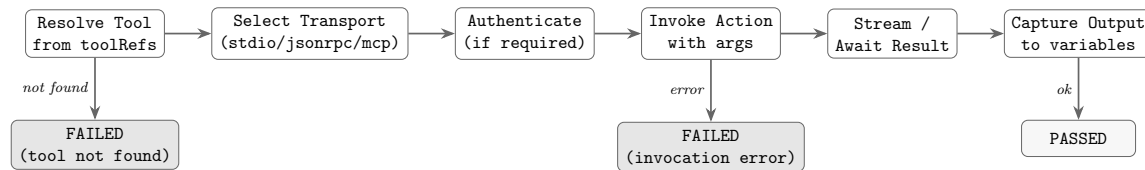


FIGURE 4.4: **type:** `tool` invocation flow: tool resolution, transport selection, authentication, action invocation, result streaming, and output capture. A missing tool or invocation error terminates the step as FAILED.

4.4.8 Step Type: `wait_for_event`

Pauses runbook execution until an inbound event arrives on a declared source. Use this step type whenever a runbook must suspend and wait for an external signal before proceeding — a Prometheus alert webhook, an approval callback from a ticketing system, a SIEM event, a message-queue message, or an OS/process signal.

When the runtime reaches a `wait_for_event` step it:

1. Opens the declared inbound endpoint (HTTP listener, queue consumer, or internal channel) for the lifetime of the wait.
2. Suspends the run record in state `waiting_for_event`.
3. When a matching event arrives and passes any declared `event.filter` and `event.payload_schema` checks, it captures payload fields into run variables (via `capture`) and resumes execution at the next step.
4. If the `timeout` expires before a matching event arrives, the step behaves according to `on_timeout`: `fail` (default), `continue` (skip and proceed), or `branch:<step-id>` (jump to the named step).

Field	Type	Required	Description
<code>event.source</code>	string	Yes	Event source type: <code>webhook</code> , <code>message</code> , <code>signal</code> , or <code>channel</code>
<code>event.id</code>	string	No	Logical event name or channel (e.g. <code>prometheus.alert.firing</code>); templates expanded
<code>event.filter</code>	map	No	Key-value pairs that must match in the incoming event payload
<code>event.payload_schema</code>	string	No	JSON Schema URL to validate the incoming payload before capture
<code>capture</code>	map	No	<code>json.<path></code> → variable; maps event payload fields to run variables
<code>timeout</code>	string	No	Wall-clock duration (e.g. <code>30m</code> , <code>2h</code>); default: no timeout
<code>on_timeout</code>	string	No	<code>fail</code> (default), <code>continue</code> , or <code>branch:<step-id></code>

Example 1 — Prometheus alert webhook

```
- step:
  id: wait_prometheus_alert
  type: wait_for_event
  title: "Wait for Prometheus alert to fire"
  event:
    source: webhook
    id: prometheus.alert.firing
    filter:
      alertname: "HighErrorRate"
      severity: critical
  capture:
    alert_value: json.commonAnnotations.value
    alert_labels: json.commonLabels
  timeout: 30m
```

```
on_timeout: fail
contract:
  effects: []
  reads: []
  writes: [alert_value, alert_labels]
```

Example 2 — Named channel / approval callback

```
- step:
  id: wait_approval_callback
  type: wait_for_event
  title: "Wait for external approval system callback"
  event:
    source: channel
    id: "approval.{{ .request_id }}"
  capture:
    approved_by: json.approver
    decision: json.decision
  timeout: 48h
  on_timeout: branch:escalate_approval
  contract:
    effects: []
    reads: [request_id]
    writes: [approved_by, decision]
```

Transport Note The runtime is responsible for exposing the inbound endpoint; the runbook schema only declares intent. The specific endpoint URL or address is runtime-determined and available to subsequent steps as `{{ .gert.event.<id>.endpoint }}`. Event source semantics:

- **webhook** — runtime opens an HTTP POST listener; the sender POSTs a JSON body.
- **message** — runtime subscribes to a message-queue topic or queue name given by `event.id`.
- **signal** — runtime registers a named OS or process signal handler.
- **channel** — runtime uses a gert-internal named channel, enabling runbook-to-runbook or tool-to-runbook event delivery.

Security Note Webhook endpoints are authenticated per run. The runtime generates a one-time HMAC token when the endpoint is opened; the token is available as `{{ .gert.event.<id>.token }}` for use in configuring the sending system (e.g. embedding in a Prometheus alertmanager receiver URL). The token is single-use and expires when the endpoint closes (on resume or timeout).

4.4.9 Step Type: include

Expands another runbook's steps *inline* into the current run path at the point of the **include** step. This is **not** a sub-procedure call: the included runbook shares the caller's variable space entirely — no isolated scope, no input/output mapping. All included steps appear in the caller's execution trace and audit log as if they were defined inline.

The **include.with** map injects variable overrides into the shared scope before expansion; these values are immediately visible to the included steps and to any steps that follow in the caller.

Field	Type	Required	Description
<code>include.runbook</code>	string	Yes	Path or ID of the runbook to include. Relative paths resolve from the including runbook's location.
<code>include.with</code>	map	No	Variable overrides injected into the shared scope before expansion (key: variable name, value: expression). Applied in order, visible to included steps.
<code>include.when</code>	string	No	Boolean expression. If false, the include is skipped entirely (no steps expanded).
<code>include.expand</code>	enum	No	Per-site override of the expansion mode: <code>eager</code> <code>lazy</code> <code>auto</code> . Empty inherits the parent runbook's mode (or the platform default). See §4.4.9.

```
- step:
  id: run_db_checks
  type: include
```

```
title: "Run database pre-flight checks"
include:
  runbook: ../shared/db-preflight.runbook.yaml
  with:
    db_host: "{{ .target_db }}"
    check_mode: read-only
contract:
  effects: [db.read]
  reads: [target_db]
  writes: []
```

Cycle Detection The runtime builds a runbook dependency graph at load time by resolving all `include.runbook` references transitively. A cycle exists when a runbook directly or transitively includes itself. Cyclic includes are a hard validation error; the runtime rejects the runbook set and reports the full include chain:

```
cyclic include detected: A -> B -> A
```

The validator reports every edge in the offending cycle so the operator can identify the exact chain.

Eager vs Lazy Expansion Each include site is materialized into the execution plan in one of two modes:

- **eager** — The planner loads, parses, validates, and expands the referenced runbook *at plan time*. Every step in the child (and its transitive includes) appears in the resolved plan before execution begins. Plan-time errors include broken references in branches that may never run.
- **lazy** — The planner records the include as an opaque flow node carrying the absolute path of the child. The child runbook is loaded, planned, and executed only when the include step is actually entered at run time. Plan time is fast and only walks the local runbook; broken references in unentered branches surface only when (and if) those branches execute, or via a separate `gert lint` pass.
- **auto** — Resolves to **lazy** when the runbook contains more than a small number of include sites (default: 5), otherwise **eager**. The threshold is configurable via the embedding application's expansion policy.

The effective mode at any include site is resolved with the following precedence (highest first):

1. `include.expand` on the site itself.
2. `expand` on the parent runbook.
3. The runtime default supplied by the CLI flag `-expand` or the embedding application.
4. The platform default: `lazy`.

Lazy is the baked-in default because real-world orchestrator runbooks fan out into many sub-runbooks of which only a few execute on any given path; eager loading punishes the common case to validate branches that will never run. Operators who want full plan-time validation opt in via `expand: eager` on the runbook or `-expand=eager` on the command line.

Lazy expansion is *recursive*: an include site reached through a lazily loaded child runbook is itself lazy by default, regardless of the child runbook's own `expand` field. This keeps an entire sub-tree opaque until it is entered. A site that explicitly sets `include.expand: eager` overrides the contagion at that point.

Cycle detection under lazy expansion Lazy expansion does *not* disable cycle detection. The planner still runs cycle detection across the eagerly expanded sub-tree at plan time, and each lazy expansion at execution time runs cycle detection on the freshly loaded child runbook against the active include chain. A cycle reached through a lazy edge surfaces as a runtime error on the offending include step with the same chain message as the plan-time error.

Difference from Sub-Procedure Call `type: include` shares the caller's variable space; there is no input mapping and no output extraction. The included steps appear inline in the run trace — there is no sub-run, no separate audit entry, and no scope boundary.

If isolation *is* needed (separate variable scope, explicit input/output mapping), that is the responsibility of a future step type (`type: call` or `type: delegate`). That capability is **not in the scope**.

Outcome Gates (gate:) A `gate` block on a `type: include` step lets the parent run react to the child runbook's completion *outcome* — specifically the `category` field written by the child's terminal `type: end` step — without requiring the parent to inspect variables explicitly.

```
- step:
  id: run_connectivity_check
  type: include
  include:
    runbook: connectivity_test
    with:
      service_name: "{{ .service_name }}"
  gate:
    stop_if: [resolved]
```

`gate.stop_if` is a list of outcome category strings. When the included runbook completes, the runtime checks whether its final outcome category appears in the list:

- **Gate matches** — the parent run terminates cleanly with the child's outcome; no error is raised. The gate *absorbs* the resolution.
- **Gate does not match** — execution continues normally to the next step in the parent.
- **Child run fails (error)** — the gate is not consulted. Execution errors propagate to the parent in the usual way.
- **No gate present** — the current behaviour: always continue to the next step regardless of child outcome.

Use case — incident triage fan-out. A parent runbook dispatches a sequence of diagnostic child runbooks. If any child resolves the incident (outcome `category: resolved`), the parent stops immediately without executing the remaining diagnostic branches, avoiding unnecessary work and producing a clean audit trail.

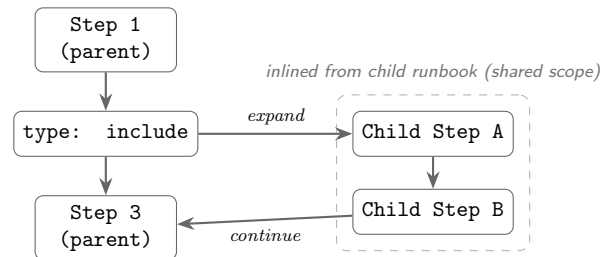


FIGURE 4.5: `type: include` inlines the child runbook's steps into the parent's execution path. All included steps share the parent variable scope; no sub-run boundary is created.

4.4.10 Step Type: `branch`

Evaluates a list of conditions and executes the first matching arm. In the schema, `branch` is an explicit first-class step type (as well as the inline `branches: sibling` on any step node).

Field	Type	Required	Description
<code>branches</code>	array	Yes	List of Branch objects
<code>branches[].condition</code>	string	Yes	Go template evaluating to <code>true/false</code>
<code>branches[].label</code>	string	No	Display label for diagram/trace
<code>branches[].steps</code>	array	No	Steps to execute if condition matches

At most one branch arm executes. If no arm matches and no arm has an `else: true` marker, execution continues past the branch node with no steps run. The `else: true` marker (new in this schema) designates a catch-all arm:

```

- step:
  id: route_on_env
  type: branch
  title: Route by environment
  branches:
    - condition: 'environment == "production"'
      label: Production path
      steps:

```

```
- step:
  id: require_approval
  type: manual
  title: Require approver confirmation
  approvals:
    min: 1
    roles: [approver]
- condition: 'environment == "staging"'
  label: Staging path
  steps:
    - step:
      id: log_staging
      type: cli
      title: Log staging deploy
      command: echo
      args: ["Deploying to staging"]
- else: true
  label: Unknown environment - abort
  steps:
    - step:
      id: abort
      type: end
      title: Unknown environment
      outcome:
        category: escalated
        code: unknown_env
```

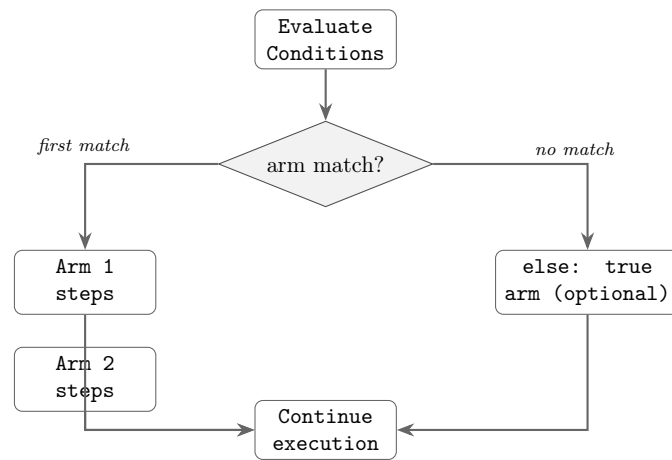


FIGURE 4.6: **type: branch** evaluates each arm’s condition in order and executes the first matching arm. At most one arm runs. If no arm matches and no **else** arm is defined, execution continues with no steps run.

4.4.11 Step Type: iterate

Repeats a block of steps over a collection or until a convergence condition is met. In the schema **iterate** is a **TreeNode** sibling (not a step type in the **step.type** enum) — it appears at the tree level alongside **step** nodes.

Field	Type	Required	Description
<code>iterate.over</code>	string	*	Template resolving to comma-separated list
<code>iterate.as</code>	string	No	Loop variable name (default: <code>item</code>)
<code>iterate.max</code>	int	*	Max iterations (convergence mode)
<code>iterate.until</code>	string	*	Convergence condition expression
<code>iterate.collect</code>	map	No	Accumulate captures across passes
<code>iterate.steps</code>	array	Yes	Steps to execute each pass

* `over` or (`max` + `until`) required; mutually exclusive.

The loop variable `{{ .iteration }}` (0-indexed) is always available. In list mode, `{{ .item }}` (or the `as` name) holds the current element.

```
# List iteration
- iterate:
  id: check_all_services
  over: "{{ .service_list }}" # e.g. "api,worker,scheduler"
  as: svc
  collect:
    results: "{{ .svc }}:{{ .http_result }}"
  steps:
    - step:
      id: probe
      type: tool
      title: "Check {{ .svc }}"
      tool:
        name: curl
        action: head
        args:
          url: "https://{{ .svc }}.internal/healthz"
      capture:
        http_result: stdout

# Convergence iteration
- iterate:
  id: wait_for_deploy
  max: 10
  until: 'operation_status == "Running"'
  steps:
    - step:
      id: poll_deploy
      type: tool
      title: "Poll operation (pass {{ .iteration }})"
      tool:
        name: kubectl
        action: get-deploy-status
        args:
          name: "{{ .operation }}"
```

```

    capture:
      operation_status: json.status.phase
- step:
  id: wait
  type: cli
  title: Wait 15s
  command: sleep
  args: ["15"]

```

Concurrent Iteration (`concurrency:`) By default `iterate` processes one item at a time. Setting `concurrency: N` ($N > 1$) runs up to N iterations simultaneously as a worker pool, useful for health checks across many services or parallel deployments to multiple regions.

```

- iterate:
  over: services
  as: svc
  concurrency: 3
  collect:
    results: "{{ .last_host }}: {{ .last_status }}"
  steps:
    - step:
      id: check_svc
      type: include
      include:
        runbook: check
        with:
          service_host: "{{ .svc }}"
      capture:
        last_status: service_status
        last_host: checked_host

```

- `concurrency: 0` or `concurrency: 1` — sequential; identical to the default behaviour.
- `concurrency: N` ($N > 1$) — worker pool of N ; up to N iterations run simultaneously.
- **Fail-fast:** if any iteration fails, remaining in-flight iterations are cancelled and the `iterate` step fails.

- **Collect ordering:** results are appended in *completion order*, not iteration order — the collected list is non-deterministic when concurrency is enabled.
- **Variable isolation:** each iteration receives its own copy of the loop variable; there are no cross-iteration data races.

Contrast with type: `parallel`. **type:** `parallel` requires statically-defined branches known at authoring time. **concurrency:** on `iterate`: handles dynamic lists whose length is not known until runtime — it is the idiomatic fan-out pattern for variable-length collections.

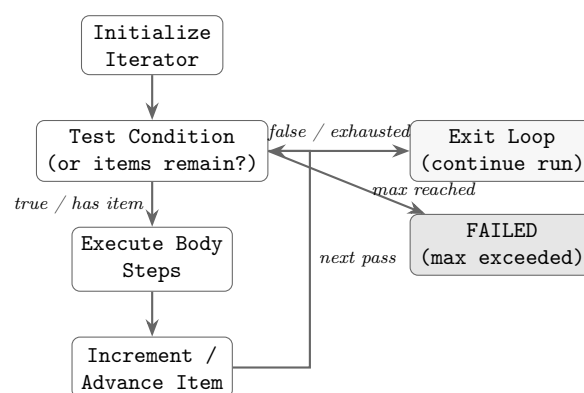


FIGURE 4.7: **type:** `iterate` loop: the condition (items remaining or `until` expression) is tested before each pass. The body executes, the iterator advances, and the loop repeats until the condition is false, items are exhausted, or `max` iterations are reached.

4.4.12 Step Type: `parallel`

Executes a set of independent step groups concurrently, then joins. New in this schema, addressing the fan-out/fan-in gap identified in Dennis’s research brief.

Field	Type	Required	Description
<code>branches</code>	array	Yes	Each branch is an independent group
<code>join.wait_for</code>	enum	No	<code>all</code> <code>any</code> <code>majority</code> (default: <code>all</code>)
<code>join.on_failure</code>	enum	No	<code>fail</code> <code>continue</code> (default: <code>fail</code>)

Parallel branches are isolated from each other's variable writes during execution. After the join, all `capture` values from all branches are merged into the caller's variable map. Write conflicts (two branches writing the same variable) are resolved by last-writer-wins and a warning is emitted.

```
- step:
  id: parallel_checks
  type: parallel
  title: Run DNS and HTTP checks concurrently
  branches:
    - label: DNS check
      steps:
        - step:
            id: dns_check
            type: tool
            tool:
              name: nslookup
              action: lookup
              args:
                host: "{{ .hostname }}"
            capture:
              dns_result: stdout
    - label: HTTP check
      steps:
        - step:
            id: http_check
            type: tool
            tool:
              name: curl
              action: head
              args:
                url: "https://{{ .hostname }}/healthz"
            capture:
              http_result: stdout
  join:
    wait_for: all
    on_failure: fail
```

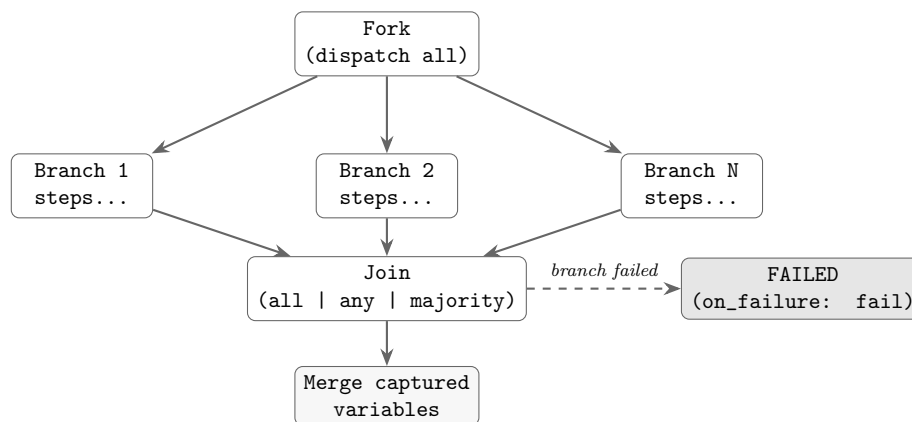


FIGURE 4.8: **type: parallel** fan-out/fan-in: all branches are dispatched concurrently. The join node waits for the configured quorum (**all**, **any**, or **majority**), then merges captured variables. A branch failure triggers the **on_failure** policy.

4.4.13 Step Type: assert

Evaluates a list of assertions against captured variables. Fails the run if any assertion does not pass.

```

- step:
  id: verify_deployment
  type: assert
  title: Verify operation succeeded
  assert:
    - type: contains
      subject: "{{ .deploy_output }}"
      expected: "successfully rolled out"
    - type: eq
      subject: "{{ .exit_code }}"
      expected: "0"
    - type: json_path
      subject: "{{ .api_response }}"
      path: "$.status"
      expected: "active"

```


Assertion types: `contains`, `not_contains`, `matches` (regex), `eq`, `ne`, `lt`, `gt`, `exit_code`, `json_path`.

4.4.14 Step Type: `compensate`

New in this schema. Registers a compensation (rollback) action that is executed in reverse order if the run fails or is cancelled after this step. Implements the saga pattern for long-running operational transactions.

Field	Type	Required	Description
<code>compensate.on</code>	enum	No	<code>failure</code> <code>cancel</code> <code>any</code> (default: <code>failure</code>)
<code>compensate.steps</code>	array	Yes	Steps to execute as compensation

The compensation steps execute in LIFO order relative to the sequence of `compensate` nodes encountered during the forward execution pass. Compensation steps share the same variable context as the point at which the `compensate` node was registered.

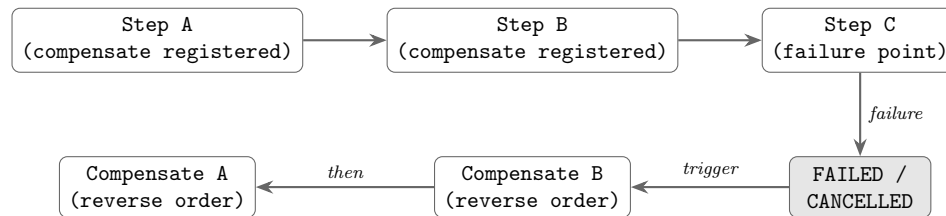
```
- step:
  id: scale_down
  type: tool
  title: Scale down service
  tool:
    name: kubectl
    action: scale
    args:
      operation: "{{ .operation }}"
      replicas: "0"

- step:
  id: register_scale_up
  type: compensate
  title: Compensation - scale back up on failure
  compensate:
    on: failure
    steps:
      - step:
```

```

id: scale_up_comp
type: tool
title: Restore service scale
tool:
  name: kubectl
  action: scale
  args:
    operation: "{{ .operation }}"
    replicas: "{{ .original_replicas }}"

```



saga pattern: undo in reverse registration order

FIGURE 4.9: `type: compensate` saga flow: each compensation step is registered during forward execution. On failure or cancellation, registered compensations fire in reverse order, implementing the saga pattern for long-running operational transactions.

4.4.15 Step Type: noop

A no-operation step that advances execution without side effects. Useful for pure delays between steps or variable accumulation without command execution.

Execution semantics:

- Applies `delay` if present (blocks for the specified duration)
- Applies `capture` if present (evaluates and binds variables)
- Records step execution event in trace
- Immediately advances to the next step

No side effects: A noop step does not execute commands, invoke tools, wait for user input, or perform any external actions. All behavior is driven by common step fields (`delay`, `capture`, `when`, `timeout`).

```
# Pure delay step
- step:
  id: wait_for_settle
  type: noop
  title: "Wait for system to settle"
  delay: 5s

# Variable accumulation without command execution
- step:
  id: accumulate_report
  type: noop
  title: "Record {{ .svc }} result"
  capture:
    report: "{{ .report }}{{ .svc }}: {{ .status }}\n"
```

Use cases:

- Delays between steps (e.g. waiting for eventual consistency)
- Variable transformations without command execution
- Report accumulation in iterate loops
- Stereotyping/rendering hints in TUI

Interaction with step-level fields: `delay`, `capture`, `when`, `timeout`, `scope`, and `export` apply as expected. `retry` and `on_error` apply only if `timeout` is set (since the only failure mode is timeout expiration). `contract` does not apply (noop has no effects, reads, or writes).

4.4.16 Step Type: end

Declares a terminal outcome and halts execution.

```
- step:
  id: resolved
  type: end
  title: Workflow complete
  outcome:
    category: resolved      # resolved / escalated / no_action / needs_rca
    code: dns_healthy
    meta:
      resolution_time: "{{ .elapsed }}"
```

4.5 Tool Definition Schema

Tool definitions describe callable tools: their actions, argument schemas, transport, and behavioral contract. In the schema the `tool/v0` (`legacy`) version is replaced by `tool/v1`.

4.5.1 Tool Definition Field Inventory

```
$schema: "https://schemas.gert.dev/tool/v1.json"
apiVersion: tool/v1

meta:
  name: kubect1
  version: "1.2"
  description: Kubernetes CLI wrapper
  binary: kubect1

transport:
  mode: stdio      # stdio / jsonrpc / mcp
  # jsonrpc options:
  # start: ["node", "server.js"]
  # port: 3000
  # mcp options:
  # start: ["python", "mcp_server.py"]

actions:
  get-pods:
    description: List pods in a namespace
```

```
argv: ["get", "pods", "-n", "{{ .namespace }}", "-o", "json"]
args:
  namespace:
    type: string
    required: true
    description: Kubernetes namespace
  output:
    type: string
    required: false
    default: json
capture:
  stdout:
    format: json    # json / text / lines
contract:
  effects: [k8s.read]
  reads: [namespace]
governance:
  action: allow
```

New

- `capture.stdout.format` — declares the output format; enables structured `json.<path>` capture in runbook steps.
- `actions.<name>.contract` — per-action behavioral contract automatically merged with any step-level contract override.
- `actions.<name>.governance` — per-action governance policy; overrides the runbook-level rules for this action only.
- `meta.version` — explicit tool version for auditing and pinning.

4.5.2 Transport Type: native

Overview

The `native` transport type enables runbooks to invoke native CLI utilities (e.g., `ping`, `curl`, `nslookup`) without requiring a JSON protocol. Unlike `stdio`, `jsonrpc`, and `mcp` transports which speak a structured protocol, native tools accept command-line arguments (`argv`), write plain text to `stdout/stderr`, and exit with a status code.

When to Use

Use `type: native` when:

1. The tool is a system binary or CLI utility that accepts argv-style arguments.
2. Output is plain text, not JSON or other structured formats.
3. The tool is stateless and invoked once per action (no persistent connection).
4. You do not control or cannot modify the tool's source code.

Use `type: stdio`, `jsonrpc`, or `mcp` when the tool can speak a structured protocol.

Schema: `transport: for native tools`

Field	Type	Description
<code>transport.type</code>	string	Must be "native"
<code>transport.command</code>	string	Binary name or full path (e.g., "ping", "/usr/bin/curl")

Schema: `actions: with argv templates`

Each action in a native tool must declare an `argv:` field: a list of Go `text/template` strings. Each element is rendered at invocation time with the step's `tool.args` as template data.

Field	Type	Description
<code>actions.<name>.argv</code>	[]string	List of argv elements (Go templates)
<code>actions.<name>.args</code>	map	Argument schema (same as <code>stdio</code>)
<code>actions.<name>.description</code>	string	Human-readable action description
<code>actions.<name>.returns</code>	string	Output format; typically "text"

Example argv with template rendering

```
argv:
- "-c"                # -c flag (literal)
- "{{ .count }}"      # count argument from step.tool.args
- "{{ .host }}"        # host argument (required)
```

When invoked with `args: {count: "4", host: "8.8.8.8"}`, the argv renders to `["-c", "4", "8.8.8.8"]`.

Complete Tool Definition Example

```
apiVersion: tool/v1
name: ping
version: "1.0"
description: Send ICMP ECHO_REQUEST packets to test network reachability

transport:
  type: native
  command: ping

actions:
  check:
    description: Ping a host with a fixed packet count
    argv:
      - "-c"
      - "{{ .count }}"
      - "{{ .host }}"
    args:
      host:
        type: string
        required: true
        description: Hostname or IP address to ping
      count:
        type: string
        required: false
        default: "4"
        description: Number of ICMP packets to send
    returns: text

  check-timeout:
    description: Ping a host with a deadline per packet
    argv:
      - "-c"
      - "{{ .count }}"
      - "-W"
      - "{{ .timeout }}"
      - "{{ .host }}"
```

```
args:
  host:
    type: string
    required: true
    description: Hostname or IP address to ping
  count:
    type: string
    required: false
    default: "4"
    description: Number of ICMP packets to send
  timeout:
    type: string
    required: false
    default: "5"
    description: Time in seconds to wait for each reply
returns: text
```

Runbook Invocation: toolRefs: and Tool Steps

Tools are declared in the runbook's `toolRefs:` section. Each ref has a `name` and a `path` (relative to the runbook file):

```
toolRefs:
- name: ping
  path: ../../tools/ping.tool.yaml
- name: curl
  path: ../../tools/curl.tool.yaml
- name: nslookup
  path: ../../tools/nslookup.tool.yaml
```

Path resolution is relative to the runbook file's directory. The loader resolves relative paths to absolute paths at load time and registers the tools into the tool registry before execution.

Complete Runbook Example

```
apiVersion: runbook/v1
id: network-check
```



```
name: Network Connectivity Check

toolRefs:
- name: ping
  path: ../../tools/ping.tool.yaml
- name: nslookup
  path: ../../tools/nslookup.tool.yaml

flow:
- step:
  id: ping-dns
  type: tool
  title: Test connectivity to Google DNS
  tool:
    name: ping
    action: check
    args:
      host: "8.8.8.8"
      count: "3"
  capture:
    output: stdout

- step:
  id: resolve-domain
  type: tool
  title: Resolve example.com
  tool:
    name: nslookup
    action: lookup
    args:
      name: "example.com"
  capture:
    dns_response: stdout

- step:
  id: check-results
  type: conditional
  title: Check if both checks passed
```

```
conditions:
- target: ping-dns.output
  op: contains
  value: "bytes from"
- target: resolve-domain.dns_response
  op: contains
  value: "Address:"
```

Execution Model When a `type: tool` step with a native tool executes:

1. The runtime looks up the tool by name in the registry (resolved from `toolRefs:`).
2. It renders each argv element as a Go template using `step.tool.args` as data.
3. It spawns the binary (`transport.command`) with the rendered argv and waits for completion.
4. It captures stdout and stderr as plain text.
5. If the process exits with status 0, the step passes; non-zero exit is an error.
6. Output is stored as `stdout` and `stderr` and can be captured via `capture:` directives.

Error Handling Native tool invocation fails if:

- The binary does not exist or is not executable.
- An argv template fails to render (e.g., undefined variable).
- The process exits with a non-zero status code.

4.6 Provider Definition Schema

Provider definitions declare external input resolution services. In the schema the `provider/v0` version is replaced by `provider/v1`.

```
$schema: "https://schemas.gert.dev/provider/v1.json"
apiVersion: provider/v1
```

```
meta:
  name: tracking
  version: "1.0"
  description: Issue tracking context provider
  kind: input-provider    # input-provider / enrichment-provider

transport:
  mode: jsonrpc
  start: ["/providers/issue-tracker-provider"]

fields:
  id:
    type: string
    description: Active request ID
  title:
    type: string
    description: Request title
  severity:
    type: string
    enum: [P1, P2, P3, P4]
  affected_service:
    type: string
    description: Primary impacted service name

capabilities:
  - input-resolution
```

The `fields` map defines the provider's schema. Each field name corresponds to a `from: <provider>.<field>` binding in a runbook input. The semantic validator checks that all `from: <provider>.*` bindings reference a field declared in a loaded provider definition.

4.7 Namespace Convention for Extensions

4.7.1 The Extension Field Problem

Strict JSON Schema validation (`additionalProperties: false`) is a core design constraint: unknown fields in a runbook must be rejected, not silently ignored. At the same

time, extensions must be able to attach metadata to steps, runbooks, and tool definitions without forking the core schema.

4.7.2 Convention: `x-<namespace>`: at the Field Level

The adopts the `x-<namespace>` prefix convention used by OpenAPI and GitHub Actions. Extension fields must:

1. Start with `x-`.
2. Be followed by a namespace component (lowercase alphanumeric and hyphens, matching `[a-z][a-z0-9-]+`).
3. Be declared in an **extension manifest** registered via `gert extension register` or in `gert.yaml`.

At the runbook level

```
$schema: "https://schemas.gert.dev/runbook/v1.json"
apiVersion: runbook/v1
id: provision-service
name: Provision Service

x-myorg-cost-center: "engineering-ops"
x-myorg-change-id: "CHG-20260418-001"
x-otel-service-name: "payment-service"
```

At the step level

```
- step:
  id: provision
  type: tool
  title: Provision to environment
  x-myorg-risk-level: high
  x-myorg-cleanup-id: "{{ .provision_id }}"
```

4.7.3 Extension Validation

1. **Unregistered extension fields** — The JSON Schema validator accepts any field matching `^x-[a-z][a-z0-9-]+$` without requiring a registered schema. This enables zero-friction annotation. The `gert validate` command emits a `WARNING` for each unregistered extension field.
2. **Registered extension fields** — An extension may ship a JSON Schema fragment that constrains its field values. Once registered, the validator applies the fragment and reports violations as errors, not warnings.
3. **Unknown non-extension fields** — Any field that does not match a core field name and does not start with `x-` is a hard validation error (`additionalProperties: false`).

4.7.4 Extension Registration Format

```
# .gert/extensions/myorg.yaml
apiVersion: extension/v1
namespace: myorg
description: MyOrg internal runbook annotations

fields:
  cost-center:
    type: string
    description: Cost center code for billing
    pattern: "^[a-z]+-[a-z]+"
  risk-level:
    type: string
    enum: [low, medium, high, critical]
  change-id:
    type: string
    pattern: "^CHG-[0-9]{8}-[0-9]{3}$"
  cleanup-id:
    type: string

applies-to: [runbook, step]
```

The extension schema is merged into the live JSON Schema when the extension is registered, enabling IDE auto-completion for extension fields as well.

4.8 Structural vs. Semantic Validation

The validation runs in two sequential phases. Both must pass for a runbook to be considered valid.

4.8.1 Phase 1: Structural Validation

Structural validation is pure JSON Schema Draft 2020-12 evaluation. It checks:

- **Required fields** present (`id`, `name`, `apiVersion`, `flow`).
- **Field types** correct (string vs. integer vs. object vs. array).
- **Enum values** valid (e.g. `kind` must be one of the four defined values).
- **Pattern constraints** satisfied (e.g. `id` must match slug pattern; `timeout` must match `[0-9]+(s|m|h)`).
- **minItems** / **minLength** constraints met (e.g. `flow` must have at least one node; `choices.options` must have at least two options).
- **additionalProperties** — no unknown core fields present.

Structural validation is performed by the generated JSON Schema artifact (see §4.9). It is stateless: it does not load tool definitions, resolve imports, or evaluate templates.

Implementation note for Brian The Go runtime uses the `invopop/jsonschema` library to generate the JSON Schema from struct tags at startup (no static file required), then validates the parsed YAML document against it. The `yaml.v3 KnownFields(true)` call provides an additional pre-JSON-Schema pass that rejects unknown fields before the struct is even unmarshalled.

4.8.2 Phase 2: Semantic Validation

Semantic validation enforces domain rules that cannot be expressed in JSON Schema. It runs after structural validation and requires the full runbook AST and loaded tool/provider definitions. Semantic checks include:

1. **Step ID uniqueness** — All step IDs in the flat walk of the tree must be unique within the runbook.
2. **Include target resolution** — Every `include.runbook` value must resolve to a relative file path that exists on disk.
3. **Branch condition variable references** — Each `condition` template references only variables that are in scope at that point in the execution tree (declared as `inputs`, `vars`, or captured by an earlier step on a reachable path).
4. **Tool reference resolution** — Every `tool.name` in a tool step must match a declared `toolRefs` entry, and the named `.tool.yaml` file must be loadable and structurally valid.
5. **Tool action existence** — The `tool.action` must be declared in the referenced tool definition.
6. **Provider binding resolution** — Every `inputs.*.from: <provider>.<field>` must resolve to a loaded provider whose field manifest declares that field name.
7. **Cyclic include detection** — The include graph must be a DAG; any cycle is a semantic error reported as `cyclic include detected: A → B → A`.
8. **Capture write-before-read** — A step that reads a captured variable via a template must have a reachable predecessor step that writes it; a warning is emitted if the write is only on some branches.
9. **Governance policy pre-check** — The static governance rules are evaluated against the step contracts declared in the runbook. Any step with an effect that would trigger `deny` is flagged as a semantic error. Steps that would trigger `require-approval` are flagged as warnings.
10. **Output declaration completeness** — Every declared `output` must reference a capture name that is written by at least one step.

11. **Compensate ordering** — Each `compensate` step must appear after the step it is compensating for in tree walk order.

Error reporting Both validation phases produce structured error reports with:

- Error code (e.g. SCH-001, SEM-007)
- File path and YAML line/column (where available)
- Human-readable message
- Suggested fix (where computable)

```
$ gert validate service-health.runbook.yaml
[SEM-003] step "check_http": tool action "head" not declared in
          tools/curl.tool.yaml (declared actions: get, post, download)
          --> service-health.runbook.yaml:34:7
[SEM-007] step "eval": template references variable "dns_result" which
          is only written on branch "DNS check" - may be unbound
          --> service-health.runbook.yaml:51:12
          hint: add a default value in vars: or guard with {{ if .dns_result }}
```

4.9 Schema Artifacts

4.9.1 Shipped Artifacts

`gert` ships the following schema artifacts:

Artifact	Description
<code>runbook.v1.schema.json</code>	JSON Schema Draft 2020-12 for runbook documents
<code>tool.v1.schema.json</code>	JSON Schema for tool definitions
<code>provider.v1.schema.json</code>	JSON Schema for provider definitions

All three schemas are generated at runtime from Go struct tags via `invopop/jsonschema` and served by `gert schema export`. They are also embedded in the binary and served at `https://schemas.gert.dev/*` for IDE tooling.

4.9.2 gert schema export

```
# Export runbook schema (default)
gert schema export

# Export specific schema
gert schema export --kind runbook    # runbook / tool / provider / bundle

# Export full bundle (all three)
gert schema export --kind bundle > gert-schemas.json

# Validate a document against the exported schema
gert schema export | check-jsonschema --schemafile - runbook.yaml
```

The bundle format is a JSON object with three keys: `runbook`, `tool`, and `provider`.

4.9.3 VS Code Integration

The gert VS Code extension configures schema-backed YAML validation via the `yaml-language-server` extension:

```
# .vscode/settings.json (auto-generated by gert vscode init)
{
  "yaml.schemas": {
    "https://schemas.gert.dev/runbook/v1.json": "**/*.runbook.yaml",
    "https://schemas.gert.dev/tool/v1.json":    "**/*.tool.yaml",
    "https://schemas.gert.dev/provider/v1.json": "**/*.provider.yaml"
  }
}
```

Because the schema URL matches the `$schema` field embedded in each document, IDEs with built-in JSON Schema support (VS Code 1.80+, IntelliJ 2024+) pick up validation and auto-completion without any per-project configuration.

The gert VS Code extension additionally provides:

- Step-type-aware field completion via the `schema/stepFields` JSON-RPC method (returns the field set for a given `type` value).

- Tool argument completion via `schema/toolArgs` (returns the `args` schema for a given tool+action pair from the live tool catalog).
- Inline governance warnings derived from `exec/dryRun` pre-check.

4.9.4 Schema Version Manifest

Each schema artifact carries a version manifest in its `$id` and a `x-gert-version` annotation:

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.gert.dev/runbook/v1.json",
  "x-gert-version": "1.0.0",
  "x-gert-released": "2026-04-18",
  "title": "gert runbook",
  ...
}
```

Automated regression testing (part of the gert CI pipeline) runs every example runbook through the exported JSON Schema validator.

Chapter 5

Extension Runtime

The gert extension runtime defines how external packages contribute tools, schema fields, governance policies, and input providers to a running gert host without requiring changes to the host binary. Extensions run as child processes; they communicate with the host over JSON-RPC 2.0 and are governed by an explicit capability grant model.

A normative reference specification is maintained at `specs/002-extension-runtime-v0/spec.md` in the repository. This chapter derives from and supersedes that document for design purposes; in any conflict between the two, the specification takes precedence for implementation.

5.1 Architecture Overview

1. The gert host discovers extension manifests on startup.
2. The host evaluates compatibility and capability policy before launching any process.
3. Each accepted extension is started as a child process; its executable communicates with the host over its declared transport.
4. The host and extension perform a versioned handshake. The host sends the set of capabilities it is willing to grant; the extension confirms receipt.
5. The extension registers its contributions (tools, schema fields, policies, providers).
6. For the remainder of the run the host dispatches invocations to the extension and enforces all capability constraints.
7. At run completion the host sends a graceful shutdown signal.

5.2 Capability Taxonomy

Every operation an extension may perform is gated by a named capability. Capabilities are requested in the manifest and granted (or denied) by host policy before the process is started. The initial capability set is given in Table 5.1.

The capability string format is `capability/<name>`. Future versions may introduce scoped variants using path notation, e.g. `capability/file-read:logs/**`.

5.3 Extension Manifest Format

Each extension ships a `gert-extension.yaml` file alongside its executable. The manifest declares identity, requested capabilities, transport, and compatibility constraints.

```
# gert-extension.yaml - Extension Manifest
apiVersion: extension/v1

meta:
  name: acme.incident-pack           # Reverse-DNS namespaced identifier
  version: "1.2.0"                   # Semver
  description: "Acme incident resolution tools and providers"
  author: "Acme Platform Team <platform@acme.example>"

compatibility:
  api-version: ">=2.0.0 <3.0.0"      # Host API version range (semver)
  protocol-version: "2.0"            # Extension protocol version

entry-point:
  executable: "./bin/acme-extension"
  args: ["--config", "acme.yaml"]

transport: stdio-jsonrpc             # stdio-jsonrpc / grpc / mcp

capabilities:
  - capability/tool-registration
  - capability/provider-registration
  - capability/schema-extension
  - capability/network
```

```
- capability/file-read

# Required when capability/file-read is declared:
file-read-paths:
- "./runbooks/**"
- "./data/**"

# Required when capability/network is declared:
network-hosts:
- "api.pagerduty.com"
- "*.acme.example"

# Required when capability/env-read is declared:
env-read-patterns:
- "ACME_API_*"
- "PD_TOKEN"

platform:                                # Optional OS/arch constraints
  os: ["linux", "darwin", "windows"]
  arch: ["amd64", "arm64"]
```

Manifest validation is performed by the host before the process is started. An extension with an unsatisfied `compatibility.api-version` range is rejected before any subprocess is created.

5.4 Extension Discovery

The host discovers extensions using the following resolution order, with later entries taking precedence:

1. **Built-in extensions:** compiled into the host binary; always available; no manifest required.
2. **Workspace config:** extensions declared in `.gert/extensions.yaml` at the workspace root.
3. **Runbook declaration:** extensions listed in the runbook's top-level `extensions:` field.

4. **CLI override:** `-extension <path>` flags provided at invocation time.

The `.gert/extensions.yaml` workspace config format:

```
# .gert/extensions.yaml
extensions:
- path: "./extensions/acme-pack"    # directory containing gert-extension.yaml
- path: "/opt/gert-extensions/audit-pack"
```

The runbook `extensions:` field:

```
apiVersion: runbook/v1
meta:
  name: incident-triage
extensions:
- path: "./extensions/acme-pack"
```

When the same extension appears in multiple discovery sources, the host deduplicates by `meta.name`. If two distinct extensions share the same name, the one with the higher resolution precedence wins; the lower one is logged as a warning and skipped.

5.5 Handshake and Lifecycle Protocol

All messages use JSON-RPC 2.0. The host is always the initiator; the extension never sends unsolicited requests.

5.5.1 extension/initialize (host → extension)

Sent immediately after the extension process starts. The host communicates the capabilities it is willing to grant and its own version identity.

Request params:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "method": "extension/initialize",
  "params": {
    "host": {
```

```
    "name": "gert",
    "version": "2.0.0",
    "apiVersion": "2.0"
  },
  "protocolVersion": "2.0",
  "grantedCapabilities": [
    "capability/tool-registration",
    "capability/network"
  ],
  "runContext": {
    "runId": "run-abc123",
    "mode": "real"
  }
}
```

Success response:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "extension": {
      "name": "acme.incident-pack",
      "version": "1.2.0"
    },
    "protocolVersion": "2.0",
    "acknowledgedCapabilities": [
      "capability/tool-registration",
      "capability/network"
    ]
  }
}
```

Error codes for this method:

- -32001: Protocol version not supported by extension.

- -32002: Required capability not granted (extension considers the capability non-negotiable).
- -32003: Host API version outside extension compatibility range.

5.5.2 contributions/list (host → extension)

Sent after a successful `extension/initialized` notification. The extension returns its contribution manifest, filtered to the granted capability set.

```
{
  "jsonrpc": "2.0",
  "id": "2",
  "method": "contributions/list",
  "params": {}
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "id": "2",
  "result": {
    "tools": [ { "name": "acme.pd-lookup", "... } ],
    "providers": [ { "name": "pd", "prefixes": ["pd."] } ],
    "schemaExtensions": [],
    "policyContributions": []
  }
}
```

5.5.3 extension/ping (host → extension)

Periodic health check. The extension **MUST** respond within the host-configured ping timeout (default 5 seconds). A non-response triggers a crash-recovery cycle (Section 5.7.3).

```
{ "jsonrpc": "2.0", "id": "42", "method": "extension/ping", "params": {} }
```

Response:

```
{ "jsonrpc": "2.0", "id": "42", "result": { "status": "ok" } }
```


5.5.4 extension/shutdown (host → extension)

Graceful termination. The extension SHOULD drain in-flight requests and release resources within the shutdown timeout (default 10 seconds).

```
{ "jsonrpc": "2.0", "id": "99", "method": "extension/shutdown", "params": {} }
```

Response (best-effort; host may not wait):

```
{ "jsonrpc": "2.0", "id": "99", "result": { "status": "ok" } }
```

If the extension does not exit within the shutdown timeout, the host sends SIGTERM followed by SIGKILL after a grace period.

5.5.5 Lifecycle State Machine

1. **discovered**: manifest found and parsed; no process yet.
2. **verified**: manifest validated; capabilities evaluated against policy.
3. **starting**: executable launched; stdin/stdout pipes attached.
4. **initializing**: `extension/initialize` sent; awaiting response.
5. **ready**: `contributions/list` completed; accepting invocations.
6. **draining**: `extension/shutdown` sent; waiting for in-flight completions.
7. **stopped**: process has exited cleanly.
8. **crashed**: process exited unexpectedly (see Section 5.7.3).

5.6 Versioning and Compatibility

The host maintains a monotonically increasing API version string following semver. The extension manifest declares a semver range in `compatibility.api-version`. Compatibility is evaluated before any process is started:

- If the host API version falls outside the declared range, the extension is rejected with a structured error logged to the run trace. The run continues with the remaining available extensions; a warning is surfaced to the operator.

- During `extension/initialize`, the host also sends its `protocolVersion`. If the extension does not support that protocol version it MUST return error code `-32001`.
- If an extension requests a capability that the host version does not implement (e.g., a capability introduced in a later host version), the capability is silently absent from `grantedCapabilities`. The extension SHOULD treat absent non-required capabilities as optional degradation rather than a fatal error.

Host API versions are tracked in `specs/api-versions.md`. Each new capability addition constitutes a minor version bump; breaking changes to existing method contracts constitute a major version bump.

5.7 Sandboxing and Process Isolation

5.7.1 Process Isolation

Extensions always run as child processes of the gert host. They are never loaded as shared libraries or plug-ins executed in-process. The host:

- Sets a restricted environment: only env vars matching `env-read-patterns` from the manifest are forwarded.
- Attaches the extension's stderr to a structured log collector; raw stderr lines are stored in the run trace under `extension.<name>.stderr`.
- Does not forward the host process's file descriptor table beyond stdin/stdout/stderr.

5.7.2 Capability Enforcement

Every RPC method the extension may call is guarded by a capability check. The host maintains a granted-capability set per extension process. Before dispatching any extension-initiated request (e.g., a tool registration during `contributions/list`) the host verifies the required capability is in the granted set. Violations return:

```
{
  "jsonrpc": "2.0",
  "id": "...",
  "error": {
```

```
"code": -32010,  
"message": "capability not granted",  
"data": {  
  "required": "capability/file-write",  
  "granted": ["capability/file-read", "capability/network"]  
}  
}  
}
```

File-path and network-host constraints are enforced by the host at the system call boundary using OS-level mechanisms where available (e.g., `pledge/unveil` on OpenBSD, `seccomp` on Linux) and by host-side validation of all path and URL arguments on other platforms.

5.7.3 Timeout and Crash Handling

- **Per-invocation timeout:** each `tools/invoke` call carries a `timeoutMs` field. If the extension does not respond within that deadline, the host sends a `tools/cancel` request and, if still unresponsive after a grace period, terminates the process.
- **Ping timeout:** if an extension fails to respond to `extension/ping` within 5 seconds it is considered unresponsive. The host logs a structured warning and terminates the process.
- **Unexpected exit:** if the extension process exits with a non-zero status or is killed by a signal, the host marks the extension as **crashed**, emits a `extension.crashed` trace event, and cancels all in-flight invocations with a structured `-32099` (extension crashed) error.
- **Host isolation:** extension crashes never crash the host. A run that loses an extension mid-execution surfaces an error on the affected step and allows the operator to decide whether to abort or continue with a degraded tool set.

Capability name	Allows	Excludes
capability/tool-registration	Register new tool definitions at runtime via <code>contributions/list</code> .	Does not allow the extension to invoke other tools or modify existing tool definitions.
capability/schema-extension	Contribute namespaced additional fields to the runbook schema (e.g. <code>x-acme.*</code> annotations).	Cannot modify or remove existing schema fields; cannot alter core validation rules.
capability/event-subscription	Subscribe to host runtime events (step-started, step-completed, run-ended, etc.) via the <code>events/subscribe</code> method.	Cannot emit synthetic events into the host event bus.
capability/policy-contribution	Register additional governance policy rules evaluated during pre-flight and approval checks.	Cannot override or disable built-in governance rules; cannot grant capabilities to other extensions.
capability/provider-registration	Register new input providers that handle <code>from:</code> prefix bindings.	Cannot intercept resolution requests destined for another registered provider.
capability/file-read	Read files within the path scope declared in the manifest (<code>file-read-paths</code>).	Cannot read files outside declared paths; cannot write or delete files.
capability/file-write	Write or create files within the path scope declared in the manifest (<code>file-write-paths</code>).	Cannot write outside declared paths; does not imply read access.
capability/network	Make outbound network calls to the hosts declared in the manifest (<code>network-hosts</code>).	Cannot connect to undeclared hosts; cannot listen for inbound connections.
capability/env-read	Read environment variables whose names match the declared <code>env-read-patterns</code> list.	Cannot read undeclared variables; cannot write or unset environment variables.
capability/run-state-read	Read current run state (step status, captured variables, trace events).	Cannot modify run state; does not imply the ability to emit output.

TABLE 5.1: Extension capability taxonomy.

Chapter 6

Tool Runtime

The tool runtime is the subsystem responsible for resolving, validating, and invoking tool definitions at runbook execution time. Tools are the primary mechanism by which gert performs side-effecting work: executing binaries, calling JSON-RPC services [18], and delegating to MCP-hosted capabilities [10].

6.1 Tool Discovery

The host resolves tool references in three phases:

1. **Static discovery:** `.tool.yaml` files are located by scanning the following directories in order:
 - (a) The built-in tool registry compiled into the host binary.
 - (b) `<workspace-root>/tools/` — the conventional location for project-level tool definitions (`tools/<name>.tool.yaml`).
 - (c) Directories listed in the workspace config (`.gert/config.yaml`, field `tool-paths`).
 - (d) Required packages declared in the runbook's `requires:` field (each package's `tools/` subdirectory is scanned).
2. **Dynamic registration:** extensions with `capability/tool-registration` may register additional tool definitions during the `contributions/list` phase (see Section 5.5). Dynamically registered tools are merged into the resolved catalog before execution begins.

3. **MCP discovery**: when the runbook declares an MCP server under `tools.mcp-servers`, the host performs a `tools/list` call to that server at startup and registers every tool reported as a dynamically discovered tool (see Section 6.7).

6.1.1 Name Resolution Order

When a runbook step references a tool by name, the host resolves as follows:

1. Exact match in dynamically registered tools (MCP and extension, in registration order).
2. Exact match in project-level `tools/` directory.
3. Exact match in required-package tool directories (in `requires:` declaration order).
4. Exact match in the built-in registry.

If two tools share the same name, the first match wins and a warning is emitted. Tool authors SHOULD use reverse-DNS namespace prefixes to avoid collisions (e.g., `acme.pd-lookup`).

6.2 Tool Definition Schema

```
# tools/my-service.tool.yaml
apiVersion: tool/v1

meta:
  name: my-service           # Unique identifier within the workspace
  version: "2.0.0"          # Tool definition version (semver)
  description: "Calls the MyService REST API"
  binary: my-service-cli     # Executable name (looked up in PATH)

transport:
  mode: stdio-jsonrpc        # stdio | stdio-jsonrpc | grpc | mcp

# For stdio-jsonrpc and mcp: startup behaviour
startup:
  argv: ["--server-mode"]    # Extra args passed to the binary
  ready-pattern: "ready"     # Regex matched against stdout/stderr to
```

```
                                # confirm the server is accepting requests
    timeout: "10s"
    shutdown-method: "shutdown" # JSON-RPC method to call on graceful stop

governance:
    requires-capabilities: []      # Host capabilities the tool needs, e.g.
                                # ["capability/network"]
    allowed-environments: ["real", "dry-run"]
    requires-approval: false      # Set true to gate invocation on operator OK

# Optional: per-platform implementation blocks (mobile execution only)
impl:
    ios:
        transport: native-sdk      # Dispatch to registered iOS handler
        handler: GertSDK.MyService.lookup
    android:
        transport: native-sdk      # Dispatch to registered Android handler
        handler: com.gert.tools.myservice.LookupHandler

actions:
    lookup:
        description: "Look up an entity by ID"
        argv: ["lookup", "--id", "{ .id }"] # Only for stdio transport
        timeout: "30s"
        args:
            id:
                type: string
                required: true
                description: "Entity identifier"
                redact: false
        output:
            format: json            # text / json / jsonl
        capture:
            stdout:
                format: json
```

6.2.1 Transport-Specific Fields

stdio The binary is spawned once per invocation. The `argv` array in the action definition is rendered as a Go template with the resolved argument values, then appended to the binary name to form the complete command. The process's stdout is captured as the tool output.

stdio-jsonrpc The binary is spawned once per run (persistent process). The host keeps a single process alive across all action calls for that tool. Each action invocation sends a JSON-RPC request to the process's stdin and reads the response from stdout. The `startup.ready-pattern` confirms the server is accepting connections.

mcp The host spawns the binary and performs the MCP initialization handshake (Section 6.7). Actions map to MCP `tools/call` requests.

6.3 Transport Layer

6.3.1 stdio Transport

Process establishment. The host calls `exec.Command(binary, resolvedArgv...)` with a controlled environment. A per-invocation timeout context wraps the call.

Invocation envelope. For stdio tools the invocation is entirely encoded in the process arguments and environment; there is no envelope. The arguments are rendered from the action's `argv` template.

Response envelope. Stdout is captured in full after process exit. The exit code is checked: non-zero is mapped to a tool error.

Error envelope.

```
{
  "error": {
    "code": "TOOL_NONZERO_EXIT",
    "message": "tool exited with code 1",
    "details": {
```



```
    "exitCode": 1,  
    "stderr": "connection refused\n",  
    "durationMs": 320  
  }  
}  
}
```

6.3.2 stdio-jsonrpc Transport

Process establishment. A single process is spawned at first use and kept alive for the duration of the run. The host waits for `startup.ready-pattern` on stdout/stderr before sending the first request.

Invocation envelope.

```
{  
  "jsonrpc": "2.0",  
  "id": "<invocationId>",  
  "method": "<actionName>",  
  "params": {  
    "args": { "id": "INC-12345" },  
    "context": {  
      "runId": "run-abc123",  
      "stepId": "lookup_incident",  
      "traceId": "trace-xyz",  
      "userId": "ops@acme.example",  
      "attempt": 1  
    }  
  }  
}
```

Response envelope.

```
{  
  "jsonrpc": "2.0",  
  "id": "<invocationId>",  
  "result": {  
    "output": { ... },  
  }  
}
```

```
    "telemetry": {
      "startedAt": "2026-04-18T10:00:00Z",
      "finishedAt": "2026-04-18T10:00:00.320Z",
      "durationMs": 320
    }
  }
}
```

Error envelope.

```
{
  "jsonrpc": "2.0",
  "id": "<invocationId>",
  "error": {
    "code": -32050,
    "message": "entity not found",
    "data": {
      "category": "runtime",
      "retryable": false,
      "details": { "entityId": "INC-12345" }
    }
  }
}
```

Error code ranges:

- -32700 to -32600: JSON-RPC parse and request errors.
- -32050: Application-level tool error (retryable per `data.retryable`).
- -32051: Capability not granted.
- -32052: Input validation failed.
- -32053: Output schema validation failed.
- -32054: Timeout.
- -32055: Cancelled by host.
- -32099: Tool process crashed.

6.3.3 MCP Transport

See Section 6.7 for the full MCP integration specification.

6.4 Invocation Context

Every tool invocation — regardless of transport — receives a structured context object. For `stdio` tools, relevant fields are injected as environment variables prefixed `GERT_`; for `stdio-jsonrpc` and `mcp` tools, the context is transmitted in the invocation envelope.

```
{
  "context": {
    "runId":      "run-abc123",           // Globally unique run identifier
    "stepId":     "lookup_incident",     // Step ID within the runbook
    "traceId":    "trace-xyz789",        // Distributed trace correlation ID
    "userId":     "ops@acme.example",    // Identity executing the run
    "mode":       "real",                // real | dry-run | replay
    "attempt":    1,                     // Retry attempt number (1-based)
    "capabilities": [                    // Effective host capability set
      "capability/network",
      "capability/file-read"
    ]
  }
}
```

6.5 Capability Gate

Before dispatching any tool invocation the host checks two conditions:

1. The tool's `governance.requires-capabilities` list is a subset of the current run's granted capability set. If not, the step fails immediately with a `CAPABILITY_DENIED` error; the run does not attempt to start the tool process.
2. The tool's `governance.allowed-environments` list includes the current execution mode. A tool not listed for `dry-run` will be skipped (with a warning) when the run mode is `dry-run`.

6.6 Output Capture

Tool output is mapped to named step captures using the `capture:` declaration in the action definition.

```
capture:
  dns_output: stdout          # Capture stdout as variable dns_output
  error_log:  stderr         # Capture stderr as variable error_log
  exit_code:  exitCode       # Capture numeric exit code
```

For `json` output format, capture paths may use dot notation to extract nested fields:

```
capture:
  incident_id: "stdout.incident.id"
  severity:   "stdout.incident.severity"
```

Redaction. Before any captured value is stored in the run state or emitted to the trace, the governance redaction pipeline is applied. Redaction rules are declared in the runbook's `meta.governance.redact` block and in the tool action's per-argument `redact: true` flag. Redacted values are replaced with `<redacted>` in all stored and displayed output.

6.7 MCP Tool Integration

MCP (Model Context Protocol) servers are long-lived processes that expose a dynamic catalog of tools. Gert treats an MCP server as a tool source, not as a single tool.

6.7.1 MCP Server Declaration

```
# In runbook meta or .gert/config.yaml
tools:
  mcp-servers:
    - name: acme-mcp
      binary: acme-mcp-server
      args: ["--port", "stdio"]
      transport: mcp
      startup:
        timeout: "15s"
```

6.7.2 MCP Discovery

On run startup the host:

1. Spawns the MCP server binary.
2. Sends the MCP `initialize` request (per MCP specification).
3. After receiving `initialized`, sends `tools/list`.
4. Registers each returned tool under the name `<server-name>/<tool-name>` in the dynamic tool catalog.

6.7.3 MCP Tool Invocation

When a runbook step calls an MCP-backed tool, the host sends:

```
{
  "jsonrpc": "2.0",
  "id": "5",
  "method": "tools/call",
  "params": {
    "name": "lookup_incident",
    "arguments": { "id": "INC-12345" },
    "_gert_context": {
      "runId": "run-abc123",
      "stepId": "triage_step",
      "traceId": "trace-xyz"
    }
  }
}
```

The `_gert_context` field is a gert extension to the MCP envelope. MCP servers that do not recognise it **MUST** ignore it. The response follows the standard MCP `tools/call` response format.

6.7.4 MCP vs stdio-jsonrpc Differences

- MCP servers expose multiple tools under a single process; stdio-jsonrpc tools are one tool per process.

- MCP uses a distinct initialization handshake (`initialize` / `initialized`) before tool registration.
- MCP tool names are discovered dynamically at runtime; stdio-jsonrpc action names are declared statically in the `.tool.yaml`.
- MCP responses use a `content` array; gert flattens text content to a single string for capture.

6.8 Cancellation and Timeout

Per-tool timeout. Declared in the action definition under `timeout`: (e.g., 30s). The default is the runbook's `meta.defaults.timeout`. The host wraps each invocation in a `context.WithTimeout`; when the deadline elapses the host sends a `tools/cancel` request (JSON-RPC transport) or sends `SIGTERM` (stdio transport), then forcibly terminates the process after a 5-second grace period.

Cancellation propagation. When a run is cancelled (via `Ctrl-C`, the debugger quit command, or an API cancellation request), the host:

1. Sets the run-level cancel context.
2. Sends `tools/cancel` to all in-flight JSON-RPC and MCP invocations with a `reason`: `"run-cancelled"` field.
3. After a 5-second drain window, forcibly terminates any unresponsive tool processes.
4. Emits a `run.cancelled` trace event recording which steps were interrupted.

The `tools/cancel` request envelope:

```
{
  "jsonrpc": "2.0",
  "id": "cancel-5",
  "method": "tools/cancel",
  "params": {
    "invocationId": "5",
    "reason": "run-cancelled"
  }
}
```

6.9 Tool Versioning

Tool definitions carry a `meta.version` field. When multiple definitions for the same tool name are present (e.g., from a required package and from a workspace override), the host selects by resolution order (Section 6.1), not by version number. Explicit version pinning is supported via the runbook's `requires:` block:

```
requires:
- package: github.com/acme/gert-tools
  version: ">=1.2.0 <2.0.0"
```

Chapter 7

Runtime Events and Determinism

Gert is an event-driven system. Every state change in the runtime—run initiation, step advancement, governance checks, tool invocations, human interactions—produces a structured event. Events serve three purposes: (1) adapters (TUI, Web, VS Code) subscribe to events to render UI state without coupling to execution logic, (2) events are appended to the JSONL trace file for audit and replay (§12), and (3) events propagate to observability systems for real-time monitoring and alerting.

This section defines the complete event system: the event envelope schema, delivery semantics, the normative event catalog, event channels, subscription API, wire format, and the required vs. optional event classification.

7.1 Event System Design

7.1.1 Role of Events

Events are **fan-out notifications to subscribers**, not a command or control channel. The Runtime Core emits events as a consequence of state transitions; events do not trigger state transitions. This design ensures that execution semantics remain centralized in the Runtime Core and adapters remain passive renderers.

Key principles:

- **One-way flow:** Events flow from the Runtime Core outward to subscribers. Subscribers never write events back to the runtime.
- **Non-blocking emission:** Event emission must not block step execution. The event bus uses a bounded channel with a discard-on-full policy for in-process subscribers to prevent slow consumers from stalling the runtime.

- **Immutable once emitted:** Events are value objects; they cannot be modified after emission.
- **Best-effort delivery to in-process subscribers:** If a subscriber channel is full, the event is dropped. The trace file write (§12) is the authoritative record.

7.1.2 Delivery Semantics

At-least-once delivery. The JSONL trace file receives every event via a synchronous write before the runtime proceeds to the next action. In-process subscribers (TUI, event forwarders) receive events on a best-effort basis; if the subscriber channel is full, the event is discarded without blocking the runtime.

In-order per run. All events within a single run are totally ordered by their **sequence** field. Events from different runs are not ordered relative to each other. Subscribers that process events asynchronously must tolerate out-of-order arrival from concurrent runs.

Replay determinism. Events written to the trace file are sufficient to reconstruct the execution timeline. Replay mode (§12.3) re-emits the same events in the same order, allowing adapters to render a historical run indistinguishably from a live run.

7.2 Event Envelope

Every event carries a mandatory envelope with the following fields. Consumers **MUST** tolerate unknown fields to support additive schema evolution.

Field	Type	Description
<code>event_id</code>	UUID v4	Globally unique event identifier
<code>run_id</code>	UUID	Execution run identifier (stable across pauses/resumes)
<code>runbook_id</code>	string	Runbook name or path from runbook YAML <code>meta.name</code>
<code>timestamp</code>	RFC3339	UTC timestamp with microsecond precision [19]
<code>kind</code>	string	Event type (e.g., <code>step/started</code> , <code>governance/approved</code>)
<code>sequence</code>	int64	Monotonically increasing integer within this run (starting from 0)
<code>payload</code>	object	Event-specific data (schema varies per kind)

Sequence numbers. The `sequence` field provides a total order over events in a single run. It is assigned by the `RunStore.WriteTrace` implementation and persists across pause/resume cycles. Sequence numbers are not globally unique; they are scoped to a single `run_id`.

Timestamps. The `timestamp` field uses RFC3339 format with at least microsecond precision (e.g., `2026-04-18T14:32:05.123456Z`). Timestamps are wall-clock times and must not be used for precise duration calculations (use event-specific `duration_ms` fields instead).

Event IDs. The `event_id` is a UUID v4 assigned at emission time. It is globally unique across all runs and all gert deployments. Event IDs are used for correlation in distributed tracing systems (§15) and for deduplication in event forwarding pipelines.

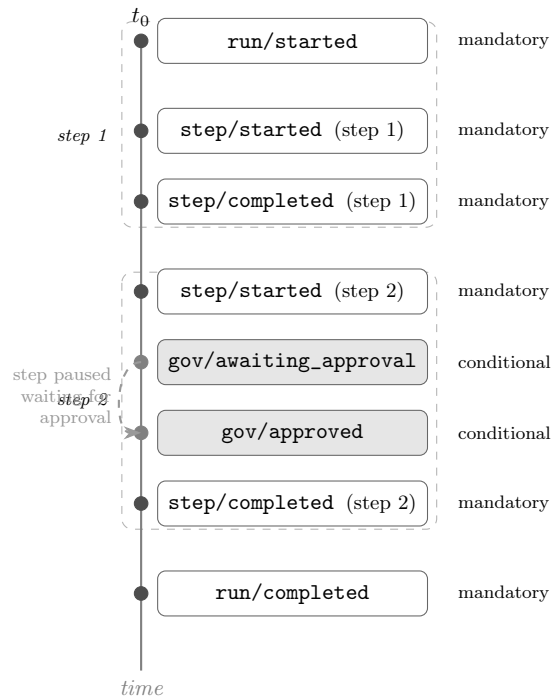


FIGURE 7.1: Typical event sequence for a two-step run where the second step requires governance approval. Mandatory events (solid dots) are emitted for every run; conditional events (hollow region) appear only when the relevant policy is triggered.

7.3 Event Catalog

The following subsections define every normative event kind. Each definition includes the payload schema, emission semantics, and whether the event is required or optional.

7.3.1 Run Lifecycle Events

run/started

Emitted: Once, as the first event in every execution. Must be the first line in the trace file.

Payload:

```
{
  "runbook_id": <string>, // from meta.name or file path
  "version": <string>, // runbook schema version (e.g., "runbook/v1")
  "actor": <string>, // identity (from --as flag or adapter session)
  "input_hash": <string>, // SHA256 of sorted input key-value pairs
  "mode": <string>, // "real" | "dry-run" | "replay"
  "client": <string>, // "cli" | "tui" | "web" | "vscode" | "mcp"
  "parent_run_id": <string> // omitted if not an invoke child
}
```

Semantics. The `input_hash` is the SHA256 hex digest of the JSON-serialized input dictionary with keys sorted lexicographically. This hash is used to detect input tampering in replay mode. The `parent_run_id` field is present only if this run was initiated by an `invoke` step in another run.

Required: YES. Every run MUST emit this event.

run/completed

Emitted: Once, as the terminal event for a successful run (all steps passed or skipped).

Payload:

```
{
  "outcome": <string>, // "success" | "failure" | "cancelled"
  "step_count": <int>, // total steps in the plan
  "passed": <int>, // steps that completed successfully
  "failed": <int>, // steps that failed
  "skipped": <int>, // steps skipped (branch/iterate)
  "duration_ms": <int64> // wall-clock duration from run/started
}
```

Semantics. The `outcome` field distinguishes clean completion (`success`), error termination (`failure`), and operator-initiated halt (`cancelled`). If `outcome` is `failure`, the last event before this one should be a `step/failed` or `governance/blocked`.

Required: YES. Every run that reaches a terminal state MUST emit this event.

run/cancelled

Emitted: When a run is cancelled by operator action (SIGINT, API call, or approval rejection).

Payload:

```
{
  "reason":      <string>, // "user_interrupt" / "approval_rejected" /
    ↪ "timeout"
  "cancelled_by": <string>, // actor identity (omitted if SIGINT)
  "step_id":     <string>  // step that was active at cancellation time
}
```

Semantics. This event is followed immediately by `run/completed` with outcome: `"cancelled"`. The `reason` field disambiguates graceful operator cancellation (`user_interrupt`) from governance-driven rejection (`approval_rejected`).

Required: YES, if a run is cancelled before natural completion.

7.3.2 Step Lifecycle Events

`step/started`

Emitted: Immediately before a step executes (after governance checks pass).

Payload:

```
{
  "step_id": <string>, // stable step identifier from runbook YAML
  "step_type": <string>, // "cli" / "manual" / "tool" / "invoke" / "branch"
    // / "iterate" / "parallel" / "compensate"
  "index": <int> // zero-based position in the ExecutionPlan
}
```

Required: YES. Every step that begins execution MUST emit this event.

`step/completed`

Emitted: After a step exits successfully.

Payload:

```
{
  "step_id": <string>,
  "outcome": <string>, // "success" / "skipped"
  "duration_ms": <int64>,
  "exit_code": <int> // for cli/tool steps; 0 for manual/invoke
}
```

Semantics. An outcome of `skipped` occurs for steps in non-taken branch paths or exhausted iterate loops. Exit code is present only for `cli` and `tool` steps; for other step types it is always 0.

Required: YES, for every step that completes without error.

`step/failed`

Emitted: When a step exits with a non-zero exit code or runtime error.

Payload:

```
{
  "step_id": <string>,
  "error_code": <string>, // "exit_nonzero" / "timeout" / "tool_error"
                        // / "governance_blocked" / "invoke_failed"
  "error_message": <string>, // human-readable (redacted)
  "retryable": <bool>, // true if retry policy allows retry
  "exit_code": <int> // for exit_nonzero; omitted otherwise
}
```

Semantics. The `retryable` field indicates whether the runtime will attempt a retry (requires a retry policy on the step). If `retryable` is true, this event will be followed by `step/retrying`.

Required: YES, for every step that fails.

`step/skipped`

Emitted: When a step is not executed due to control flow.

Payload:

```
{
  "step_id": <string>,
  "reason": <string> // "branch_not_taken" / "iterate_exhausted"
                // / "dry_run" / "cancelled"
}
```

Required: YES, for every step that is planned but not executed.

step/retrying

Emitted: Before each retry attempt (feature; requires retry policy).

Payload:

```
{
  "step_id":    <string>,
  "attempt":    <int>,      // 1 = first retry, 2 = second, etc.
  "max_attempts": <int>,
  "backoff_ms": <int64>    // delay before this retry
}
```

Required: YES, if retry is attempted. Optional in the (retry not yet implemented).

7.3.3 Governance Events

All governance events are normatively defined in §11. The following is a summary for completeness.

governance/command_checked

Emitted: When a command is evaluated against allowlist and denylist.

Payload:

```
{
  "step_id":    <string>,
  "command":    <string>,  // base name (argv[0] stripped of path)
  "verdict":    <string>,  // "allowed" | "denied"
  "rule_matched": <string> // "allowlist" | "denylist" | "default_allow"
}
```

Required: MAY be emitted. Recommended for audit trails but not required for basic operation.

governance/approval_requested

Emitted: When a step's approval gate is entered.

Payload:

```
{
  "step_id":      <string>,
  "approver_roles": [<string>], // roles that can approve
  "min_approvals": <int>,
  "timeout_seconds": <int>,    // 0 if no timeout
  "message":      <string>    // prompt text for approvers (redacted)
}
```

Required: YES, when approval gates are used.

governance/approval_received

Emitted: Each time an approver submits a decision.

Payload:

```
{
  "step_id": <string>,
  "approver": <string>, // actor identity
  "role": <string>,    // role the approver is acting in
  "decision": <string>, // "approved" | "rejected"
  "comment": <string>  // optional justification (redacted)
}
```

Required: YES, for every approval decision (including rejections).

governance/redaction_applied

Emitted: When redaction rules match and modify captured output.

Payload:

```
{
  "step_id": <string>,
  "field_path": <string>, // "stdout" | "stderr" | "capture.var_name"
  "pattern_count": <int>  // number of patterns that matched (not the
    ↪ patterns)
}
```

Semantics. The actual redaction patterns are not logged to avoid accidentally revealing what sensitive data looks like. Only the fact that redaction occurred is recorded.

Required: MAY be emitted. Useful for audit compliance.

7.3.4 Extension and Tool Events

extension/loaded

Emitted: When an extension successfully completes the handshake.

Payload:

```
{
  "extension_id": <string>, // from extension manifest
  "version":      <string>, // semver version
  "capabilities": [<string>] // list of capability names granted
}
```

Required: MAY be emitted. Useful for debugging extension issues.

extension/unloaded

Emitted: When an extension process exits or is terminated.

Payload:

```
{
  "extension_id": <string>,
  "reason":      <string> // "clean_exit" / "crash" / "timeout" / "killed"
}
```

Required: MAY be emitted.

tool/invoked

Emitted: When a tool call is dispatched (before the tool executes).

Payload:

```
{
  "step_id":      <string>,
  "tool_name":    <string>,
  "transport":    <string>, // "stdio" / "jsonrpc" / "mcp"
  "correlation_id": <string> // UUID v4, links to tool/completed
}
```

Required: MAY be emitted. Recommended for observability.

tool/completed

Emitted: When a tool call returns.

Payload:

```
{
  "step_id":      <string>,
  "tool_name":    <string>,
  "correlation_id": <string>, // matches tool/invoked
  "exit_code":    <int>,
  "duration_ms":  <int64>
}
```

Required: MAY be emitted.

7.3.5 Human-in-the-Loop Events

input/prompted

Emitted: When a manual step or input provider displays a prompt.

Payload:

```
{
  "step_id":      <string>, // omitted if prompt is from provider
  "prompt_id":    <string>, // UUID v4 for correlation
  "input_type":   <string>, // "text" | "checklist" | "attachment"
  "prompt":       <string>  // prompt text (redacted)
}
```

Required: YES, for manual steps. MAY be emitted for provider prompts.

input/received

Emitted: When the operator submits a response.

Payload:

```
{
  "step_id":      <string>,
  "prompt_id":    <string>, // matches input/prompted
  "redacted":     <bool>    // true if redaction rules matched the response
}
```

Semantics. The actual input value is not included in this event (it is recorded in the trace via `manual/completed` or `step/completed`). The `redacted` flag indicates whether the input passed through the redaction pipeline.

Required: YES, for manual steps.

7.3.6 Saga and Compensation Events (+)

The following events are reserved for the saga/compensation pattern, which is targeted for . They are included here for forward compatibility.

`saga/compensation_triggered`

Emitted: When a step fails and has registered compensation handlers.

Payload:

```
{
  "failed_step_id": <string>,
  "compensation_steps": [<string>] // step IDs to execute in reverse order
}
```

Required: YES, if compensation is triggered (+).

`saga/compensation_completed`

Emitted: After all compensation steps complete.

Payload:

```
{
  "outcome": <string> // "success" | "partial" | "failed"
}
```

Required: YES, if compensation is triggered (+).

7.4 Event Channels

Events flow through three channels, each with distinct delivery semantics and subscribers.

7.4.1 In-Process Event Bus

Purpose. The in-process event bus delivers events from the Runtime Core to adapters (TUI, `gert serve` session handlers) running in the same process.

Implementation. A Go `chan Event` with a fixed buffer size (default: 100 events). The Runtime Core writes to the channel; adapters read from it. If the channel is full, new events are discarded (the trace file is the authoritative record).

Subscriber API. Adapters call `RunHandle.Events()` to obtain a read-only event channel:

```
type RunHandle interface {  
    // Events returns a read-only channel of events for this run.  
    // The channel is closed when the run enters a terminal state.  
    Events() <-chan Event  
    // ... other methods ...  
}
```

Lifecycle. The event channel is created at `Runtime.Start()` and closed when the run emits `run/completed` or `run/cancelled`. Subscribers must drain the channel to avoid blocking the runtime (though the runtime will discard events if the subscriber is too slow).

7.4.2 WebSocket Broadcast (HTTP Adapter)

Purpose. For `gert serve -http`, events are broadcast to WebSocket clients for real-time UI updates.

Implementation. The HTTP adapter subscribes to `RunHandle.Events()` and forwards events to all connected WebSocket clients for the given `run_id`. The `/events` endpoint uses WebSocket.

Wire format. JSON-encoded event envelopes, one per WebSocket message frame.

Delivery semantics. Best-effort. If a WebSocket client is slow, the adapter may drop events to prevent backpressure. Clients should use the `/runs/:id/trace` HTTP endpoint to fetch the authoritative trace file if they detect gaps.

7.4.3 JSONL Trace File

Purpose. The append-only JSONL trace file (§12) is the durable, authoritative record of all events. It is the only channel with guaranteed at-least-once delivery.

Implementation. The Runtime Core writes every event synchronously to `.runbook/runs/<run-id>/trace.jsonl` before proceeding to the next action. File writes use `O_APPEND` mode to ensure atomicity on Unix systems.

Format. One JSON object per line, with the envelope fields from §6.2. See §12.1 for the complete trace file specification.

7.5 Event Subscriptions

7.5.1 Programmatic Subscription API

Extensions and adapters can subscribe to events programmatically. The subscription API supports filtering by event kind prefix and by `run_id`.

```
// EventSubscriber is the interface for subscribing to runtime events.
type EventSubscriber interface {
    // Subscribe registers a subscription with optional filters.
    // Returns a channel that will receive matching events.
    Subscribe(ctx context.Context, opts SubscribeOptions) (<-chan Event, error)
}

type SubscribeOptions struct {
    // KindPrefix filters events to those matching the prefix.
    // Examples: "step/", "governance/", "run/started"
    KindPrefix string

    // RunID filters events to a specific run. If empty, all runs.
    RunID string

    // BufferSize is the channel buffer size. Default: 100.
    BufferSize int
}
```

Example use cases.

- A TUI adapter subscribes with KindPrefix: "step/" to render step progress.
- A governance dashboard subscribes with KindPrefix: "governance/" to display policy violations.
- An observability forwarder subscribes with no filters to send all events to an Open-Telemetry collector (§15).

7.6 Wire Format

Events are serialized as JSON. The wire format is identical for the JSONL trace file, WebSocket broadcasts, and JSON-RPC notifications.

7.6.1 Concrete Example: step/started

```
{
  "event_id":    "a3f9c1e2-8d4b-4e1f-9a2c-3d5e6f7a8b9c",
  "run_id":      "20260418T143022-a3f9c1",
  "runbook_id":  "deploy-frontend",
  "timestamp":   "2026-04-18T14:30:25.123456Z",
  "kind":        "step/started",
  "sequence":    3,
  "payload": {
    "step_id":    "build_assets",
    "step_type":  "cli",
    "index":      2
  }
}
```

7.6.2 Concrete Example: governance/approval_requested

```
{
  "event_id":    "b1c2d3e4-5f6a-7b8c-9d0e-1f2a3b4c5d6e",
  "run_id":      "20260418T143022-a3f9c1",
  "runbook_id":  "deploy-frontend",
  "timestamp":   "2026-04-18T14:32:10.456789Z",
```

```

"kind":      "governance/approval_requested",
"sequence":  12,
"payload": {
  "step_id":      "deploy_to_prod",
  "approver_roles": ["SRE", "PM"],
  "min_approvals": 2,
  "timeout_seconds": 3600,
  "message":      "Deploying to production. Confirm change window."
}
}

```

7.7 Required vs. Optional Events

The following table classifies every event as REQUIRED (MUST be emitted) or OPTIONAL (MAY be emitted). Required events form the minimal contract for adapter compatibility; optional events provide observability and debugging enhancements.

Event Kind	Status	Notes
run/started	REQUIRED	First event in every trace
run/completed	REQUIRED	Terminal event
run/cancelled	REQUIRED	If cancelled
step/started	REQUIRED	For every executed step
step/completed	REQUIRED	For every successful step
step/failed	REQUIRED	For every failed step
step/skipped	REQUIRED	For every skipped step
step/retrying	OPTIONAL	+ retry feature
governance/command_checked	OPTIONAL	Useful for audit trails
governance/approval_requested	REQUIRED	If approval gates used
governance/approval_received	REQUIRED	If approval gates used
governance/redaction_applied	OPTIONAL	Compliance logging
extension/loaded	OPTIONAL	Debugging
extension/unloaded	OPTIONAL	Debugging
tool/invoked	OPTIONAL	Observability
tool/completed	OPTIONAL	Observability
input/prompted	REQUIRED	For manual steps
input/received	REQUIRED	For manual steps
saga/compensation_triggered	REQUIRED	+ if saga used
saga/compensation_completed	REQUIRED	+ if saga used

Adapter compatibility. Adapters **MUST** handle all **REQUIRED** events. They **SHOULD** gracefully ignore unknown event kinds to support forward compatibility (e.g., a the adapter rendering a trace with saga events).

7.8 Determinism and Replay Semantics

7.8.1 Replay Mode

In replay mode (§12.3), the Runtime Core reads events from a historical trace file and re-emits them in sequence order, respecting the original timing and causality. Adapters (TUI, Web, VS Code) subscribe to the replayed event stream and render the historical run indistinguishably from a live run.

Replay guarantees.

- Every event in the original trace is re-emitted with the same **sequence**, **timestamp**, **kind**, and **payload**.
- New **event_id** values are assigned (event IDs are not stable across replays).
- Step execution does not occur; the runtime only emits events. No subprocesses are forked, no tools are invoked, no governance checks are evaluated.

7.8.2 Non-determinism Boundaries

The following actions introduce non-determinism and are not captured in events:

- System time (**timestamp** fields use wall-clock time, not logical time).
- External network calls (DNS resolution, HTTP responses, database queries).
- File system state outside **.runbook/**.
- Random UUIDs (**event_id** and **correlation_id** fields are regenerated in replay).

To achieve full deterministic replay (for testing and debugging), use dry-run mode (§8.2) or snapshot the external environment.

Chapter 8

Security and Trust

Gert is designed to operate in regulated, high-trust environments where operational actions must be auditable, accountable, and governed by explicit policy. Security is not an add-on; it is foundational to the system's architecture. This chapter defines the threat model, trust boundaries, process isolation mechanisms, credential handling, transport security, RBAC model, audit non-repudiation, and supply chain security.

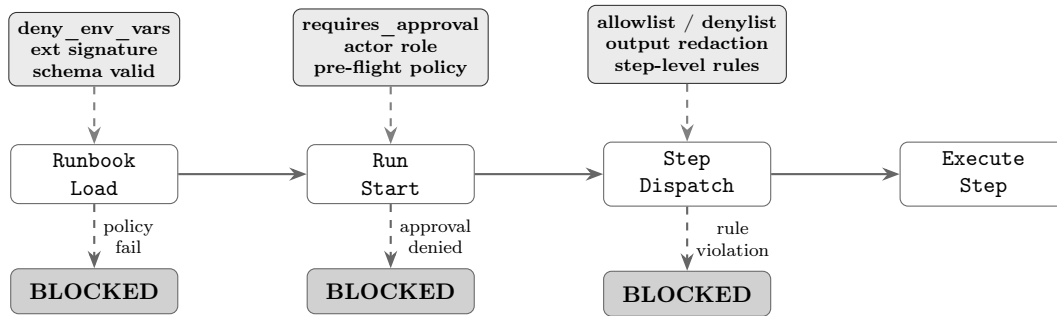
8.1 Threat Model

Gert addresses the following threat categories, which are common in operational runbook systems and have real-world precedent in systems like AWS Systems Manager, PagerDuty Runbook Automation, and Rundeck [8, 23, 24].

This threat model is derived from NIST SP 800-53 security controls [1] and ISO/IEC 27001 guidance on privileged access management and audit logging [4].

Threat	Mitigation	Section Ref
Runbook author executes unauthorized commands	Command allowlist/denylist enforced at runtime core	§8.1, §11
Malicious extension injects code into host process	Out-of-process extension isolation; cryptographic signature verification	§8.2, §4
Tool or extension leaks credentials into trace	Mandatory redaction for sensitive fields; <code>sensitive: true</code> input marking	§8.4
Unauthorized user triggers runbook execution	RBAC enforced at <code>gert serve</code> ingress	§8.6
Man-in-the-middle attack on JSON-RPC transport	TLS required for all non-localhost listeners	§8.5
Replay attack on approval decision	Approval tokens signed with Ed25519; <code>approvalId</code> is single-use	§8.6, §11
Compromised tool binary executes untrusted code	Tool binary checksum verification before invocation	§8.8

TABLE 8.1: Threat model and mitigations for gert.

FIGURE 8.1: Governance enforcement points across the gert execution pipeline. Each stage has dedicated policy checks; failure at any checkpoint halts execution and emits a `governance/blocked` trace event.

8.2 Extension Trust Model

Extensions operate under one of three trust levels, declared explicitly in the extension manifest and evaluated by the host before any process is started.

8.2.1 Trust Levels

Local trusted. Installed by the machine owner (via package manager or explicit installation to a system-wide extension directory). These extensions run as child processes of the current user, inherit the user’s capability set (subject to manifest restriction), and have access to all host capabilities declared in their manifest. No signature verification is performed; trust is conferred by the installation act.

Example: an extension installed via `apt-get` or `brew` into `/usr/local/lib/gert-extensions/`.

Project-scoped. Declared in the runbook project directory (e.g., `.gert/extensions.yaml`) or inline in the runbook’s `extensions:` field. These extensions run in a restricted sandbox: only capabilities explicitly granted by the workspace governance policy are forwarded. File and network access are constrained to declared paths and hosts. Signature verification is *optional* but recommended.

Use case: a team-maintained extension for Acme-specific incident tooling, distributed via the runbook repository.

Remote/signed. Extensions distributed via package repository or downloaded from the internet. These extensions *must* carry a cryptographic signature verified against a trust store. The signature covers the extension manifest and binary. If verification fails, the extension is rejected before any subprocess is started.

Signature format is defined in Section 8.2.2.

8.2.2 Extension Signature Format

Remote/signed extensions must include a `signature:` block in `gert-extension.yaml`:

```
# gert-extension.yaml - signed extension
apiVersion: extension/v1

meta:
  name: acme.incident-pack
```

```
version: "1.2.0"

signature:
  algorithm: ed25519
  publicKeyFingerprint: "sha256:abcd1234..."
  signatureData: "base64-encoded-signature"
  signedFields: ["meta", "capabilities", "entry-point", "transport"]

trust_chain:
- fingerprint: "sha256:abcd1234..."
  issuer: "Acme Platform Team CA"
  notBefore: "2024-01-01T00:00:00Z"
  notAfter: "2025-01-01T00:00:00Z"
```

Signature verification algorithm. The signature is computed over the canonical JSON representation of the fields listed in `signedFields`. The host:

1. Extracts the public key fingerprint from the `signature` block.
2. Looks up the corresponding public key in the trust store (`$HOME/.gert/trusted-keys/` or `/etc/gert/trusted-keys/`).
3. Reconstructs the canonical JSON of the signed fields (sorted keys, no whitespace).
4. Verifies the Ed25519 signature over the canonical JSON bytes.
5. Checks that the current time falls within the `notBefore` and `notAfter` window of the trust chain.

If any step fails, the extension is rejected with a `governance/extensionRejected` trace event and reason `signature_verification_failed`.

8.2.3 Extension Verification Command

The `gert extension verify` command allows operators to manually verify an extension before installation:

```
$ gert extension verify ./acme-extension/gert-extension.yaml \
  --trust-store /etc/gert/trusted-keys/
```

```
Extension: acme.incident-pack v1.2.0
Signature: valid (ed25519, expires 2025-01-01)
Issuer: Acme Platform Team CA
Capabilities: tool-registration, network
Trust level: remote/signed
Verification: PASSED
```

8.3 Process Isolation

Extensions and tools always run as separate OS processes. They are never loaded as shared libraries or executed in-process. This isolation boundary ensures that a crash, hang, or malicious extension cannot compromise the host process.

8.3.1 Isolation Mechanisms

Separate OS process. As defined in §4, extensions are forked as child processes with their own address space, file descriptor table, and signal handlers.

Filesystem access control. Extensions with `capability/file-read` or `capability/file-write` declare the set of filesystem paths they may access via `file-read-paths` and `file-write-paths` in the manifest. The host enforces these constraints:

- On OpenBSD: via `unveil(2)` to restrict visible paths.
- On Linux: via `seccomp-bpf` to filter `open(2)` and `openat(2)` syscalls.
- On all platforms: by validating all path arguments in `file-read` and `file-write` RPC methods before dispatch.

Attempts to access undeclared paths return a `capability-denied` error.

Network access control. Extensions with `capability/network` declare the set of hostnames they may connect to via `network-hosts`. The host validates outbound connections:

- Before dispatching an extension RPC that specifies a target URL, the host checks that the hostname matches one of the declared `network-hosts` patterns.
- Extensions may not listen for inbound connections.

- Wildcard patterns are supported: `*.acme.example` permits `api.acme.example`, `prod.acme.example`, etc.

Environment variable isolation. Extensions receive ONLY the environment variables explicitly forwarded by the runbook's `governance.allowed_env` list. Host environment variables that match `governance.deny_env_vars` patterns are never forwarded, even if they appear in `allowed_env`.

The denylist takes precedence: if `AWS_SECRET_ACCESS_KEY` is denied, it is stripped from the extension's environment regardless of other policy.

Crash isolation. If an extension process crashes (exits with non-zero status or is killed by a signal), the host:

1. Emits an `extension.crashed` trace event with exit code and signal number.
2. Marks the extension as unavailable for the remainder of the run.
3. Cancels all in-flight invocations to that extension with error code `-32099` (`extension_crashed`).
4. Continues the run: the host process does not terminate.

8.4 Credential and Secret Handling

Gert does not store secrets. It acts as a pass-through: secrets are resolved from input providers at runtime, used for a single step invocation, and then discarded. Secrets never appear in trace logs or captured output.

8.4.1 Secret Resolution and Redaction

Input provider sensitivity marking. Input provider definitions declare which resolved values are sensitive via `sensitive: true` in the provider schema:

```
# vault.provider.yaml
outputs:
  credentials:
    type: object
    sensitive: true
```

When the host resolves `from: vault.credentials`, it marks the returned value as sensitive in memory. This flag is respected by all downstream consumers.

Tool sensitive inputs. Tool action definitions declare which input parameters accept sensitive data via `sensitive_inputs`:

```
# my-tool.tool.yaml
actions:
  deploy:
    args:
      api_key:
        type: string
        required: true
    sensitive_inputs: [api_key]
```

When the host prepares the tool invocation envelope, it marks the `api_key` field as sensitive. The tool runtime redacts this field in all trace events.

Mandatory redaction. All values marked `sensitive: true` are redacted before being written to the trace or displayed in any adapter UI. Redaction replaces the value with the string `<redacted>`.

In addition, the runbook's `governance.redact` block defines regex patterns applied to all captured output (stdout, stderr, JSON responses) before storage. See §11 for the full redaction specification.

8.4.2 Secret Storage Policy

Gert does NOT:

- Store secrets in the runbook YAML.
- Write secrets to disk (trace files, state snapshots, etc.).
- Cache secrets across steps (unless explicitly declared by the input provider's `cache_scope`).
- Transmit secrets over unencrypted transports.

Secrets must be resolved via input providers that integrate with secret management systems (e.g., HashiCorp Vault, AWS Secrets Manager, 1Password).

8.5 Transport Security

Gert supports two primary transport modes for JSON-RPC communication:

1. **stdio**: JSON-RPC over stdin/stdout of a child process. No network exposure; inherently local. No TLS required.
2. **HTTP/WebSocket**: JSON-RPC over HTTP for `gert serve` mode. Network exposure possible; TLS and authentication required for non-localhost.

8.5.1 TLS Requirements

Localhost exception. If `gert serve` binds to `127.0.0.1` or `localhost`, TLS is optional. Connections are assumed to be local and trusted.

Non-localhost listeners. If `gert serve` binds to `0.0.0.0`, a public IP, or a hostname, TLS is *mandatory*. The host will refuse to start without a valid TLS certificate and private key.

Configuration:

```
$ gert serve --http 0.0.0.0:8443 \  
  --tls-cert /etc/gert/tls/cert.pem \  
  --tls-key /etc/gert/tls/key.pem
```

The host performs standard TLS 1.3 handshake with the client. TLS 1.2 is supported for compatibility but TLS 1.1 and earlier are rejected.

8.5.2 Client Authentication

The host supports two authentication modes for `gert serve -http`:

Bearer token authentication. Clients include an `Authorization: Bearer <token>` header on all HTTP requests. The token is validated against a configured token store (file-based, environment variable, or integration with an IdP).

Example:

```
$ gert serve --http :8443 --auth-mode bearer \  
  --bearer-tokens /etc/gert/tokens.yaml
```


The `tokens.yaml` file maps tokens to actor identities:

```
# tokens.yaml
tokens:
- token: "abc123..."
  actor: "ops@acme.example"
  roles: ["operator", "viewer"]
```

Mutual TLS (mTLS). The host requires clients to present a valid X.509 client certificate signed by a trusted CA. The client certificate's subject CN or SAN is used as the actor identity.

Configuration:

```
$ gert serve --http :8443 --auth-mode mtls \
  --tls-client-ca /etc/gert/tls/client-ca.pem
```

mTLS is the recommended authentication mode for production deployments and enterprise environments.

8.5.3 WebSocket Security

WebSocket connections for the event stream (§13) inherit the security context of the HTTP server:

- If the HTTP server uses TLS, WebSocket connections are encrypted (`wss://`).
- Client authentication applies: bearer tokens or client certificates are validated before the WebSocket upgrade.
- Once upgraded, the WebSocket connection is bound to the authenticated actor identity for the lifetime of the connection.

8.5.4 Extension JSON-RPC Transport

Extensions communicate with the host via stdio JSON-RPC (§4). This transport is not exposed to the network and does not require TLS. The process boundary is the trust boundary: the host trusts that OS-level process isolation prevents interception of stdin/stdout pipes.

8.6 RBAC for gert serve

When gert runs in server mode (`gert serve -http`), it enforces role-based access control (RBAC) on all API operations. Actors are assigned roles, and each role grants a specific set of permissions.

8.6.1 Actor Identity

An **actor** is the authenticated identity making a request. Actor identity is derived from:

- The bearer token (mapped to an actor ID via the token store), or
- The client certificate subject CN/SAN (in mTLS mode).

The actor identity is recorded in all trace events where a human or external system initiates an action (run start, approval submission, cancellation).

8.6.2 Roles

Gert defines three standard roles:

Viewer. Read-only access. Permissions:

- List runs (`GET /runs`)
- Get run details (`GET /runs/:id`)
- Subscribe to event stream (`GET /ws` or `GET /events`)
- Read trace files

Cannot start runs, approve, or cancel.

Operator. Execution permissions. Includes all **viewer** permissions plus:

- Start a run (`POST /exec/start`)
- Cancel a run (`POST /exec/cancel`)
- Submit approvals (`POST /approvals`)

Cannot modify governance policy or manage roles.

Admin. Full control. Includes all **operator** permissions plus:

- Modify workspace governance policy
- Manage role assignments (add/remove actors from roles)
- Configure extension trust store

8.6.3 Role Assignment

Roles are assigned in the workspace config (`gert-project.yaml`):

```
# gert-project.yaml
access:
  roles:
    - role: admin
      actors:
        - "alice@acme.example"
        - "bob@acme.example"
    - role: operator
      actors:
        - "ops-team@acme.example"
        - "oncall@acme.example"
    - role: viewer
      actors:
        - "*@acme.example" # All Acme employees
```

Actor patterns support wildcards for organization-wide grants.

8.6.4 Approval Authorizer Role

The `governance.require_approval` field in the runbook declares the role required to approve a step:

```
steps:
  - id: delete_database
    cli: "kubectl delete database prod-db"
    approvals:
      min: 1
      roles: ["dba", "admin"]
```

When the run reaches this step, it emits an **approval/required** trace event. The step is paused. Only actors with the **dba** or **admin** role may submit a valid approval.

The approval submission includes:

```
POST /approvals
{
  "runId": "run-abc123",
  "stepId": "delete_database",
  "decision": "approve",
  "actorId": "alice@acme.example",
  "signature": "<ed25519-signature-of-approval-payload>",
  "timestamp": "2026-04-18T12:00:00Z"
}
```

The host verifies:

1. The actor is authenticated and has one of the required roles.
2. The signature is valid (Ed25519 over the canonical JSON of **runId**, **stepId**, **decision**, **actorId**, **timestamp**).
3. The **timestamp** is within a 5-minute clock skew window.
4. The **runId** and **stepId** match an active approval gate.
5. This approval has not been submitted before (replay protection).

If all checks pass, the approval is recorded in the trace (**approval/received**) and the step proceeds if the minimum approval count is reached.

8.7 Audit Non-Repudiation

Non-repudiation ensures that governance decisions — especially approvals and rejections — cannot be denied by the actor who made them. This is critical for regulatory compliance (SOC2, ISO27001, HIPAA) [4, 7, 28].

8.7.1 Governance Decision Recording

Every governance decision event (`governance/approval_received`, `governance/approval_rejected`, `governance/commandBlocked`) must record:

- Actor identity (authenticated user or service principal)
- Timestamp (RFC 3339 with timezone) [19]
- Decision payload (what was approved/rejected)
- Signature (Ed25519 over the canonical decision payload)

Example trace event:

```
{
  "eventType": "governance/approval_received",
  "timestamp": "2026-04-18T12:00:00Z",
  "runId": "run-abc123",
  "stepId": "delete_database",
  "actor": "alice@acme.example",
  "decision": "approve",
  "signature": "ed25519:abcd1234...",
  "signatureAlgorithm": "ed25519",
  "signaturePublicKey": "sha256:pubkey-fingerprint"
}
```

The signature covers the fields `runId`, `stepId`, `actor`, `decision`, and `timestamp`. The public key is looked up from the actor's identity record or client certificate.

8.7.2 Trace File Integrity (HMAC Chaining)

The JSONL trace file (§12) is append-only, but this alone does not prevent tampering. To provide cryptographic integrity, the host optionally computes an HMAC chain over the trace:

1. At run start, initialize $H_0 = \text{HMAC-SHA256}(\text{key}, \text{runId})$.
2. After writing each trace event E_i , compute $H_i = \text{HMAC-SHA256}(\text{key}, H_{i-1} || E_i)$.

3. Store `H_i` in the event as `integrityHash`.
4. At run completion, emit a final `run.completed` event with `finalHash: H_N`.

Verification: a verifier recomputes the HMAC chain from `H_0` forward. If any event is modified or deleted, the chain breaks.

The HMAC key is derived from the workspace master key or provided via `-integrity-key`.

8.7.3 SHA256 Evidence Capture

As defined in §12, gert captures SHA256 hashes of all evidence artifacts (attachments, screenshots, CLI output files). These hashes are written to the trace event immediately after the artifact is captured, providing a cryptographic anchor for the artifact's integrity.

The combination of HMAC trace chaining + SHA256 evidence hashes provides end-to-end non-repudiation for the entire run.

8.8 Supply Chain Security

Tools and extensions are executable code distributed via package managers, repositories, or direct download. Supply chain attacks (e.g., compromised binaries, backdoored dependencies) are a real threat in operational environments. Gert provides checksum verification and signature validation to mitigate these risks.

8.8.1 Tool Binary Checksums

Tool definitions may declare a `checksum` field in `.tool.yaml`:

```
# kubectl.tool.yaml
meta:
  name: kubectl
  binary: kubectl

checksum:
  algorithm: sha256
  digest: "a3b4c5d6e7f8..."
```

Before invoking the tool, the host:

1. Resolves the binary path (via `PATH` lookup or absolute path).
2. Computes the SHA256 hash of the binary file.
3. Compares the computed hash to `checksum.digest`.
4. If the hashes do not match, the tool invocation is blocked with a `governance/toolChecksumMismatch` trace event.

8.8.2 gert verify Command

The `gert verify` command validates all tools and extensions in the workspace before execution:

```
$ gert verify --workspace .
Verifying tools...
✓ kubectl: checksum verified (sha256:a3b4c5...)
✓ curl: no checksum declared (skipped)
× jq: checksum mismatch (expected sha256:abc123, got sha256:def456)

Verifying extensions...
✓ acme.incident-pack: signature valid (ed25519, expires 2025-01-01)
× debug-tools: no signature (required for remote extensions)

Verification: FAILED (2 errors)
```

This command should be run in CI/CD pipelines before deploying runbooks to production.

8.8.3 Extension Signature Verification

As defined in §8.2.2, remote extensions must carry a cryptographic signature. The signature verification process ensures that:

- The extension binary and manifest have not been tampered with.
- The extension was signed by a trusted authority (public key in the trust store).
- The signature has not expired (`notAfter` check).

If verification fails, the extension is rejected before any subprocess is started.

8.9 Security Recommendations

For production deployments, the following security practices are recommended:

1. **Enable TLS for all gert serve deployments** — even internal networks can be compromised.
2. **Use mTLS for client authentication** — bearer tokens are vulnerable to theft; client certificates provide stronger identity.
3. **Enable HMAC trace chaining** — provides cryptographic integrity for audit trails.
4. **Declare tool binary checksums** — prevents execution of tampered binaries.
5. **Require signatures for all remote extensions** — do not trust unsigned code from the internet.
6. **Run gert verify in CI** — catch integrity violations before production.
7. **Use least-privilege capability grants** — extensions should request only the capabilities they need; admins should audit and deny excessive requests.
8. **Regularly rotate approval keys** — the Ed25519 keys used for signing approvals should be rotated every 90 days.

These recommendations align with NIST SP 800-53 controls for privileged access management (AC-2, AC-6), audit and accountability (AU-2, AU-9), and system integrity (SI-7) [1].

Chapter 9

Testing and Acceptance

This section specifies the complete testing strategy for gert: the test framework, unit and contract test patterns, integration test strategy, the `gert test` feature, regression testing against the runbook corpus, performance acceptance criteria, the extension test harness, and CI/CD gate requirements.

9.1 Test Framework

9.1.1 Standard Library and testify

All gert tests are written in Go using the standard `testing` package as the test runner and github.com/stretchr/testify [26] for assertions:

- `testify/assert` — non-fatal assertions; test continues on failure.
- `testify/require` — fatal assertions; test halts on failure.
- `testify/mock` — interface mocking for unit tests (engine, event bus, trace writer).

Table-driven tests are the preferred pattern for all logic that has well-defined input/output pairs (e.g. expression evaluation, schema validation, governance rule matching):

```
func TestGovernanceAllowlist(t *testing.T) {
    cases := []struct {
        name      string
        cmd       []string
        allowed   bool
    }{
```

```
    {"exact match", []string{"kubectl", "get", "pods"}, true},
    {"partial match", []string{"kubectl", "delete", "ns"}, false},
    {"empty allowlist", []string{"echo", "hi"}, false},
  }
  for _, tc := range cases {
    t.Run(tc.name, func(t *testing.T) {
      result := allowlist.Check(tc.cmd)
      assert.Equal(t, tc.allowed, result)
    })
  }
}
```

9.1.2 Test Packages

Each Go package in the module ships a co-located `_test.go` file (white-box tests) and optionally an `<pkg>_integration_test.go` file (integration tests, gated by a build tag). The naming convention:

```
pkg/engine/engine.go
pkg/engine/engine_test.go           // white-box unit tests
pkg/engine/engine_integration_test.go // build tag: //go:build integration
```

9.2 The gert test Feature

9.2.1 Scenario Replay Testing

`gert test <runbook.yaml>` discovers and executes all scenario tests for a runbook. A scenario test consists of:

- `inputs.yaml` — resolved input values for the run.
- `scenario.yaml` (optional) — pre-recorded command responses and manual evidence.
- `test.yaml` — assertions over the resulting trace and captures.

Scenario directories are located by convention:

```
{runbook-dir}/scenarios/{runbook-name}/{scenario-name}/
  inputs.yaml
  scenario.yaml  # optional; omit for runbooks with no CLI/tool steps
  test.yaml
```

The test runner executes the runbook in replay mode using the scenario file as the command/evidence fixture, then evaluates assertions from `test.yaml`.

9.2.2 Test Assertions (`test.yaml`)

```
assertions:
  outcome: passed          # "passed" / "failed" / "skipped"
  steps_passed: 5
  steps_failed: 0

captures:
  db_host:
    equals: "prod-db.example.com"
  migration_status:
    contains: "complete"

trace:
- type: step/completed
  step_id: run-migration
- type: governance/blocked
  step_id: drop-table
  data.rule: denylist
```

The `trace` assertions match events in order of occurrence. The `data.*` dotted paths dereference into the event's `data` object.

9.3 Contract Tests

Contract tests verify the interface boundaries between the schema's five components. Each contract test lives in the *consumer* package and uses a conformance helper that validates any implementation satisfying the interface.

9.3.1 Core Runtime Contract

The runtime `Executor` interface must satisfy:

- Emits a `run/started` event before executing any step.
- Emits a `step/started` before and `step/completed/step/failed` after each step.
- Emits `run/completed` or `run/failed` as the final event.
- Does not emit events after the terminal event.
- Does not call any adapter method directly (events only).

```
// ConformanceTest validates any Runtime implementation.
func ConformanceTest(t *testing.T, rt Runtime) {
    t.Helper()
    bus := newTestEventBus()
    err := rt.Execute(context.Background(), fixtureRunbook, bus)
    require.NoError(t, err)
    events := bus.Events()
    require.Equal(t, "run/started", events[0].Type)
    require.Equal(t, "run/completed", events[len(events)-1].Type)
    assertNoEventsAfterTerminal(t, events)
}
```

9.3.2 Schema Contract Tests

- Every `apiVersion: runbook/v1` document in `testdata/` must pass structural validation without errors.
- Every invalid fixture in `testdata/invalid/` must fail validation with a structured error (message + path).
- Structural and semantic validation are tested independently: a structurally valid document may fail semantic validation (e.g. referencing an undefined capture).

9.3.3 Extension Manifest and Handshake Tests

The extension host verifies:

- An extension process that responds to the initialisation handshake within the configured timeout is accepted.
- An extension process that fails to respond is terminated and its capabilities are not registered.
- Capability grants in the manifest are enforced: an extension without `cap:schema.register` cannot register schema.
- A manifest with missing required fields is rejected at load time with a structured error.

9.3.4 Tool Invocation Envelope Tests

- Tool invocations carry the correct `run_id` and `step_id` correlation fields.
- Tool outputs are captured and available as step captures after the step completes.
- A tool that exits with non-zero triggers `step/failed` with `error_type: tool_error`.
- Timeout cancellation: a tool that exceeds its timeout receives a `context.DeadlineExceeded` and emits `step/failed` with `error_type: timeout`.

9.3.5 Event Bus Determinism Tests

- Events delivered to the event bus maintain the `seq` ordering guaranteed by `RunStore.WriteTrace`.
- Slow adapter consumers do not cause the engine's execution loop to block (bounded channel or dropping policy required).
- Event replay (`ReadTraceSince`) returns events in ascending `seq` order.

9.4 Integration Test Strategy

9.4.1 End-to-End Fixture Runs

Integration tests spin up the gert runtime against fixture runbooks stored in `testdata/runbooks/` and assert on the resulting trace files. They are gated by the `integration` build tag and run in CI separately from unit tests.

```
//go:build integration

func TestE2E_BranchingRunbook(t *testing.T) {
    dir := t.TempDir()
    runID, err := exec.RunFixture(dir, "testdata/runbooks/branch.runbook.yaml",
        WithInputs(map[string]string{"env": "prod"}),
        WithScenario("testdata/scenarios/branch-prod.yaml"),
    )
    require.NoError(t, err)
    trace := readTrace(t, dir, runID)
    assertEvent(t, trace, "step/completed", map[string]any{
        "data.step_id": "deploy-prod",
    })
    assertNoEvent(t, trace, "step/completed", map[string]any{
        "data.step_id": "deploy-staging",
    })
}
```

9.4.2 Fixture Runbook Corpus

The integration test corpus includes fixtures covering:

- Linear runbook (all step types: cli, manual, tool, invoke).
- Branching runbook (both branch arms; governance-blocked branch).
- Iterate runbook (list mode, convergence mode, parallel concurrency).
- Nested invoke (parent + two child runbooks).
- Retry policy (transient failure followed by success).
- Resumption (crash after step 2, resume, verify trace continuity).
- Governance: allowlist violation, denylist violation, approval gate.

9.4.3 Trace Assertion Helpers

A shared test package `pkg/testutil` provides helpers:

```
package testutil

// ReadTrace reads and parses all events from a run's trace.jsonl.
func ReadTrace(t *testing.T, runDir, runID string) []engine.DurableEvent

// AssertEvent asserts that at least one event matches type and data predicates.
func AssertEvent(t *testing.T, events []engine.DurableEvent,
    eventType string, predicates map[string]any)

// AssertEventOrder asserts events appear in the given type order.
func AssertEventOrder(t *testing.T, events []engine.DurableEvent,
    types ...string)

// AssertNoEvent asserts no event matches the given type and predicates.
func AssertNoEvent(t *testing.T, events []engine.DurableEvent,
    eventType string, predicates map[string]any)
```

9.5 Regression Testing

9.5.1 Runbook Corpus as Fixtures

The runbook corpus (located in `examples/` and operator-contributed runbooks) serves as a regression suite for schema stability. The regression test runner:

1. Reads each runbook from the corpus.
2. Runs `gert validate` (schema) and asserts it passes.
3. For runbooks with associated scenario files, runs `gert test` and asserts the outcome matches the golden output.
4. Fails the build if any previously passing corpus runbook regresses.

The schema acceptance target is $\geq 95\%$ of corpus runbooks passing `gert validate` unmodified between minor releases.

9.5.2 Golden Trace Comparisons

For deterministic scenarios, the expected trace can be stored as a golden file. The golden comparison test:

1. Runs the scenario.
2. Normalises timestamps and run IDs (replaces with fixed values).
3. Compares the normalised JSON-lines output to `testdata/golden/<scenario>.jsonl`.
4. Fails if any event differs. Golden files are updated by running with `-update` flag.

9.6 Performance Acceptance Criteria

The following latency budgets must pass on a reference machine (4-core, 8 GB RAM, SSD) with the integration test suite:

Operation	p50 target	p99 target
<code>gert validate</code> (single runbook, <50 steps)	<50 ms	<200 ms
Step startup (cli step)	<10 ms	<50 ms
Event delivery to adapter (local channel)	<1 ms	<5 ms
Trace event write + fsync	<5 ms	<20 ms
Snapshot write (100 steps history)	<20 ms	<80 ms
<code>gert test</code> (10-step replay scenario)	<500 ms	<1 s
Tool invocation round-trip (stdio, local process)	<50 ms	<200 ms

Performance benchmarks are written using Go's `testing.B` and run in CI weekly (not on every PR). A regression of $>2\times$ on any metric blocks the nightly build.

9.7 Extension Test Harness

9.7.1 Overview

Extension developers can test their extension logic without spinning up the full gert runtime. The `pkg/exttest` package provides a test harness:


```
import "github.com/ormasoftchile/gert/pkg/exttest"

func TestMyExtension(t *testing.T) {
    h := exttest.NewHarness(t,
        exttest.WithManifest("testdata/my-ext.manifest.yaml"),
        exttest.WithCapabilities("cap:schema.register", "cap:tool.register"),
    )
    defer h.Close()

    // Start the extension process
    h.Start("./my-extension")

    // Assert it registered expected tools
    tools := h.RegisteredTools()
    require.Contains(t, tools, "my-tool")

    // Invoke a tool and assert the response
    resp := h.InvokeTool("my-tool", map[string]any{"arg1": "value"})
    assert.Equal(t, 0, resp.ExitCode)
    assert.Contains(t, resp.Stdout, "expected output")
}
```

9.7.2 Harness Capabilities

The test harness provides:

- A mock extension host that implements the full handshake protocol.
- Assertion helpers for registered tools, schema registrations, and event subscriptions.
- A `FakeRunContext` that provides a realistic run context without executing a runbook.
- Automatic process cleanup on test completion.
- Configurable capability grants (to test capability enforcement).

9.7.3 Mocking the Extension Transport

For pure unit tests of extension logic (without spawning a subprocess), the transport layer is mockable:

```
// InProcessTransport runs the extension handler in-process for unit testing.
type InProcessTransport struct {
    Handler ExtensionHandler
}
```

9.8 Acceptance Criteria

The following acceptance criteria must be satisfied before a release is considered shippable:

1. Core runtime has no direct adapter dependencies — verified by the `no_import` contract test.
2. Extensions register schema and tools through declared contracts — verified by the extension handshake conformance suite.
3. Capability violations fail fast with structured errors — verified by the governance contract test for each capability type.
4. Adapters integrate using the same stable core contracts — verified by running each adapter’s integration test against a mock core.
5. Every step type (`cli`, `manual`, `tool`, `invoke`, `branch`, `iterate`) has ≥ 1 end-to-end integration test covering the happy path and at least one failure path.
6. The runbook corpus passes `gert validate` at $\geq 95\%$ without modification.
7. All performance targets in § 8.6 pass on the reference machine.
8. `gert test` successfully replays all scenario fixtures in `testdata/scenarios/`.

9.9 CI/CD Integration

9.9.1 PR Gate (every pull request)

1. **Unit tests:** `go test ./...` (no build tags). Target: <60 s.
2. **Schema validation:** validate all YAML fixtures in `testdata/` against JSON Schema. Target: <10 s.

3. **Contract tests:** `go test -run Contract ./....` Target: <30 s.
4. **Linters:** `golangci-lint` run with the project's `.golangci.yaml` config.
5. **Vet:** `go vet ./....`

All five gates must pass for a PR to be mergeable.

9.9.2 Integration Gate (merge to main)

1. **Integration tests:** `go test -tags=integration ./....` Target: <5 min.
2. **Regression corpus:** `gert test` against the corpus. Target: <2 min.
3. **Golden trace comparison:** normalised trace comparison for all deterministic fixtures.

9.9.3 Nightly Gate

1. **Performance benchmarks:** `go test -bench=. -benchtime=10s ./...` — compare against stored baseline; fail on >2× regression.
2. **Race detector:** `go test -race ./....`
3. **Extension conformance suite:** run all built-in extensions through the conformance harness.

Chapter 10

Open Questions

The questions below are unresolved architectural decisions. Each entry states **(a)** why the question matters for the design, **(b)** the options under consideration, and **(c)** what additional information is needed to close it. Decisions must be recorded in `.squad/decisions.md` before core contract freeze.

Q1. What is the concurrency model for the core runtime?

Why it matters: Whether the runtime supports multiple concurrent runs in a single process affects the state isolation model, the trace writer design, the governance enforcer (is it per-run or global?), and whether `gert serve` can multiplex concurrent VS Code sessions.

Options: (a) Single-run-per-process; each `gert exec` or `gert tui` is a separate OS process. (b) Multiple concurrent runs in-process, with run-scoped state isolation via Go context and per-run goroutine groups. (c) Multiple runs via a long-lived daemon (`gert serve` as the host), with runs as logical sessions.

Information needed: Whether the VS Code extension use case requires concurrent runs in a single `gert serve` process (e.g., multiple open runbook panels), and what the goroutine safety requirements are for the trace writer and governance layer.

Q2. How do extension-contributed policies compose with host-level governance policies?

Why it matters: The host enforces a governance policy (command allowlist, env blocking, redaction). An extension may also wish to contribute policies—e.g., a security extension that adds additional command denylists or redaction patterns. The composition model determines whether extensions can weaken host policies

(prohibited), extend them (likely desired), or operate in a separate policy namespace (safest but most limited).

Options: (a) Additive composition only: extensions may only add restrictions (denylist entries, redaction patterns); they may never remove host-level restrictions. (b) Policy namespacing: each extension has an isolated policy scope; the host applies all active policies via intersection. (c) Policy delegation: the host can explicitly delegate a policy domain to an extension (e.g., redaction rules owned by a data-classification extension).

Information needed: A threat model analysis of what happens when a malicious or compromised extension contributes a policy that attempts to weaken host restrictions. This should feed into §8.

Q3. Does the `.provider.yaml` input provider framework survive unchanged?

Why it matters: The input provider model (`.provider.yaml`, JSON-RPC resolution, `from:` bindings, workspace config) is a complete and working framework. It must either be carried forward as-is, redesigned to fit the component model, or replaced by an extension-based resolution mechanism. The choice affects the schema (§4), the extension runtime (§5), and any operator who has a custom provider binary.

Options: (a) Preserve as-is: `.provider.yaml` format and JSON-RPC protocol are unchanged; providers are a stable first-class concept. (b) Redesign as extension capability: providers become a named capability that extensions may implement; the `.provider.yaml` format is deprecated. (c) Hybrid: `.provider.yaml` providers continue to work as a built-in extension type, with a migration path to the full extension protocol.

Information needed: Whether any existing provider binaries are in active production use that would be broken by a protocol change, and whether the extension handshake protocol can be made a strict superset of the provider JSON-RPC protocol.

Q4. Is the `gert serve` JSON-RPC contract a versioned breaking change or an evolutionary extension?

Why it matters: The VS Code extension consumes the `gert serve` JSON-RPC 2.0 API over stdio. If the runtime changes the method names, parameter shapes, or notification payloads, the extension must be updated simultaneously. A versioned negotiation protocol (client declares supported API version at handshake) allows the extension to evolve independently of the binary.

Options: (a) Hard break: define a new JSON-RPC method set; the VS Code extension is updated as part of the launch. (b) Versioned negotiation: the `initialize` handshake includes a protocol version field; the host advertises capabilities and the client selects a compatible version. (c) No change: the current JSON-RPC surface is preserved unchanged.

Information needed: A complete method inventory of the JSON-RPC API surface (all methods and notification shapes currently used by the VS Code extension), and a determination of which methods require changes to expose new features (e.g., saga events, OTel trace IDs).

Q5. What is the extension supply chain and signing model?

Why it matters: Extensions run in separate processes but are granted capabilities that include tool registration, schema field contribution, and event subscription. A malicious or tampered extension binary could exfiltrate captured secrets, inject commands, or suppress governance events. An extension signing model (verified publisher key) raises the bar for supply chain attacks, but adds operational friction for local development.

Options: (a) No signing required; extensions are trusted if they are present in the declared extension directory. (b) Optional signing: extensions may carry a detached signature; the host warns but does not block unsigned extensions. (c) Mandatory signing for capability classes above a threshold (e.g., extensions that request `governance.policy` capability must be signed). (d) Keyless signing via Sigstore/cosign for developer convenience.

Information needed: A threat model for the extension trust boundary (see Q4 above), and a survey of whether the target operator community has PKI infrastructure that could issue extension signing keys.

Q6. Is a gRPC transport for the VS Code extension viable for v0, or should it be deferred?

Why it matters: The current JSON-RPC 2.0 over stdio transport works but limits the VS Code extension to a single active connection and requires the extension to manage the gert process lifecycle. A gRPC transport would enable streaming (server-push events without polling), multiple concurrent clients, and connection multiplexing. However, it adds protocol complexity and requires a port allocation model.

Options: (a) Retain JSON-RPC stdio as the sole transport; gRPC deferred to a future version. (b) Add gRPC as an opt-in alternative transport alongside JSON-RPC stdio.

(c) Replace JSON-RPC stdio with gRPC as the primary VS Code transport; provide a backward-compat shim for the transition period.

Information needed: Benchmarking of event delivery latency over JSON-RPC stdio versus gRPC under realistic VS Code workloads (step execution at 100ms/step, TUI refresh at 60fps equivalent). Also requires a determination of whether VS Code's Node.js extension host supports gRPC client libraries without native module issues.

Q7. What is the parallel execution model for independent steps within a single run?

Why it matters: A purely sequential executor is the simplest model, but many real runbooks contain independent diagnostic steps that could execute in parallel, reducing total wall-clock time. However, parallelism complicates the trace ordering guarantee, the governance enforcer (can two parallel steps both consume the same approval token?), and the TUI rendering model.

Options: (a) No parallelism for now; fan-out deferred to a future version. (b) Explicit parallel blocks via a new `parallel:` step type that joins on completion of all children. (c) Implicit parallelism inferred from the step dependency graph (steps with no data dependency on preceding steps execute concurrently).

Information needed: Whether any runbooks in active use have independent steps that would benefit from parallelism, and what the governance implications are for concurrent step execution (e.g., two CLI steps executing simultaneously under the same allowlist).

Q8. How should the saga compensation registry interact with the trace and approval model?

Why it matters: A compensation registered for a step must itself be subject to governance (it may execute privileged commands), must appear in the trace (compensations are audit events), and may require its own approval gate if the original step had one. The interaction between the compensation executor and the governance layer is non-trivial and must be specified before implementation.

Options: (a) Compensations execute with the same governance context as the original step (same allowlist, same approver roles). (b) Compensations are exempt from approval gates (compensations are emergency rollback actions; blocking them on approval would defeat their purpose). (c) Compensation governance is separately configurable per compensation definition.

Information needed: Industry precedent from Temporal (compensation activities run in the same workflow context) and AWS Step Functions [9] (catch/retry blocks have no separate IAM policy). A decision here feeds into §4 (compensation field design) and §8 (governance scope).

Q9. What is the schema namespace convention for extension-contributed fields, and how are they validated?

Why it matters: The schema design (§4) states that extension fields are namespaced, but the namespace convention and validation mechanism are undefined. If two extensions contribute fields under the same namespace, or if the host does not validate extension-contributed fields against a schema, runbooks using those fields will have unpredictable behavior at runtime.

Options: (a) Extension ID as namespace prefix (`x-myext.fieldName`); validated against a JSON Schema fragment published by the extension at registration time. (b) Dot-notation namespace under a reserved top-level key (`extensions.myext.fieldName`); host validates at parse time. (c) Extension fields are opaque to the host; the extension validates its own fields at execution time.

Information needed: Whether JSON Schema Draft 2020-12 supports dynamic `$ref` resolution at validation time, which would allow the host to assemble a composite schema from the host schema plus all registered extension schema fragments.

Q10. What is the timeout and escalation model for human approval steps?

Why it matters: Approval gates may block indefinitely waiting for a role-holder to approve. In production, a stalled approval can block an incident response indefinitely. SLA enforcement—timeout after N minutes, escalate to the next role, optionally auto-approve or auto-reject—is a standard pattern in ITIL emergency change processes and is implemented by PagerDuty [23] and AWS Step Functions [9] human approval tasks. Without it, gert is not usable for SLA-bound incident response (*Dennis Research Brief §5.2*).

Options: (a) No timeout; approval escalation deferred. (b) Optional `timeout` field on approval steps; on expiry, run is suspended and operator is notified. (c) Full escalation chain: `timeout` triggers escalation to a secondary role list; if that also times out, the step enters a configurable terminal state (auto-approve, auto-reject, or suspend-for-manual-review).

Information needed: Whether the runtime concurrency model (Q2 above) supports timer-based step suspension without blocking the process, and what notification mechanism is available to alert escalation targets (stdout, webhook, or an extension-contributed notifier).

Chapter 11

Governance and Policy

Governance is Gert’s core differentiator: allowlists, denylists, environment variable blocking, output redaction, and approval gates. This section defines the governance model, specifying which policy fields are first-class, how policy is evaluated at runtime, whether policy is host-enforced or extension-contributed, and how approval gates integrate with the event model.

Governance features are a first-class design requirement and must not be treated as optional or defer-to-extension.

11.1 Governance Primitives

The following primitives form gert’s policy vocabulary. Each is defined precisely below, with its evaluation point in the execution pipeline, its schema representation, and its trace event.

11.1.1 Command Allowlist (`allowed_commands`)

Definition. A whitelist of executable names (`argv[0]`, without path components) that may be invoked by `cli` steps and tool subprocess invocations. If the allowlist is non-empty, any command whose base name is not in the list is blocked before the subprocess is forked.

Semantics. Path-stripping is applied before matching: `/usr/bin/curl` matches the allowlist entry `curl`. Glob patterns are *not* supported in the (exact match only). The empty list (`[]`) means all commands are permitted (permissive default).

Schema.

```
meta:
  governance:
    allowed_commands: [kubect1, curl, jq, aws]
```

Evaluation point. Runtime Core, immediately before subprocess fork (after template evaluation). See §11.2.

Trace event. `governance/commandBlocked` — written when a command is denied. Fields: `stepId`, `command`, `reason`: `"not_in_allowlist"`.

11.1.2 Command Denylist (`denied_commands`)

Definition. An explicit list of executable names that are unconditionally blocked. The denylist takes precedence over the allowlist: if a command appears in both, it is denied.

Use case. Operators use the denylist to forbid dangerous commands (`rm`, `dd`, `shutdown`) without needing to enumerate all permitted commands.

Schema.

```
meta:
  governance:
    denied_commands: [rm, dd, shutdown, reboot]
```

Evaluation point. Runtime Core, same checkpoint as allowlist. Denylist check runs first.

Trace event. `governance/commandBlocked` — fields: `stepId`, `command`, `reason`: `"in_denylist"`.

11.1.3 Environment Variable Blocking (`deny_env_vars`)

Definition. A list of glob patterns matched against environment variable names. Any environment variable whose name matches a pattern is stripped from the subprocess environment before the command is executed.

Rationale. Prevents accidental or malicious exfiltration of secrets (e.g., `AWS_SECRET_ACCESS_KEY`, `GITHUB_TOKEN`) by commands in the runbook.

Schema.

```
meta:
  governance:
    deny_env_vars:
      - "*_SECRET*"
      - "*_TOKEN"
      - "AWS_*"
      - "GITHUB_*
```

Evaluation point. Runtime Core, immediately before subprocess fork. Applied to the full process environment (not just explicitly passed vars).

Trace event. `governance/envVarBlocked` — fields: `stepId`, `varNames`: `[string]` (names of blocked variables, not values).

11.1.4 Output Redaction (redact)

Definition. A list of redaction rules. Each rule has a `pattern` (a RE2 regular expression) and a `replace` string. After a `cli` or `tool` step produces output, all redaction rules are applied to the captured stdout/stderr before the output is stored in run variables or written to the trace.

Semantics. Rules are applied in declaration order. The replace string may use RE2 backreferences (`$1`). The special replace value `[REDACTED]` is the conventional replacement for secrets.

Schema.

```
meta:
  governance:
    redact:
      - pattern: "token=[A-Za-z0-9_\\-]{20,}"
        replace: "token=[REDACTED]"
      - pattern: "password: .+"
        replace: "password: [REDACTED]"
      - pattern: "(?i)bearer\\s+[A-Za-z0-9._\\-]+"
        replace: "Bearer [REDACTED]"
```

Evaluation point. Runtime Core, immediately after subprocess/tool output is captured, before capture variables are set and before any trace write.

Trace event. `governance/redactionApplied` — fields: `stepId`, `ruleCount` (number of rules that matched, not the patterns themselves, to avoid logging the pattern that matched sensitive data).

11.1.5 Approval Gates

Approval gates are first-class step-level governance primitives. They are defined on any step type and pause execution until a minimum number of approvals from specified roles have been recorded. See §11.3 for the complete approval gate design.

11.2 Policy Evaluation Model

Policy primitives are evaluated at specific, well-defined points in the execution pipeline. The evaluation order within each checkpoint is fixed and must not be reordered by extensions.

Checkpoint	Primitives Evaluated	Phase
Parser validation	Schema structural rules	Parse
Planner validation	Import cycles, tool existence	Plan
Run pre-flight	Extension policy rules	Run init
Step pre-flight	Denylist check	Runtime Core
Step pre-flight	Allowlist check	Runtime Core
Step pre-flight	Approval gate	Runtime Core
Subprocess fork	Env var blocking	Runtime Core
Output captured	Redaction	Runtime Core
Post-step	Contract risk warning	Runtime Core

Order of precedence. Within the pre-flight checkpoint for a `cli` or `tool` step, the evaluation order is:

1. Denylist check (hard block, step fails immediately if matched).
2. Allowlist check (hard block, step fails immediately if not matched and list is non-empty).
3. Approval gate (soft pause; step is held until approvals arrive).
4. Env var blocking (applied at fork time, not a failure condition — silently strips).

A policy violation at any hard-block checkpoint writes the corresponding trace event and returns an error from `RunHandle.Next()`. The run is placed in the `policy_violation` terminal state. The operator must resolve the policy (e.g., add the command to the allowlist) and re-run; resumption from a policy violation is not supported.

11.3 Approval Gate Design

11.3.1 Schema Declaration

Approval gates are declared in the step's `approvals` block:

```

steps:
- id: delete_database
  kind: cli
  command: "kubectl delete namespace {{ .targetNamespace }}"
  approvals:
    min: 2
    roles: ["approver", "reviewer"]
    timeout: 24h
    on_timeout: fail    # or: escalate
    escalate_to: ["senior-sre", "director-of-eng"]
    message: "Deleting {{ .targetNamespace }} is irreversible. Confirm change
    ↪ window."

```

11.3.2 UX Contract

When the Runtime Core reaches a step with an `approvals` block, it:

1. Writes `governance/approvalRequired` trace event (stepId, message, roles, min, timeout).
2. Emits an `ApprovalRequired` runtime event on the event channel.
3. Returns from `Next()` with a `StepResult` of kind `PendingApproval`. The step does *not* execute yet.
4. Subsequent calls to `Next()` on the same step return `ErrAwaitingApproval` until the approval quorum is met.

The adapter (VS Code extension, TUI, or `gert serve` client) is responsible for presenting the approval request to authorized users and collecting decisions.

11.3.3 Approval Decision Protocol

An approver submits a decision by calling `RunHandle.Approve(ctx, ApprovalDecision)`:

```

type ApprovalDecision struct {
  StepID  string    // step to approve (must match current paused step)
  Actor   string    // identity of the approver (e.g., "alice@corp.com")
  Role    string    // role the approver is acting in (must be in
  ↪      step.approvals.roles)

```

```

    Approved bool      // true = approved, false = rejected
    Comment  string    // optional justification
    At       time.Time // timestamp of decision
}

```

The runtime validates:

- The Actor identity is established (either from `-as` flag or, in a multi-user context, from the adapter's authenticated session).
- The Role is in the step's declared `approvals.roles`.
- The same actor may not approve the same gate twice.
- If `Approved = false`, the gate is immediately rejected (any single rejection vetos the gate), and the run enters the `rejected` terminal state.

When the approval count reaches `min`, the runtime writes `governance/approvalGranted` to the trace and the next call to `Next()` proceeds with normal step execution.

11.3.4 Approval Recording in the Trace

Every approval decision is written to the trace as an immutable record:

```

// governance/approvalRequired
{
    "sequence": 14,
    "event": "governance/approvalRequired",
    "stepId": "delete_database",
    "message": "Deleting prod-ns is irreversible. Confirm change window.",
    "roles": ["approver", "reviewer"],
    "min": 2,
    "timeout": "24h",
    "at": "2026-04-18T10:00:00Z"
}

// governance/approvalDecision (written once per decision)
{
    "sequence": 15,

```

```
"event": "governance/approvalDecision",
"stepId": "delete_database",
"actor": "alice@corp.com",
"role": "approver",
"approved": true,
"comment": "Change window confirmed with customer.",
"at": "2026-04-18T10:05:00Z"
}

// governance/approvalGranted (written when quorum is met)
{
  "sequence": 16,
  "event": "governance/approvalGranted",
  "stepId": "delete_database",
  "approvals": ["alice@corp.com", "bob@corp.com"],
  "at": "2026-04-18T10:07:00Z"
}
```

The trace provides a complete, tamper-evident audit record of who approved what and when.

11.3.5 Timeout and Escalation

If `timeout` is set and the approval quorum is not met within the deadline:

- `on_timeout: fail`: The run enters the `timed_out` terminal state. `governance/approvalTimedOut` is written to the trace.
- `on_timeout: escalate`: The runtime emits an `ApprovalEscalated` event. If `escalate_to` roles are declared, a new `ApprovalRequired` event is emitted targeting those roles. The timeout clock restarts. Escalation is attempted once; if the escalated request also times out, the run fails.

Timeout enforcement is implemented via a `context.WithDeadline` passed to the approval wait loop. The adapter (or the `gert serve` session timer) is responsible for surfacing the timeout to the approver.

11.4 Policy-as-Code Direction

11.4.1 Structured YAML Policy Blocks

Gert uses **structured YAML policy blocks** embedded in the runbook's `meta.governance` section. The block is extended with the richer primitives defined in §11.3.

Rationale:

- The policy vocabulary is small and closed (six primitives at present). A full policy DSL would be over-engineered for this set.
- YAML policy blocks are self-contained: the runbook declares its own governance constraints without requiring external policy files or a running policy server.
- Structured blocks are straightforward to validate with JSON Schema at authoring time, giving operators early feedback in the VS Code extension.
- The governance model (allowlists, redaction, env blocking) translates directly to structured blocks with no loss of expressiveness.

11.4.2 Future: OPA Integration Hooks

Dennis's research confirms that Open Policy Agent (OPA) is the CNCF standard for policy-as-code in operational tooling. For , gert will introduce OPA integration at the following evaluation hooks:

```
// OPA evaluation hook interface (target)
type PolicyEngine interface {
    // EvaluatePreStep is called before each step pre-flight.
    // The policy engine receives the full step context and returns
    // allow/deny with a reason.
    EvaluatePreStep(ctx context.Context, input PolicyInput) (PolicyDecision,
        ↪ error)

    // EvaluatePreRun is called once after plan construction.
    // Used for runbook-level access control (RBAC).
    EvaluatePreRun(ctx context.Context, input PolicyInput) (PolicyDecision,
        ↪ error)
}
```

```
type PolicyInput struct {
    Runbook    RunbookMeta
    Step       ResolvedStep
    Actor      string
    Vars       map[string]string
    Contract   *contract.Contract
}

type PolicyDecision struct {
    Allow    bool
    Reason   string
    // For approval gates: required roles computed by policy engine
    RequiredApprovers []string
}
```

The PolicyEngine interface will have two implementations in a future release:

1. **Built-in:** evaluates the structured YAML policy blocks (backward-compatible with the runbooks).
2. **OPA:** calls an OPA bundle loaded from a project policy directory (`.gert/policy/*.rego`).

The two implementations compose: the built-in engine runs first; if it allows, the OPA engine runs second. A denial from either blocks the step.

Recommendation: Do not block the the launch on OPA integration. Ship the structured YAML model first, then introduce OPA as an opt-in engine in a future release. This allows the governance interface to be designed correctly without the complexity of Rego authoring in the initial release.

11.5 RBAC Model

11.5.1 Identity Establishment

Actor identity is established via the `-as <identity>` flag passed at invocation time. The identity string is an opaque label (conventionally an email address or a role name) recorded in the trace and used for approval gate matching.

the limitation: Identity is not authenticated. The `-as` flag is self-asserted; gert does not verify that the caller is who they claim to be. Enforcement of identity authenticity is the responsibility of the surrounding infrastructure (CI/CD pipeline, VS Code extension SSO, MCP server authentication).

target: An `IdentityProvider` interface will allow adapters to supply verified identity tokens (e.g., OIDC tokens from VS Code GitHub Copilot authentication), replacing self-asserted identity for governed environments.

11.5.2 Runbook-Level Access Control

Runbooks may declare a `requires_role` field to restrict who may execute them:

```
meta:
  governance:
    requires_role: "responder"
    # or: requires_any_role: ["approver", "responder"]
    # or: requires_all_roles: ["approver", "reviewer"]
```

The Runtime Core evaluates this constraint during run initialization (pre-flight, before any step executes). If the actor's identity does not match, `governance/accessDenied` is written to the trace and the run fails immediately.

11.5.3 Step-Level Role Restriction

Individual steps may declare `requires_role` to restrict who may trigger that step. This is distinct from approval gates (which require external approval) — step role restriction *prevents* a step from executing unless the actor themselves holds the required role.

```
steps:
- id: promote_to_production
  kind: cli
  requires_role: "release-manager"
  command: "kubectl set image deployment/app app={{ .imageTag }}"
```

11.5.4 Separation of Duties

The approval gate model supports separation of duties: the `same-actor-cannot-approve` invariant (defined in §11.3) ensures that a step author cannot self-approve their own gate.

In environments requiring formal approval processes, the actor who initiates the run and the actor(s) who approve high-risk steps must be different.

11.6 Redaction

11.6.1 Pattern-Based Redaction

The primary redaction mechanism is pattern-based: operators declare RE2 regular expressions in `meta.governance.redact`. Redaction is applied eagerly: captured output is redacted *before* any capture variable assignment or trace write.

11.6.2 Explicit Field Marking ()

For structured tool outputs (JSON objects returned by MCP tools), will introduce explicit field marking: a tool definition may declare that a returned field contains sensitive data. The runtime will redact the field value without requiring a regex pattern.

```
# In .tool.yaml (target)
outputs:
  - name: api_key
    sensitive: true    # value redacted from trace and captures
```

11.6.3 Redaction Guarantees

- Redaction is applied to all output paths: captured variables, trace events, and event channel payloads.
- Redaction rules are compiled (regex pre-compiled) during run initialization, not re-compiled per step.
- Redaction is applied unconditionally in all run modes (real, dry-run, replay).
- The original (pre-redaction) value is *never* written to disk in any path.

11.7 Audit Trail Requirements

The append-only JSONL trace file is gert's primary audit artifact. Every governance event is a first-class trace entry. The following governance events are guaranteed to be written:

Trace Event	Written When
<code>governance/commandBlocked</code>	A command is blocked by denylist or allowlist
<code>governance/envVarBlocked</code>	Env vars are stripped before subprocess fork
<code>governance/redactionApplied</code>	Redaction rules matched captured output
<code>governance/approvalRequired</code>	A step's approval gate is entered
<code>governance/approvalDecision</code>	An approver submits approve or reject
<code>governance/approvalGranted</code>	Approval quorum is met
<code>governance/approvalRejected</code>	An approver rejects the gate
<code>governance/approvalTimedOut</code>	Approval timeout expires
<code>governance/accessDenied</code>	RBAC check fails (run or step level)
<code>governance/contractWarning</code>	Step contract risk level is critical
<code>governance/policyViolation</code>	Generic policy violation (OPA engine,)

11.7.1 Trace Integrity

The trace file is protected by:

- **Append-only write discipline:** the trace writer never seeks backward or truncates. Each write is followed by `fsync`.
- **SHA256 evidence hash:** at run completion, the SHA256 of the entire trace file is written as the final record and also stored in a sidecar `trace.sha256` file.
- **Monotonic sequence numbers:** every trace event has a `sequence` field (incrementing integer). Gaps in sequence numbers signal corruption or tampering.

11.7.2 Compliance Alignment

These audit trail guarantees are designed to satisfy the requirements of:

- **SOC 2 Type II** — Approval and authorization controls: every command execution and approval is logged with actor identity and timestamp.
- **HIPAA** — Access controls and audit logging: env var blocking and redaction prevent PHI from appearing in traces; approval gates enforce separation of duties.
- **ITIL Process Management** — Normal/standard/emergency workflow paths can be modeled via `requires_role` and multi-approver gates.

11.8 Extension-Contributed Policy

Extensions may contribute governance rules through the `ContributedPolicyRules()` method of the Extension Host (see §02). These rules compose with host policy under the following invariants:

1. **Additive only.** Extension rules may add restrictions; they may never remove or override host-defined policy. An extension cannot remove a command from the denylist, relax a required role, or suppress a redaction rule.
2. **Host policy takes precedence on conflict.** If an extension rule and a host rule address the same primitive and disagree, the more restrictive interpretation applies.
3. **Extension rules are visible in the trace.** A `governance/extensionPolicyApplied` trace event is written when an extension-contributed rule is evaluated, naming the extension and rule ID.
4. **Extension rules are not trusted with approval authority.** Extensions may not grant approvals on behalf of human approvers. Approval decisions must originate from a human actor via the `RunHandle.Approve()` API.

11.8.1 Extension Policy Rule Schema

```
// PolicyRule is contributed by an extension and evaluated at pre-step
↪ checkpoint.
type PolicyRule struct {
    ID          string          // unique within the extension's namespace
    Description string
    // Evaluate returns Deny if the rule blocks this step, Allow otherwise.
    // Extensions communicate rules as static declarations (); dynamic
    // rule evaluation via the extension RPC transport is .
    StepKinds   []StepKind // step types this rule applies to (empty = all)
    CommandGlob string     // if set, only applies to commands matching this
    ↪ glob
    Action      PolicyAction // Deny or Warn
    Reason      string       // human-readable reason written to trace event
}
```

Extension-contributed policy is loaded during the Extension Host's `Load()` phase, before the first `Next()` call. Policy rules are static for the lifetime of a run; extensions may not add or remove rules after the run has started.

Chapter 12

Evidence, Tracing, and Resumption

Gert captures a complete, durable record of every execution for audit, compliance, incident response, and operational replay. This section specifies the append-only JSONL trace format, evidence capture model, per-step snapshot contract, run resumption semantics, replay mode, and OpenTelemetry integration. Together these mechanisms ensure that every run produces a self-contained artifact that answers three questions: *what happened*, *what evidence was collected*, and *how can execution be recovered or replicated*.

12.1 Trace File Format

12.1.1 File Location and Naming

Each run produces a single trace file located at:

```
.runbook/runs/<run-id>/trace.jsonl
```

The <run-id> is a URL-safe string assigned at run creation time (e.g. 20260418T143022-a3f9c1). All run artefacts are co-located under the run directory:

```
.runbook/runs/<run-id>/
  trace.jsonl           # append-only JSONL event log
  snapshots/           # per-step state checkpoints
    step-0000.json
    step-0001.json
    ...
  attachments/         # binary evidence referenced from trace events
    <sha256>.bin
```



```
...
run.yaml          # completion manifest (written on terminal event)
annotations.jsonl # operator notes (append-only)
lock              # PID lock (removed on clean exit)
```

12.1.2 JSONL Envelope

Every line in `trace.jsonl` is a self-contained JSON object with a common envelope. Consumers MUST tolerate unknown fields to allow additive schema evolution.

```
{
  "event_id": <string>,    // UUID v4 for correlation
  "sequence": <int64>,     // monotonically increasing within this run
  "kind": <string>,        // event kind (see §12.1.3)
  "timestamp": <RFC3339>,  // UTC timestamp with sub-second precision
  "run_id": <string>,      // run identifier
  "path": <array>,         // tree-path to the active node (may be null)
  "payload": <object>     // kind-specific payload (see below)
}
```

The `sequence` field provides a total order over events in a single run. It is assigned by the `RunStore.WriteTrace` implementation and is not trusted for cross-run ordering. The `event_id` field is a UUID v4 assigned at event emission time for correlation with external systems.

12.1.3 Event Type Catalogue

The following event kinds are normative in the schema. All user-supplied string fields (command arguments, outputs, prompt responses) pass through the redaction pipeline before being written.

`run/started`

Emitted once, as the first event in every trace file.

```
"payload": {
  "runbook_id": <string>, // runbook name/path
```

```

"user":      <string>, // actor identity
"mode":      <string>, // "real" | "replay" | "dry-run"
"inputs":    <object>, // resolved input values (redacted)
"client":    <string>, // "cli" | "server" | "mobile-ios"
               // | "mobile-android"
"parent_run_id": <string> // omitted if not an invoke child
}

```

step/started

Emitted immediately before executing a step.

```

"payload": {
  "step_id": <string>, // stable step identifier
  "step_type": <string>, // "cli" | "manual" | "tool" | "invoke"
  "step_name": <string> // human-readable name from runbook YAML
}

```

step/completed

Emitted after a step exits cleanly.

```

"payload": {
  "step_id": <string>,
  "exit_status": <int>, // exit code for cli steps; 0 for others
  "duration_ms": <int64>, // wall-clock duration in milliseconds
  "captures": <object>, // captured output variables (redacted)
  "evidence": [ // zero or more evidence items
    {
      "name": <string>,
      "kind": "text" | "checklist" | "attachment",
      "value": <string>, // for kind=text
      "items": <object>, // for kind=checklist
      "path": <string>, // relative path for kind=attachment
      "sha256": <string>, // hex SHA256 for kind=attachment
      "size": <int64> // bytes for kind=attachment
    }
  ]
}

```

step/failed

Emitted when a step exits with a non-zero status or returns a runtime error.

```
"data": {
  "step_id": <string>,
  "error_type": <string>, // "exit_code" / "timeout" / "governance"
                        // / "tool_error" / "invoke_failed"
  "error": <string>, // human-readable message (redacted)
  "exit_status": <int>, // for error_type=exit_code
  "duration_ms": <int64>
}
```

step/skipped

Emitted when a step is not executed due to branch evaluation or iterate exhaustion.

```
"data": {
  "step_id": <string>,
  "reason": "branch_not_taken" || "iterate_exhausted" || "dry_run"
}
```

step/retrying

Emitted before each retry attempt (requires the retry policy feature).

```
"data": {
  "step_id": <string>,
  "attempt": <int>, // attempt number (1 = first retry)
  "max_attempts": <int>,
  "backoff_ms": <int64>
}
```

tool/invoked

Emitted when the tool runtime dispatches a tool call.

```
"data": {
  "tool_name": <string>,
}
```

```

"transport":      "stdio" | "jsonrpc" | "mcp",
"correlation_id": <string>, // UUIDv4, links to tool/responded
"inputs":         <object>  // sanitized inputs
}

```

tool/responded

Emitted when the tool call returns.

```

"data": {
  "correlation_id": <string>,
  "exit_code":      <int>,
  "duration_ms":    <int64>,
  "truncated":      <bool> // true if output was capped
}

```

manual/prompted

Emitted when a manual step displays its prompt to the operator.

```

"data": {
  "step_id":      <string>,
  "prompt_text":  <string>, // the prompt shown (redacted)
  "sla_seconds":  <int>     // 0 if no SLA configured
}

```

manual/completed

Emitted when the operator submits a response to a manual step.

```

"data": {
  "step_id":      <string>,
  "response":     <string>, // operator's text response (redacted)
  "attestation":  <string>, // "operator_confirmed" | "skipped"
  "approver":     <string>, // identity of responding actor
  "duration_ms":  <int64>
}

```

governance/blocked

Emitted when a governance rule blocks execution.

```
"data": {
  "step_id":      <string>,
  "rule":         <string>, // allowlist / denylist / env_block
                  // / output_redact / approval_gate
  "action_attempted": <string>, // description of blocked action
  "policy":       <string> // policy name or path
}
```

governance/approved

Emitted when an approval gate is cleared.

```
"data": {
  "step_id":      <string>,
  "approver":     <string>, // identity of approver
  "approval_method": <string>, // "interactive" / "mcp" / "api"
  "duration_ms":  <int64>
}
```

run/completed

Terminal event for a successful run.

```
"data": {
  "final_status": "passed" | "failed" | "skipped",
  "duration_ms":  <int64>,
  "step_count":   <int>,
  "passed":       <int>,
  "failed":       <int>,
  "skipped":      <int>
}
```

run/failed

Terminal event for a run that halted due to an unrecoverable error.

```
"data": {
  "failing_step_id": <string>,
  "error": <string>,
  "duration_ms": <int64>
}
```

checkpoint

Internal event written by `RunStore.SaveCheckpoint` to link a snapshot file to the trace sequence. Not user-facing but required for resumption.

```
"data": {
  "snapshot_file": <string> // e.g. "step-0003.json"
}
```

12.1.4 Crash Safety

The trace file is opened with `O_APPEND` and each event is written as a single `write(2)` call followed by `fsync`. On POSIX systems, a write of fewer than `PIPE_BUF` bytes (4096 bytes on Linux) to an `O_APPEND` file is atomic. Because the JSON encoding of a single trace event is always well under 4096 bytes for typical runbooks, partial-line corruption on crash is avoided in practice. For large payloads (e.g. tool output), the runtime truncates at a configurable limit (default 4 KB per field) before encoding.

The Go implementation:

```
// WriteTrace appends one event atomically and fsyncs.
func (d *DirRunStore) WriteTrace(event DurableEvent) error {
    p := filepath.Join(d.baseDir, "trace.jsonl")
    f, err := os.OpenFile(p, os.O_CREATE|os.O_WRONLY|os.O_APPEND, 0644)
    if err != nil {
        return fmt.Errorf("open trace: %w", err)
    }
    defer f.Close()

    data, err := json.Marshal(event)
    if err != nil {
        return fmt.Errorf("marshal event: %w", err)
    }
}
```

```
data = append(data, '\n') // JSONL delimiter
if _, err := f.Write(data); err != nil {
    return fmt.Errorf("write trace: %w", err)
}
return f.Sync() // durability guarantee
}
```

On recovery, any line that fails `json.Unmarshal` is silently skipped. The last clean `checkpoint` event determines the recoverable state.

12.1.5 Tamper Evidence

The trace format supports optional HMAC-SHA256 signing of each event. When a signing key is configured (via `GERT_TRACE_KEY` or the workspace config), the runtime appends a `"sig"` field to the envelope containing the HMAC-SHA256 of the canonical JSON of all other fields. Verifiers can replay the trace and recompute signatures. This does not prevent deletion of events (append-only at the filesystem level provides that property) but detects modification of existing events.

12.2 Evidence Capture

12.2.1 What Constitutes Evidence

In the schema, “evidence” is any artefact that attests to a step’s outcome:

- **Command output** — stdout/stderr of a `cli` step, captured and embedded in `step/completed`.
- **Manual attestations** — operator-provided text, checklist responses, and explicit confirmations from `manual` steps.
- **Tool responses** — structured JSON returned by a `tool` step, captured in `step/completed.captures`.
- **File attachments** — binary or text files explicitly attached by an operator or produced by a step, stored in the `attachments/` directory and referenced by SHA256 digest.

12.2.2 SHA256 Hashing

All file attachments are content-addressed by their SHA256 digest. The `evidence` package exposes:

```
// HashFile returns hex SHA256 and file size.
func HashFile(path string) (sha256hex string, size int64, err error)

// NewAttachmentEvidence creates an EvidenceValue with hash and size.
func NewAttachmentEvidence(path string) (*EvidenceValue, error)
```

The hash is stored in the `step/completed` event alongside the relative path under `attachments/`. Consumers can verify integrity at any time by re-hashing the referenced file.

Text evidence fields (command output, operator responses) are not individually hashed — the overall trace event integrity is provided by the optional HMAC-SHA256 event signature described in § 12.1.5.

12.2.3 File Attachments

When a step produces a binary artefact, the runtime:

1. Copies the file to `.runbook/runs/<run-id>/attachments/<sha256>.<ext>`.
2. Emits an `EvidenceValue` of kind `attachment` in the `step/completed` event with fields `path`, `sha256`, and `size`.

Attachments are deduplicated by content: if two steps produce identical file content, they share one attachment file. The trace events may reference the same `sha256` value independently.

12.2.4 Evidence Retention

Evidence is retained for the lifetime of the run directory. The workspace-level retention policy (configurable in `gert.yaml`) specifies:

- `retention.runs` — number of completed runs to retain (default: 50).
- `retention.max_age_days` — maximum age in days (default: 90).

- `retention.attachments` — whether to include attachments in retention (default: `true`).

The `gert gc` command enforces retention policy by removing the oldest run directories, deleting their `attachments/` content first.

12.3 Run State Snapshots

12.3.1 Checkpoint Model

A checkpoint is a complete serialisation of `RunState` written to `snapshots/step-NNNN.json` after every successfully completed step. The snapshot captures the minimum state needed to resume execution from the next step:

```
type RunState struct {
    RunID           string
    RunbookPath     string
    Mode            string           // "real" | "replay" | "dry-run"
    StartedAt       time.Time
    Actor           string
    CurrentStepIndex int           // flat-mode index (legacy compat)
    Path            TreePath      // tree-mode cursor (authoritative)
    Vars            map[string]string // resolved input bindings
    Captures        map[string]string // accumulated step outputs
    History          []*StepResult    // completed step records
    InvokeStack     []InvokeFrame    // nested invoke context stack
    IterateState     map[string]*IterateProgress
    ParallelSlots    map[string][]SlotState
    LastCheckpointSeq int64           // trace seq at checkpoint time
}
```

12.3.2 Atomic Snapshot Write

Snapshots are written atomically: the runtime marshals the state to a `.tmp` file, fsyncs it, then renames it to the final path. An incomplete snapshot (e.g. from a crash mid-write) retains the `.tmp` suffix and is skipped during recovery.

```
func (d *DirRunStore) SaveCheckpoint(state *RunState) (string, error) {
```

```

filename := fmt.Sprintf("step-%04d.json", len(state.History))
finalPath := filepath.Join(d.baseDir, "snapshots", filename)
tmpPath   := finalPath + ".tmp"

data, _ := json.MarshalIndent(state, "", " ")
os.WriteFile(tmpPath, data, 0644)           // write to temp
tf, _ := os.Open(tmpPath); tf.Sync(); tf.Close() // fsync temp
os.Rename(tmpPath, finalPath)               // atomic rename
// ... then write checkpoint trace event
}

```

12.3.3 Minimum Resumption State

The state required to resume a run from step N is exactly the snapshot at `step-NNNN.json`. It contains:

- All resolved input values (**Vars**).
- Outputs captured by completed steps (**Captures**).
- The tree cursor (**Path**) identifying which node executes next.
- Branch decisions already taken (encoded in **History** step statuses).
- Iterate progress counters and parallel slot states.
- The invoke stack for nested runbooks.

The runbook YAML itself is not stored in the snapshot; it is re-read from `RunbookPath` at resume time. Operators must not modify the runbook between the original run and resumption.

12.4 Run Resumption

12.4.1 The `gert exec -resume` Contract

```
gert exec --resume <run-id>
```

The runtime executes the following recovery sequence:

1. Locate `.runbook/runs/<run-id>/`.
2. Attempt to acquire the `lock` file. Fail if a live process holds it.
3. Call `RunStore.LoadLatestCheckpoint()` — scan `snapshots/` descending, skip `.tmp` files, return the newest valid JSON.
4. Re-open `trace.jsonl` for append.
5. Advance `CurrentStepIndex` by one (or advance the tree `Path` cursor past the last completed node).
6. Re-build the governance engine from the runbook's policy block.
7. Resume the execution loop from the restored `Path`.

The resumption does *not* re-execute any step recorded in `History`. It continues from the step *after* the last checkpoint.

12.4.2 Resumability by Step Type

- **cli steps** — safe to resume. The step is re-executed from scratch. Runbook authors should ensure cli steps are idempotent (e.g. `kubectl apply` over `kubectl create`).
- **manual steps** — the operator is re-prompted. The prior response (if any) is lost. The prompt is re-displayed and the operator must re-attest.
- **tool steps** — resumable if idempotent. If the tool call was in-flight when the process crashed (i.e. a `tool/invoked` event exists with no matching `tool/responded`), the runtime re-issues the call. Authors must declare `idempotent: true` on tools where safe; others are re-executed with a warning log.
- **invoke steps** — if the child run completed (its terminal event is in the parent trace), the invoke is treated as completed. If the child run was in progress, the child is itself resumed via the nested run ID stored in the `InvokeStack`.
- **branch/iterate** — structural nodes; not executed directly. Cursor advancement handles re-entry correctly.

12.4.3 Idempotency Guidance

Runbook authors should write resumable runbooks by following these guidelines:

- Use `kubectl apply` and `terraform apply` instead of imperative create commands.
- Use `-if-not-exists` or `-create-or-replace` flags where available.
- For tool steps, declare `idempotent: true` in the tool definition when the operation is safe to call multiple times.
- For manual steps, write prompts as questions that remain valid on a second ask (“Has the database migration completed?” rather than “Confirm you have started the migration”).
- Avoid steps whose side effects depend on the current time or a random seed without an explicit override mechanism.

12.4.4 Partial Step Resumption

If a `tool/invoked` event has no matching `tool/responded` event, the tool call was interrupted mid-flight. On resumption:

1. The runtime logs a warning identifying the orphaned correlation ID.
2. If `idempotent: true`: the tool call is re-issued from the beginning.
3. If `idempotent: false` (default): the step is treated as **failed** and execution follows the normal failure path (error branch, if defined, or halt).

12.5 Replay Mode

12.5.1 Overview

```
gert exec --mode replay --scenario <scenario.yaml>
```

Replay mode executes a runbook against a set of pre-recorded command responses and evidence values. No real commands are executed and no real tools are invoked. The mode is used for:

- **Testing** — validating that a runbook’s branching logic, captures, and governance rules behave correctly for a given set of inputs and command outputs (see § 13 Testing and Acceptance).
- **Demonstration** — showing a runbook’s execution flow without real infrastructure.
- **Regression** — verifying that a refactored runbook produces the same trace as the original.

12.5.2 Scenario File Format

A scenario file is a YAML document that maps commands to recorded responses and step IDs to recorded evidence:

```

commands:
- argv: ["kubectl", "get", "pods", "-n", "prod"]
  stdout: "NAME          READY   STATUS    RESTARTS\npod-abc    1/1     Running
↪  0\n"
  stderr: ""
  exit_code: 0

evidence:
check-pod-health:      # step_id
  observation:         # evidence name
    kind: text
    value: "pod is healthy, proceeding"

```

Commands are matched in order of declaration (first match wins). Unmatched commands fail the scenario unless `allow_unmatched: true` is set.

12.5.3 Replay Semantics

In replay mode:

- `cli` steps — the command executor returns the pre-recorded stdout/stderr/exit_code.
- `manual` steps — the evidence collector returns the pre-recorded evidence values instead of prompting the operator.

- **tool** steps — tool invocations are intercepted; if a matching recorded response exists, it is returned. Otherwise the step fails.
- **invoke** steps — the child runbook is executed in replay mode using the same scenario (or a nested scenario file if `scenario_file:` is specified on the invoke step).
- Governance rules — evaluated normally; replay does not bypass governance.
- Trace events — written normally; the replay trace can be compared against a golden trace for regression testing.

12.5.4 Determinism Requirement

For replay to produce the same outcome across runs, the runbook must be deterministic given fixed inputs and command outputs. Non-determinism sources to avoid:

- Timestamps in expressions (use captured values instead of `time.Now()`).
- Random identifiers in step names or captures.
- File system state that is not reproduced by the scenario.
- External API calls in tool steps that are not replayed (tool steps should be declared `idempotent: true` and covered by scenario fixtures).

12.6 OpenTelemetry Integration

12.6.1 Rationale

The JSONL trace format is optimised for durability, human readability, and offline analysis. OpenTelemetry (OTel) spans are optimised for real-time observability, distributed tracing, and integration with platforms such as Jaeger, Zipkin, Datadog, and Grafana Tempo. Both serve complementary purposes and supports both simultaneously.

12.6.2 Span Hierarchy

When OTel is enabled, the runtime creates a span hierarchy that mirrors the runbook execution tree:

```

Span: gert.run           // root span, covers entire run
Span: gert.step          // one per step executed
    Span: gert.tool.invoke // one per tool call within a step
Span: gert.step          // next step
...

```

The Go implementation uses `go.opentelemetry.io/otel/trace`:

```

func (e *Engine) runWithOTel(ctx context.Context) error {
    ctx, runSpan := otel.Tracer("gert").Start(ctx, "gert.run",
        trace.WithAttributes(
            attribute.String("gert.run_id",    e.State.RunID),
            attribute.String("gert.runbook",   e.Runbook.Meta.Name),
            attribute.String("gert.actor",     e.State.Actor),
            attribute.String("gert.mode",      e.State.Mode),
        ),
    )
    defer runSpan.End()
    return e.execute(ctx)
}

func (e *Engine) executeStep(ctx context.Context, step Step) error {
    ctx, stepSpan := otel.Tracer("gert").Start(ctx, "gert.step",
        trace.WithAttributes(
            attribute.String("gert.step_id",    step.ID),
            attribute.String("gert.step_type",  step.Type),
            attribute.String("gert.step_name",  step.Name),
        ),
    )
    defer stepSpan.End()
    // ...
}

```

12.6.3 Span Attributes

The following attributes are set on all spans:

Attribute	Spans	Description
<code>gert.run_id</code>	all	Run identifier
<code>gert.runbook</code>	all	Runbook name
<code>gert.actor</code>	run	Actor identity
<code>gert.mode</code>	run	Execution mode
<code>gert.step_id</code>	step, tool	Step identifier
<code>gert.step_type</code>	step	Step type
<code>gert.tool_name</code>	tool	Tool name
<code>gert.transport</code>	tool	Transport (stdio/jsonrpc/mcp)

Trace context is propagated into tool invocations via the W3C **traceparent** header where the transport supports it (HTTP-based MCP tools). For stdio tools, context is passed as an environment variable `OTEL_TRACEPARENT`.

12.6.4 Relationship to JSONL Trace

OTel spans and the JSONL trace are complementary, not redundant:

- The JSONL trace is the authoritative durable record. It contains evidence hashes, governance events, snapshot references, and redacted output that OTel spans do not.
- OTel spans are ephemeral, push-based, and optimised for latency-sensitive observability dashboards.
- The trace event **"seq"** is embedded as the `gert.trace_seq` span attribute, enabling correlation between spans and trace events.

12.6.5 Configuration

OTel is opt-in and configured via standard OTEL environment variables:

```
OTEL_EXPORTER_OTLP_ENDPOINT=http://localhost:4317
OTEL_SERVICE_NAME=gert
OTEL_TRACES_SAMPLER=always_on
```

When `OTEL_EXPORTER_OTLP_ENDPOINT` is not set, the OTel SDK is initialised with a no-op tracer and zero overhead is incurred. There is no `-otel` flag; configuration is entirely via the environment. This ensures that CI and offline use work without modification.

12.7 Implementation Notes for Go

The key Go interfaces for this subsystem are:

```
// TraceWriter is the interface the engine uses to record events.  
// Implementations must be safe for concurrent use from a single goroutine  
// (the engine's execution loop); multi-goroutine safety is optional.  
type TraceWriter interface {  
    Write(event DurableEvent) error  
    Close() error  
}  
  
// RunStore manages all artefacts for a single run.  
type RunStore interface {  
    RunID() string  
    WriteTrace(event DurableEvent) error  
    ReadTrace() ([]DurableEvent, error)  
    ReadTraceSince(afterSeq int64) ([]DurableEvent, error)  
    SaveCheckpoint(state *RunState) (string, error)  
    LoadLatestCheckpoint() (*RunState, int64, error)  
    AcquireLock() error  
    ReleaseLock() error  
    WriteManifest(manifest *RunManifest) error  
}  
  
// EvidenceCollector is the interface for gathering evidence at manual steps.  
type EvidenceCollector interface {  
    Collect(ctx context.Context, stepID string,  
        prompts []Prompt) (map[string]*EvidenceValue, error)  
}
```

The `DirRunStore` is the production implementation. A `MemRunStore` (storing events in a slice) is provided for unit tests where filesystem I/O is undesirable. The `ReplayEvidenceCollector` returns pre-recorded evidence for scenario tests.

12.8 Run Ingest API

The run ingest API enables mobile and offline executors to upload completed runs to a central gert server for audit, compliance, and operational visibility. The API accepts runs in the standard trace JSONL format and reconstructs them server-side.

12.8.1 Endpoint: POST /api/v1/runs/ingest

Accepts a stream of JSONL trace events (identical format to `trace.jsonl`).

Request format:

POST /api/v1/runs/ingest

Content-Type: application/x-ndjson

Authorization: Bearer <token>

```
{"event_id":"...","sequence":0,"kind":"run/started",...}
{"event_id":"...","sequence":1,"kind":"step/started",...}
{"event_id":"...","sequence":2,"kind":"step/completed",...}
...
```

Response:

```
{
  "run_id": "20260418T143022-a3f9c1",
  "status": "ingested",
  "events_received": 47,
  "attachments_pending": ["a3f9c1...", "b2e4d5..."]
}
```

The `attachments_pending` field lists SHA256 digests of attachments referenced in the trace but not yet uploaded.

12.8.2 Endpoint: POST /api/v1/runs/{run-id}/attachments/{sha256}

Uploads a single attachment file.

Request format:

```
POST /api/v1/runs/20260418T143022-a3f9c1/attachments/a3f9c1...
Content-Type: application/octet-stream
Authorization: Bearer <token>
```

```
<binary file content>
```

Response:

```
{
  "sha256": "a3f9c1...",
  "size": 204800,
  "status": "stored"
}
```

The server verifies the SHA256 checksum of the uploaded content before storing. If the checksum does not match the URL path, the request is rejected with HTTP 400.

12.8.3 Idempotency

Both endpoints are idempotent. Uploading the same trace or attachment multiple times has no adverse effect. The server deduplicates attachments by SHA256 digest.

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Accepts a stream of JSONL trace events (identical format to `trace.jsonl`).

Request format:

POST /api/v1/runs/ingest

Content-Type: application/x-ndjson

Authorization: Bearer <token>

```
{"event_id":"...","sequence":0,"kind":"run/started",...}
{"event_id":"...","sequence":1,"kind":"step/started",...}
{"event_id":"...","sequence":2,"kind":"step/completed",...}
...
```

Response:

```
{
  "run_id": "20260418T143022-a3f9c1",
  "status": "ingested",
  "events_received": 47,
  "attachments_pending": ["a3f9c1...", "b2e4d5..."]
}
```

The `attachments_pending` field lists SHA256 digests of attachments referenced in the trace but not yet uploaded.

12.9.2 Endpoint: POST /api/v1/runs/{run-id}/attachments/{sha256}

Uploads a single attachment file.

Request format:

POST /api/v1/runs/20260418T143022-a3f9c1/attachments/a3f9c1...

Content-Type: application/octet-stream

Authorization: Bearer <token>

<binary file content>

Response:

```
{
  "sha256": "a3f9c1...",
  "size": 204800,
}
```

```
"status": "stored"  
}
```

The server verifies the SHA256 checksum of the uploaded content before storing. If the checksum does not match the URL path, the request is rejected with HTTP 400.

12.9.3 Idempotency

Both endpoints are idempotent. Uploading the same trace or attachment multiple times has no adverse effect. The server deduplicates attachments by SHA256 digest.

Chapter 13

Adapter Contracts

Gert decouples the core runtime from presentation adapters (TUI, Web, VS Code) and integration systems (CI/CD platforms, observability backends, approval workflow systems) through explicitly versioned contracts. An **adapter contract** is a stable, versioned interface that gert commits to maintain across minor and patch releases. Adapters are consumers of these contracts: they observe, render, or orchestrate gert runs, but they own no execution logic.

This chapter defines the four primary adapter surfaces exposed by gert: JSON-RPC execution contract, WebSocket event stream contract, tool invocation contract, and extension handshake contract.

13.1 What is an Adapter Contract?

An **adapter contract** is a versioned API specification that defines:

1. The set of RPC methods, request/response schemas, and error envelopes.
2. The stability guarantee: which fields are required, which are optional, and which may be added in future minor versions.
3. The versioning scheme: how breaking changes are communicated, and how long deprecated versions are supported.
4. The transport layer: HTTP, WebSocket, stdio JSON-RPC, etc.

Adapter contracts are the integration surface of gert. They are designed for external systems to consume. Breaking changes to a contract require a major version bump and a deprecation period.

Example consumers:

- VS Code extension (consumes JSON-RPC execution contract and WebSocket event stream) [20]
- GitLab CI runner (consumes JSON-RPC execution contract to trigger runs)
- Datadog integration (consumes event stream to emit metrics)
- Slack approval bot (consumes event stream, submits approvals via JSON-RPC)

13.2 Contract Categories

Gert exposes four distinct adapter surfaces, each with its own versioning and stability guarantees.

13.2.1 JSON-RPC Execution Contract (exec/v1)

The execution contract is the primary interface for starting, monitoring, and controlling gert runs. It is served over `gert serve -stdio` (stdio JSON-RPC) or `gert serve -http` (HTTP POST to `/rpc`).

Contract version: `exec/v1`

Transport: `stdio` JSON-RPC 2.0 or HTTP POST (JSON-RPC over HTTP)

Methods:

Request schema — `exec/start`:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "method": "exec/start",
  "params": {
    "runbookPath": "./runbooks/incident-triage.yaml",
    "inputs": {
      "incident_id": "INC-12345",
```

Method	Description
<code>exec/start</code>	Start a new runbook execution. Params: <code>runbookPath</code> , <code>inputs</code> , <code>mode</code> , <code>actorId</code> . Returns: <code>runId</code> , <code>status</code> .
<code>exec/next</code>	Advance the run by one step or until it reaches a pause point (approval, manual, error). Params: <code>runId</code> . Returns: <code>stepId</code> , <code>status</code> , <code>output</code> .
<code>exec/cancel</code>	Cancel an in-progress run. Params: <code>runId</code> , <code>reason</code> . Returns: <code>cancelled: bool</code> .
<code>exec/status</code>	Get the current status of a run. Params: <code>runId</code> . Returns: <code>status</code> , <code>currentStepId</code> , <code>progress</code> .
<code>run/list</code>	List all runs in the workspace. Params: <code>filter</code> , <code>limit</code> . Returns: <code>runs: [RunSummary]</code> .
<code>run/get</code>	Get full details of a run. Params: <code>runId</code> . Returns: <code>run: RunDetails</code> .

TABLE 13.1: JSON-RPC execution contract methods (`exec/v1`).

```

    "severity": "high"
  },
  "mode": "real",           // real / dry-run / replay
  "actorId": "ops@acme.example"
}
}

```

Response schema — `exec/start`:

```

{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "runId": "run-abc123",
    "status": "running",
    "createdAt": "2026-04-18T12:00:00Z",
    "traceFile": "./traces/run-abc123.jsonl"
  }
}

```


Error envelope: All errors follow JSON-RPC 2.0 error format:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "error": {
    "code": 1001,
    "message": "run not found",
    "data": {
      "runId": "run-xyz789",
      "category": "client_error",
      "retryable": false
    }
  }
}
```

See Section 13.4 for the canonical error code registry.

Stability guarantee: Within the `exec/v1` contract:

- All required fields in request/response schemas are stable.
- Optional fields may be added in minor versions (clients MUST ignore unknown fields).
- Required fields will not be removed or renamed (breaking change → `exec/v2`).
- Error codes are stable; new codes may be added but existing codes will not change semantics.

13.2.2 WebSocket Event Stream Contract (`events/v1`)

The event stream contract allows adapters to subscribe to real-time events from a running or completed gert execution. It is served over `gert serve -http` at the `/ws` or `/events` endpoint.

Contract version: `events/v1`

Transport: WebSocket (`wss://` or `ws://`)

Connection lifecycle:

1. Client establishes WebSocket connection to `/ws`.
2. Client sends a `subscribe` message with `runId`.
3. Server streams all events for that run in order.
4. Client may disconnect at any time; server closes gracefully.

Subscribe message:

```
{
  "action": "subscribe",
  "runId": "run-abc123",
  "sinceSequence": 0 // Optional: replay from sequence number
}
```

Event envelope: Events are transmitted as JSON objects, one per WebSocket message. The event envelope is identical to the trace event format defined in §12:

```
{
  "eventType": "step/started",
  "timestamp": "2026-04-18T12:00:00Z",
  "runId": "run-abc123",
  "stepId": "lookup_incident",
  "sequence": 42,
  "payload": {
    "stepType": "tool",
    "toolName": "acme.pd-lookup"
  }
}
```

All events carry:

- `eventType`: canonical event type (e.g., `step/started`, `governance/approval_required`).
- `timestamp`: RFC 3339 timestamp.
- `runId`: globally unique run identifier.

- **sequence**: monotonically increasing event sequence number (starts at 0).
- **payload**: event-specific data.

Reconnection protocol: If the WebSocket connection is lost, the client may reconnect and provide **sinceSequence** to resume from the last received event:

```
{  
  "action": "subscribe",  
  "runId": "run-abc123",  
  "sinceSequence": 42 // Resume from event 43  
}
```

The server will replay all events with **sequence** > 42.

Stability guarantee: Within the **events/v1** contract:

- Event envelope fields (**eventType**, **timestamp**, **runId**, **sequence**) are stable.
- New event types may be added in minor versions.
- Existing event types will not change their **eventType** string.
- Payload schemas may add optional fields; clients **MUST** ignore unknown fields.

13.2.3 Tool Invocation Contract (**tool/v1**)

The tool invocation contract defines the stdio JSON-RPC protocol that tools must implement to be invoked by gert. All tools — whether built-in, project-level, or extension-contributed — implement this contract.

Contract version: **tool/v1**

Transport: stdio JSON-RPC 2.0

Invocation request:

```
{
  "jsonrpc": "2.0",
  "id": "invocation-5",
  "method": "invoke",
  "params": {
    "action": "lookup",
    "inputs": {
      "id": "INC-12345"
    },
    "context": {
      "runId": "run-abc123",
      "stepId": "lookup_incident",
      "traceId": "trace-xyz789",
      "userId": "ops@acme.example",
      "mode": "real",
      "attempt": 1
    }
  }
}
```

Success response:

```
{
  "jsonrpc": "2.0",
  "id": "invocation-5",
  "result": {
    "outputs": {
      "incident": {
        "id": "INC-12345",
        "severity": "high",
        "status": "open"
      }
    },
    "exitCode": 0,
    "telemetry": {
      "durationMs": 320,

```

```
        "startedAt": "2026-04-18T12:00:00Z",
        "finishedAt": "2026-04-18T12:00:00.320Z"
    }
}
```

Error response:

```
{
  "jsonrpc": "2.0",
  "id": "invocation-5",
  "error": {
    "code": -32050,
    "message": "incident not found",
    "data": {
      "category": "runtime",
      "retryable": false,
      "details": {
        "incidentId": "INC-12345",
        "statusCode": 404
      }
    }
  }
}
```

Cancellation request:

```
{
  "jsonrpc": "2.0",
  "id": "cancel-5",
  "method": "cancel",
  "params": {
    "invocationId": "invocation-5",
    "reason": "run-cancelled"
  }
}
```

MCP tool adapter: When an MCP server is used as a tool source (§5), the gert host wraps the MCP `tools/call` protocol to satisfy the tool invocation contract. The adapter:

1. Translates the `invoke` request to an MCP `tools/call` request.
2. Forwards the `context` object as `_gert_context` in the MCP request (extension to MCP protocol).
3. Translates the MCP response back to the tool invocation contract response.
4. Flattens MCP `content` arrays to a single string for capture.

Stability guarantee: Within the `tool/v1` contract:

- Request/response envelope fields are stable.
- The `context` object fields are stable; new fields may be added.
- Error code ranges are stable (see Section 13.4).
- Tools MUST ignore unknown fields in requests.

13.2.4 Extension Handshake Contract (`extension/v1`)

The extension handshake contract defines the initialization and capability negotiation protocol between the gert host and an extension process. It is defined in §4 and summarized here for completeness.

Contract version: `extension/v1`

Transport: `stdio JSON-RPC 2.0`

Initialize request:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "method": "extension/initialize",
  "params": {
```

```
"host": {
  "name": "gert",
  "version": "2.0.0",
  "apiVersion": "2.0"
},
"protocolVersion": "2.0",
"grantedCapabilities": [
  "capability/tool-registration",
  "capability/network"
],
"runContext": {
  "runId": "run-abc123",
  "mode": "real"
}
}
```

Initialize response:

```
{
  "jsonrpc": "2.0",
  "id": "1",
  "result": {
    "extension": {
      "name": "acme.incident-pack",
      "version": "1.2.0"
    },
    "protocolVersion": "2.0",
    "acknowledgedCapabilities": [
      "capability/tool-registration",
      "capability/network"
    ]
  }
}
```

Heartbeat (ping/pong):

Request:

```
{
  "jsonrpc": "2.0",
  "id": "42",
  "method": "extension/ping",
  "params": {}
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "id": "42",
  "result": {
    "status": "ok"
  }
}
```

Stability guarantee: Within the `extension/v1` contract:

- The `extension/initialize` request/response schema is stable.
- New capabilities may be added; extensions **MUST** ignore unknown capabilities in `grantedCapabilities`.
- The `extension/ping` method signature is stable.
- Extension error codes (§4) are stable.

13.3 Contract Versioning

Each adapter contract has a version declared in the initial handshake or HTTP header. The version format is `{surface}/{major}` (e.g., `exec/v1`, `events/v1`). There is no minor version: any breaking change bumps the major version.

13.3.1 Version Declaration

HTTP-based contracts (JSON-RPC, WebSocket): Clients **MUST** send an `X-Gert-Contract` header on the initial HTTP request:


```
GET /ws HTTP/1.1
Host: gert.acme.example:8443
X-Gert-Contract: events/v1
```

The server responds with:

```
HTTP/1.1 200 OK
X-Gert-Contract: events/v1
```

If the client sends an unsupported contract version, the server responds with HTTP 400 and error:

```
{
  "error": {
    "code": 4001,
    "message": "unsupported contract version",
    "data": {
      "requestedVersion": "events/v3",
      "supportedVersions": ["events/v1"]
    }
  }
}
```

stdio-based contracts (tool, extension): The contract version is declared in the first message exchanged (initialize or handshake):

```
{
  "protocolVersion": "2.0"
}
```

If the protocol version is unsupported, the responder returns error code -32001 (protocol version not supported).

13.3.2 Breaking vs. Non-Breaking Changes

Non-breaking changes (minor version, no contract version bump):

- Adding optional fields to requests or responses.

- Adding new methods to the contract.
- Adding new error codes.
- Adding new event types.

Consumers **MUST** implement forward compatibility: ignore unknown fields, ignore unknown methods, ignore unknown event types.

Breaking changes (major version bump):

- Removing or renaming required fields.
- Changing the semantics of existing methods.
- Removing methods.
- Changing error code meanings.
- Incompatible transport changes (e.g., switching from JSON-RPC to gRPC).

Breaking changes require a new contract version (e.g., `exec/v1` → `exec/v2`).

13.3.3 Deprecation Policy

- When a new major contract version is released, the previous version is supported for 12 months after the new version's GA date.
- During the deprecation period, both versions are supported simultaneously.
- After 12 months, the old version is removed. Clients must upgrade.

Example timeline:

- 2026-04-01: `exec/v1` released.
- 2027-06-01: `exec/v2` released; `exec/v1` enters deprecation.
- 2028-06-01: `exec/v1` removed; only `exec/v2` supported.

13.3.4 Forward and Backward Compatibility

Forward compatibility (clients): Clients MUST ignore unknown fields in responses. This allows the server to add optional fields without breaking existing clients.

Example:

```
// Server adds a new optional field in exec/v1
{
  "result": {
    "runId": "run-abc123",
    "status": "running",
    "newOptionalField": "value" // Clients ignore this
  }
}
```

Backward compatibility (servers): Servers MUST accept requests that omit optional fields added in later minor versions. Servers MUST NOT require new optional fields.

Example:

```
// Client sends an old request format (before a new optional field was added)
{
  "method": "exec/start",
  "params": {
    "runbookPath": "./runbook.yaml"
    // Missing new optional field "labels"
  }
}
// Server accepts and defaults the missing field
```

13.4 Error Codes

All adapter contracts use a canonical error code registry. Error codes are integers organized by category. Codes are stable across contract versions.

Error envelope structure:

```
{
```

```
"error": {
  "code": 2001,
  "message": "command denied: not in allowlist",
  "data": {
    "category": "governance",
    "retryable": false,
    "details": {
      "command": "rm",
      "reason": "in_denylist"
    }
  }
}
```

Error code stability:

- Existing error codes will not change their meaning.
- New error codes may be added in minor versions.
- Clients MUST handle unknown error codes gracefully (fallback to generic error handling).

13.5 Contract Test Suite

The gert repository includes a contract test suite that validates adapter implementations against the canonical contracts. Adapter authors SHOULD run these tests to ensure compatibility.

13.5.1 Test Suite Structure

Contract tests are located in `testdata/contracts/`:

```
testdata/contracts/
  exec-v1/
    01-start-run.test.yaml
    02-cancel-run.test.yaml
    03-list-runs.test.yaml
```

```
events-v1/  
  01-subscribe.test.yaml  
  02-reconnect.test.yaml  
tool-v1/  
  01-invoke.test.yaml  
  02-cancel.test.yaml  
extension-v1/  
  01-initialize.test.yaml  
  02-ping.test.yaml
```

Each test case is a YAML file with:

- **description**: human-readable test case description.
- **request**: the JSON-RPC request or WebSocket message.
- **expectedResponse**: the expected response (or error).
- **preconditions**: setup required before the test (e.g., "a run with id run-abc123 exists").

13.5.2 Running Contract Tests

```
$ gert contract test --adapter=vscode  
Running contract tests for adapter: vscode  
✓ exec-v1/01-start-run (passed)  
✓ exec-v1/02-cancel-run (passed)  
× events-v1/01-subscribe (failed: missing 'sequence' field)
```

Contract **test** results: 2 passed, 1 failed

Adapter implementations SHOULD achieve 100% pass rate before release.

13.5.3 Contract Test CI Integration

Contract tests are part of the CI gate for all adapter implementations:

- The VS Code extension CI runs `gert contract test -adapter=vscode`.
- Third-party adapters are encouraged to use the same test suite.
- Failed contract tests block merges to the main branch.

13.6 Reference Implementations

VS Code extension. The official VS Code extension lives in its own repository at <https://github.com/ormasoftchile/gert-vscode>. It is the reference implementation of the HTTP transport surface exposed by `gert serve`: it manages a child `gert serve` process, attaches a webview that loads `/preview/` from the local server, and consumes the SSE event streams (`/events`, `/runs/{id}/state`, `/runs/{id}/interactions`) plus the REST/JSON-RPC endpoints. Adapter authors should study its implementation for best practices. The extension is packaged with `vsce` and ships a `.vsix` artifact from CI.

Built-in tools. The built-in tools (`ext/builtin/tools/` in the repository) are the reference implementation of the tool invocation contract (`tool/v1`). They demonstrate how to handle `invoke`, `cancel`, and error responses.

MCP adapter. The MCP tool adapter (`ext/toolruntime/mcp_adapter.go`) demonstrates how to wrap an external protocol (MCP) to satisfy the tool invocation contract.

13.7 Contract Stability Guarantees

Gert commits to the following stability guarantees for all adapter contracts:

1. **Semantic versioning:** major version bumps indicate breaking changes; minor/patch versions are backward compatible.
2. **12-month deprecation period:** old contract versions are supported for 12 months after a new major version is released.
3. **Additive changes only:** within a major version, only non-breaking changes (adding optional fields, new methods, new event types) are permitted.
4. **Error code stability:** error codes do not change meaning within a major version.
5. **Forward compatibility:** clients must ignore unknown fields; servers must accept omitted optional fields.

These guarantees are designed to minimize adapter churn and allow external systems to depend on gert contracts with confidence.

13.8 Contract Documentation and Tooling

13.8.1 OpenAPI Specifications

All HTTP-based contracts (JSON-RPC over HTTP, WebSocket event stream) are documented as OpenAPI 3.1 specifications [3] in `specs/contracts/`:

- `specs/contracts/exec-v1.openapi.yaml`
- `specs/contracts/events-v1.openapi.yaml`

These specifications are the source of truth for request/response schemas and are used to generate client libraries.

13.8.2 JSON Schema Validation

All JSON-RPC request/response schemas are defined as JSON Schema (Draft 2020-12) [29] in `specs/contracts/schemas/`. These schemas are used for:

- Runtime validation of requests/responses.
- Contract test validation.
- Client library generation (TypeScript, Python, Go).

13.8.3 Client Library Generation

The gert project provides code generation tools to generate client libraries from the OpenAPI and JSON Schema definitions:

```
$ gert codegen client --language=typescript --output=./clients/ts/  
Generated TypeScript client for exec/v1 and events/v1  
Files written to ./clients/ts/
```

Official client libraries are provided for:

- TypeScript (for VS Code and web adapters)
- Python (for CI/CD integrations)
- Go (for extension authors)

Code	Description	Retryable
1xxx — Execution Errors		
1001	Run not found (invalid <code>runId</code>)	No
1002	Step failed (runtime error in step execution)	Maybe
1003	Run cancelled (user or timeout cancellation)	No
1004	Run already exists (duplicate <code>runId</code>)	No
2xxx — Governance Errors		
2001	Command denied (not in allowlist or in denylist)	No
2002	Approval timeout (no approval received in time)	No
2003	Approval rejected (approver rejected the request)	No
2004	Insufficient approvals (min approval count not reached)	No
3xxx — Schema Errors		
3001	Schema validation failed (runbook does not match schema)	No
3002	Unknown step type (unrecognized <code>stepType</code>)	No
3003	Import cycle detected (circular runbook dependencies)	No
4xxx — Integration Errors		
4001	Tool not found (tool name does not resolve)	No
4002	Extension handshake failed (protocol error)	Maybe
4003	Input resolution failed (provider error or missing input)	Maybe
4004	Unsupported contract version	No
5xxx — Server Errors		
5001	Internal error (unexpected host failure)	Maybe
5002	Resource exhausted (rate limit or quota exceeded)	Yes
5003	Service unavailable (host shutting down or overloaded)	Yes

TABLE 13.2: Canonical error code registry for adapter contracts.

Chapter 14

Input Provider Framework

The input provider framework governs how gert resolves dynamic input values at run time. When a runbook declares an input with a `from:` binding, the provider framework locates the appropriate provider, dispatches a resolution request, and returns the concrete value before the first step that needs it begins.

14.1 Design Goals

- **Extensibility:** third-party systems (PagerDuty, ServiceNow, Vault, etc.) integrate as provider plugins without changes to the host binary.
- **Transparency:** every resolved value is recorded in the run trace so the operator can audit where each input came from.
- **Fallback safety:** resolution failures degrade gracefully to an interactive prompt rather than aborting the run silently.
- **Consistency with tools:** providers share the same transport layer as tools (`stdio`, `stdio-jsonrpc`, `mcp`), minimising implementation surface.

14.2 Provider Definition Schema

Each provider is declared in a `.provider.yaml` file. By convention, project-level providers live in `providers/<name>.provider.yaml`.

```
# providers/pagerduty.provider.yaml
apiVersion: provider/v1
```

```
meta:
  name: pd                                # Prefix used in from: bindings (e.g. pd.*)
  version: "1.0.0"
  description: "Resolves inputs from PagerDuty incidents and services"
  binary: pd-provider                     # Executable name (looked up in PATH)

transport:
  mode: stdio-jsonrpc                     # stdio / stdio-jsonrpc / mcp
  startup:
    argv: ["--config", "pd.yaml"]
    ready-pattern: "provider ready"
    timeout: "10s"
    shutdown-method: "shutdown"

resolution:
  method: provider/resolve                 # JSON-RPC method the host calls
  prefixes:
    - "pd."                               # All from: values starting with pd.
                                           # are dispatched to this provider

# Schema for what the provider needs in each resolution request
input-schema:
  type: object
  properties:
    bindings:
      type: object
      description: "Map of input name to from: binding string"
    context:
      type: object
      description: "Execution context (runId, userId, mode)"
  required: ["bindings"]

# Schema for what the provider returns
output-schema:
  type: object
  properties:
    resolved:
```

```
    type: object
    description: "Map of input name to resolved string value"
    warnings:
      type: array
      items:
        type: string
    required: ["resolved"]

cache:
  enabled: true
  scope: run                # run / step / none
  ttl: "60s"

governance:
  requires-capabilities:
    - capability/network    # This provider needs outbound network
  env-read-patterns:
    - "PD_API_KEY"
    - "PD_*
```

14.3 Resolution Protocol

14.3.1 Resolution Flow

When the run engine encounters an input with a `from:` binding (e.g., `from: pd.incident.title`), the following steps occur:

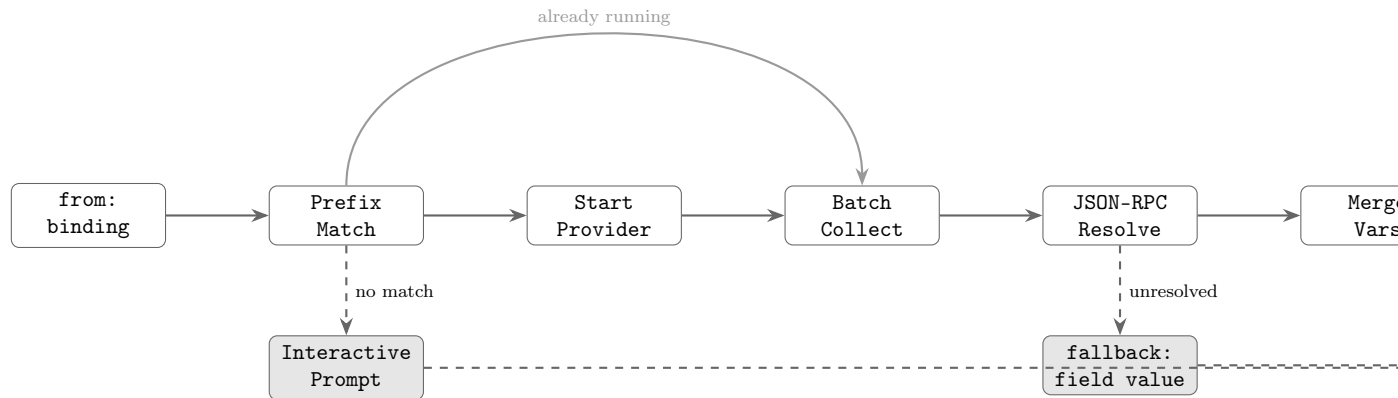


FIGURE 14.1: Input provider resolution pipeline. A `from: binding` triggers a prefix lookup; if a matching provider is found its process is started (or reused), inputs are batched, and a single `provider/resolve` JSON-RPC call returns all values. Unresolved bindings fall back to the `fallback:` field or an interactive prompt.

1. **Prefix match:** the framework scans registered providers for one whose declared prefix matches the start of the binding string. The longest matching prefix wins.
2. **Provider start:** if the matched provider's process is not yet running, the host spawns it and waits for the ready signal (same mechanics as `stdio-jsonrpc` tools; see Section 6.3).
3. **Batch collection:** all inputs whose binding prefix maps to the same provider are grouped into a single resolution request to minimise round-trips.
4. **JSON-RPC call:** the host sends a `provider/resolve` request.
5. **Result merge:** resolved values are merged into the run's variable map.
6. **Fallback:** any binding that was not resolved (absent from the response's `resolved` map, or where the provider returned a warning) falls back according to the input's `fallback:` field (default: interactive prompt).

14.3.2 Resolution Request Envelope

```
{
```

```
"jsonrpc": "2.0",
"id": "prov-1",
"method": "provider/resolve",
"params": {
  "bindings": {
    "incident_title": "pd.incident.title",
    "incident_severity": "pd.incident.severity",
    "service_name": "pd.service.name"
  },
  "context": {
    "runId": "run-abc123",
    "stepId": "__pre-flight__",
    "userId": "ops@acme.example",
    "mode": "real",
    "traceId": "trace-xyz"
  }
}
}
```

14.3.3 Resolution Response Envelope

```
{
  "jsonrpc": "2.0",
  "id": "prov-1",
  "result": {
    "resolved": {
      "incident_title": "Database connection pool exhausted",
      "incident_severity": "P1",
      "service_name": "payments-api"
    },
    "warnings": [],
    "telemetry": {
      "startedAt": "2026-04-18T10:00:00Z",
      "finishedAt": "2026-04-18T10:00:00.080Z",
      "durationMs": 80
    }
  }
}
```

14.3.4 Resolution Error Handling

If the provider returns a JSON-RPC error, or if a binding key is absent from the **resolved** map, the host applies the input's fallback strategy:

```
# In runbook meta.inputs
inputs:
  incident_title:
    from: pd.incident.title
    description: "Active incident title"
    fallback: prompt          # prompt / fail / default
    default: "Unknown incident" # Used when fallback: default
```

- **fallback: prompt** (default): the host issues an interactive prompt for the value. In non-interactive modes (dry-run, CI) this causes the run to fail.
- **fallback: default**: the declared **default:** value is used without prompting. A warning is written to the trace.
- **fallback: fail**: the run aborts immediately with a structured error.

14.3.5 Resolution Caching

Providers may declare a cache policy in their definition. The host respects this policy:

- **scope: run** — the resolved value is reused for all subsequent steps in the same run that reference the same binding. This is the default for external providers.
- **scope: step** — the value is resolved fresh on each step that references it.
- **scope: none** — caching is disabled; every binding triggers a new RPC call.

Cache entries are keyed by (**runId**, **bindingString**). Cached values are stored in the run's in-memory variable map only; they are not persisted across run resumptions.

14.4 Built-in Providers

The following providers are compiled into the gert host binary and are always available without a `.provider.yaml` file.

14.4.1 env

Reads values from environment variables.

```
inputs:
  api_key:
    from: env.ACME_API_KEY
    description: "API key for the Acme service"
```

The `env.` prefix is followed by the exact environment variable name. If the variable is not set, resolution fails and the fallback strategy is applied. The `env` provider never contacts an external process; it reads directly from the host's environment.

14.4.2 file

Reads values from files on the local file system.

```
inputs:
  ssh_key:
    from: "file ~/.ssh/id_rsa"
    description: "SSH private key"
  config_value:
    from: "file ./etc/gert/config.json#/database/host"
    description: "Database host from config file"
```

The path after `file.` is the file path. An optional JSON Pointer fragment (`#/path/to/field`) extracts a nested value from a JSON or YAML file. Without a fragment, the entire file contents are returned as a string.

14.4.3 prompt

The interactive prompt provider is the default resolution strategy. It is also the fallback for any failed provider resolution (unless overridden).

```
inputs:
  server_name:
    from: prompt
    description: "Server to check"
```

In **real** and **debug** modes, the host presents a terminal prompt to the operator. In **dry-run** and **replay** modes, the prompt provider reads from the scenario's `inputs.yaml` instead. In non-interactive environments where no scenario is available, the prompt provider fails with an informative error directing the operator to supply the value via `-var` or an explicit provider.

14.4.4 workspace

Reads values from the workspace configuration file (`.gert/config.yaml`).

```
inputs:
  default_region:
    from: workspace.defaults.region
    description: "AWS region from workspace config"
```

The path after `workspace.` is a dot-notation key into the workspace configuration object. This allows teams to centralise defaults without repeating them in every runbook.

14.5 Interactive Step Type Contracts

Input providers not only resolve `from:` bindings for runbook inputs, they also power the three interactive step types: **choice**, **decision**, and **collector**. When the run engine executes one of these steps, it dispatches a step-specific request to the configured input provider (or falls back to the built-in prompt provider).

14.5.1 Choice Step Contract

A **choice** step presents the operator with a fixed set of labeled options and stores the selected value as a variable. This is analogous to a dropdown menu or radio button group in a graphical form.

Request envelope. When the run engine executes a **choice** step, it sends a `provider/choice` JSON-RPC request to the configured provider:

```
{
  "jsonrpc": "2.0",
  "id": "choice-1",
```



```
"method": "provider/choice",
"params": {
  "stepId": "select-region",
  "prompt": "Select the target AWS region",
  "options": [
    { "value": "us-east-1", "label": "US East (N. Virginia)" },
    { "value": "us-west-2", "label": "US West (Oregon)" },
    { "value": "eu-west-1", "label": "Europe (Ireland)" }
  ],
  "default": "us-east-1",
  "context": {
    "runId": "run-abc123",
    "stepId": "select-region",
    "userId": "ops@acme.example",
    "mode": "real"
  }
}
```

Response envelope. The provider returns the selected value:

```
{
  "jsonrpc": "2.0",
  "id": "choice-1",
  "result": {
    "selected": "us-west-2"
  }
}
```

Contract requirements.

- The `options` array must contain at least two elements.
- Each option must have both `value` (the string stored in the variable) and `label` (the human-readable text displayed to the operator).
- The `default` field is optional. If present, it must match one of the `value` fields in the `options` array.

- The provider's **selected** response must match one of the option values exactly. If the provider returns a value not in the options list, the run engine fails the step with a **choice.invalid_selection** error.
- Providers MAY render options in the order provided, or MAY sort them alphabetically by label for improved usability. The canonical order is declared in the runbook YAML.
- **multiple: true** (optional, default **false**) — when set, the provider MUST allow the operator to select one or more values. The **selected** response field MUST be a JSON array of strings rather than a single string. The run engine stores the result as a JSON array of strings; see §4.4.5 for storage semantics.
- **options_from** (optional) — mutually exclusive with the static **options** array. When present, the run engine calls **inputProvider/getOptions** to fetch options dynamically before dispatching the **provider/choice** request; see §14.5.4.

Built-in prompt provider behaviour. The built-in **prompt** provider presents a numbered list in the terminal and waits for the operator to enter the number or value corresponding to their selection. In **dry-run** mode, it reads the value from the scenario's **inputs.yaml** or uses the **default** if provided.

14.5.2 Decision (Router) Step Contract

A **decision** step presents the operator with a set of routing choices, where each choice directs execution to a different runbook or labeled section. Unlike **choice**, a **decision** is a control-flow primitive: the selected route determines which node the run engine transitions to next.

Request envelope. When the run engine executes a **decision** step, it sends a **provider/decision** JSON-RPC request:

```
{
  "jsonrpc": "2.0",
  "id": "decision-1",
  "method": "provider/decision",
  "params": {
    "stepId": "triage-severity",
    "prompt": "Select the incident severity level",
```

```
{
  "routes": [
    { "route": "p1-critical", "label": "P1: Critical (immediate)" },
    { "route": "p2-high",     "label": "P2: High (1 hour SLA)" },
    { "route": "p3-normal",   "label": "P3: Normal (24 hour SLA)" }
  ],
  "context": {
    "runId": "run-abc123",
    "stepId": "triage-severity",
    "userId": "ops@acme.example",
    "mode": "real"
  }
}
```

Response envelope. The provider returns the selected route label:

```
{
  "jsonrpc": "2.0",
  "id": "decision-1",
  "result": {
    "route": "p2-high"
  }
}
```

Contract requirements.

- The `routes` array must contain at least two elements.
- Each route must have both `route` (the target node label or runbook ID) and `label` (the human-readable text).
- The provider does *not* need to know about the runbook graph structure. It only returns the selected route label. The run engine is responsible for resolving the route label to a target node and validating that the target exists.
- If the provider returns a route label not in the `routes` array, the run engine fails the step with a `decision.invalid_route` error.

- The decision result is *not* stored as a variable. It only controls execution flow. If the operator needs the decision recorded, the runbook author must add an explicit variable assignment in the target node.

Built-in prompt provider behaviour. The built-in `prompt` provider displays the routes as a numbered list and waits for the operator to select one. In `dry-run` mode, the route is read from the scenario file or the run fails if no scenario is provided.

14.5.3 Collector Step Contract

A `collector` step gathers unstructured input from the operator: free-form text, multi-line notes, file attachments, screenshots, URLs, or other evidence artifacts. The collected data is stored as named variables or artifacts in the run's state.

Request envelope. When the run engine executes a `collector` step, it sends a `provider/collect` JSON-RPC request:

```
{
  "jsonrpc": "2.0",
  "id": "collect-1",
  "method": "provider/collect",
  "params": {
    "stepId": "gather-evidence",
    "instructions": "Upload a screenshot of the error and describe the issue",
    "fields": [
      {
        "name": "screenshot",
        "type": "file",
        "label": "Error screenshot",
        "required": true,
        "visible": true,
        "accept": ["image/png", "image/jpeg"],
        "maxSizeBytes": 10485760
      },
      {
        "name": "description",
        "type": "text",
```

```
    "label": "Describe what you observed",
    "required": true,
    "visible": true,
    "multiline": true,
    "validation": {
      "minLength": 10,
      "maxLength": 2000,
      "pattern": "^[\\s\\S]+$",
      "patternHint": "Description must not be empty"
    }
  },
  {
    "name": "log_url",
    "type": "url",
    "label": "Link to related logs (optional)",
    "required": false,
    "visible": true
  }
],
"context": {
  "runId": "run-abc123",
  "stepId": "gather-evidence",
  "userId": "ops@acme.example",
  "mode": "real"
}
}
```

Response envelope. The provider returns the collected values and any uploaded artifacts:

```
{
  "jsonrpc": "2.0",
  "id": "collect-1",
  "result": {
    "values": {
      "description": "Database connection pool exhausted...",
      "log_url": "https://logs.acme.example/trace/xyz"
    }
  },
}
```

```
"artifacts": [  
  {  
    "field": "screenshot",  
    "filename": "error-2026-04-18.png",  
    "contentType": "image/png",  
    "sizeBytes": 245631,  
    "sha256": "a3c7d1f...",  
    "storagePath": "artifacts/run-abc123/screenshot.png"  
  }  
]  
}
```

Contract requirements.

- The `fields` array declares the form structure. Each field has:
 - `name`: the variable name where the value will be stored.
 - `type`: one of `text`, `number`, `integer`, `date`, `datetime`, `boolean`, `select`, `file`, `image`, or `url`. Full per-type validation and variable-storage rules are defined in §4.4.5.
 - `label`: human-readable prompt text.
 - `required`: boolean flag.
 - `visible`: boolean flag, pre-evaluated by the executor from the field's `when` expression before the request is sent. `true` means the field should be rendered; `false` means the field should be hidden or skipped. CLI providers SHOULD skip fields where `visible: false`; rich UI providers SHOULD show or hide the field dynamically as the user fills in earlier fields, re-evaluating `when` client-side so the form responds in real time without a round-trip to the executor. If `when` was absent on the field, `visible` is always `true`. See §3.3.7 for the full conditional field evaluation contract.
 - `validation` (optional object): client-side validation constraints that providers MAY enforce before submitting values, reducing unnecessary round-trips. The executor always re-validates server-side regardless of whether the provider enforced them:

- * `validation.pattern` — RE2 regex that the value must fully match (`text` fields only).
 - * `validation.patternHint` — human-readable message shown when `pattern` fails (e.g. “*Must be a valid ticket reference: ABC-1234*”).
 - * `validation.minLength`, `validation.maxLength` — character-count bounds for `text` fields.
 - * `validation.min`, `validation.max`, `validation.step` — numeric bounds for `number` and `integer` fields.
 - * `validation.min`, `validation.max` — date bounds for `date` and `datetime` fields (ISO 8601 strings; lexicographic comparison is correct for ISO dates).
 - * `options` array or `options_from` binding — required for `select` fields; see §14.5.4.
 - * `multiple: true|false` (default `false`) — enables multi-select for `select` fields.
 - * `validation.min_selections`, `validation.max_selections` — array length bounds for multi-select fields.
 - * `multiline: true|false` — UI hint for `text` fields; no effect on validation or variable storage.
 - * `accept` (MIME types) and `maxSizeBytes` — for `file` and `image` fields.
- Providers MUST return all required fields in the `values` map. If a required field is missing, the run engine fails the step with a `collector.missing_required_field` error.
 - For `type: file` fields, providers MUST upload the file to the artifact store and return metadata in the `artifacts` array. The artifact metadata includes:
 - `field`: the field name from the request.
 - `filename`: the original filename.
 - `contentType`: MIME type.
 - `sizeBytes`: file size in bytes.
 - `sha256`: SHA-256 hash for integrity verification.
 - `storagePath`: path where the artifact is stored (relative to the run’s artifact root).

- File size limits are enforced by the provider. If an uploaded file exceeds `maxSizeBytes`, the provider **MUST** reject it with a `collector.file_too_large` error before storing it.
- Supported MIME types for file uploads:
 - Images: `image/png`, `image/jpeg`, `image/gif`
 - Documents: `application/pdf`, `text/plain`, `text/markdown`
 - Archives: `application/zip`, `application/x-tar`, `application/gzip`
 - Logs: `text/plain`, `application/json`
- Providers **MAY** support additional MIME types, but **MUST** respect the `accept` constraint if declared in the field definition.
- The artifact storage path is provider-specific. The built-in `prompt` provider stores artifacts in `.gert/runs/<runId>/artifacts/`. External providers (e.g., S3, Azure Blob) **MAY** store artifacts in remote storage and return a signed URL or storage reference.

Built-in prompt provider behaviour. The built-in `prompt` provider displays each field as a terminal prompt. For `type: text` with `multiline: true`, it opens the operator's configured editor (`$EDITOR`). For `type: file`, it prompts for a file path and copies the file to the local artifact store. For `type: select`, it presents a numbered list (single-select) or a checkbox list (multi-select). For `type: boolean`, it prompts with a `[y/N]` confirmation. For `type: number`, `integer`, `date`, and `datetime`, it accepts free-text input and validates the format before accepting. In `dry-run` mode, all values are read from the scenario file.

14.5.4 Dynamic Options Protocol

When a `select` field in a `collector` step, or a `choice` step, declares `options_from.provider`, the run engine must fetch the option list dynamically from that provider before rendering the field or dispatching the interactive step request.

Fetch Sequence

1. **Provider lookup** — the run engine resolves the provider ID from `options_from.provider` in `ExecutionPlan.Providers`. If the provider is not registered, the step fails immediately with `options_from.unknown_provider`.

2. **Provider start** — if the provider process is not yet running, it is started using the normal `stdio-jsonrpc` startup sequence.
3. **Options request** — the run engine sends an `inputProvider/getOptions` JSON-RPC request to the provider, including the current variable scope so the provider can filter or compute options contextually.
4. **Option list received** — the run engine treats the returned `options` array as if it had been declared statically in the runbook. Validation and rendering proceed identically to the static-options path.

Failure to fetch options (provider error, network error, or timeout) fails the step immediately with `options_from.fetch_failed`. There is no silent fallback to an empty options list.

`inputProvider/getOptions` Request

```
{
  "jsonrpc": "2.0",
  "id": "getopts-1",
  "method": "inputProvider/getOptions",
  "params": {
    "providerId": "employee-directory",
    "field":      "managers",
    "stepId":     "assign-manager",
    "variables": {
      "department": "engineering",
      "region":     "us-west"
    },
    "context": {
      "runId": "run-abc123",
      "stepId": "assign-manager",
      "userId": "ops@acme.example",
      "mode": "real"
    }
  }
}
```

Request fields.

- `providerId` — the provider ID declared in `options_from.provider`.
- `field` — the `options_from.field` value; allows a single provider to serve multiple distinct option lists (e.g. `managers`, `engineers`, `directors`).
- `stepId` — the step making the request (for audit and debugging).
- `variables` — the current run variable scope at the time the step is reached. The provider MAY use these to filter or compute options contextually (e.g. return only managers in the operator's department).
- `context` — standard execution context (run ID, actor, mode).

inputProvider/getOptions Response

```
{
  "jsonrpc": "2.0",
  "id": "getopts-1",
  "result": {
    "options": [
      { "value": "alice",    "label": "Alice Chen (Staff Eng)" },
      { "value": "bob",    "label": "Bob Okafor (Principal)" },
      { "value": "charlie", "label": "Charlie Park (Director)",
        "hint": "Manages infrastructure" }
    ],
    "cacheTtlSeconds": 300
  }
}
```

Response fields.

- `options` — array of `{value, label}` objects. Each `value` is the string stored in the run variable on selection; `label` is the human-readable text. An optional `hint` field provides secondary descriptive text.
- `cacheTtlSeconds` (optional) — if non-zero, the run engine caches the returned option list in memory for the remainder of the run (capped at the declared TTL). This avoids repeated RPC calls when the same dynamic field appears in multiple steps of a single run. A value of 0 disables caching for this response.

Capability declaration. Providers that implement `inputProvider/getOptions` MUST declare the capability in their handshake response. If the run engine encounters an `options_from` binding and the matched provider does not advertise `getOptions: true`, the step fails with `options_from.capability_not_supported`.

14.5.5 Autocomplete Search Protocol

The `type: autocomplete` field in a `collector` step requires live-search support from the input provider. The run engine issues `inputProvider/search` RPC calls as the operator types, with debouncing handled by the UI layer.

`inputProvider/search` Request

```
{
  "jsonrpc": "2.0",
  "id": "search-1",
  "method": "inputProvider/search",
  "params": {
    "provider": "service-registry",
    "field": "query",
    "query": "pay",
    "minChars": 2,
    "runId": "20260418T142300-abc12345",
    "variables": {
      "region": "us-west"
    }
  }
}
```

Request fields.

- `provider` — the provider ID from `options_from.provider` (or the field's dedicated `autocomplete.provider` binding).
- `field` — the field ID within the step, allowing a single provider to serve multiple autocomplete fields.
- `query` — the current text typed by the operator. May be an empty string `""` on initial focus or during CLI degradation (see below).

- **minChars** — the minimum character count declared on the field. The run engine does not issue the RPC until `len(query) >= minChars`; this field is informational for the provider.
- **runId** — the current run ID (for audit and logging on the provider side).
- **variables** — the current run variable scope at dispatch time. Providers MAY use this to filter results contextually (e.g., return only services in the operator's selected region).

inputProvider/search Response

```
{
  "jsonrpc": "2.0",
  "id": "search-1",
  "result": {
    "results": [
      { "value": "payments-api", "label": "Payments API" },
      { "value": "payroll-svc", "label": "Payroll Service" }
    ],
    "hasMore": false
  }
}
```

Response fields.

- **results** — array of {value, label} objects. **value** is the string stored in the run variable on selection; **label** is the human-readable text shown in the UI.
- **hasMore** (boolean) — if **true**, the UI SHOULD display a hint such as *“Showing top N results — keep typing to narrow.”*. A **false** value indicates the full matching set is returned.

Timeout contract. Providers MUST return within **500 ms**. If no response is received within this window, the run engine treats the call as timed out, returns an empty results list, and allows the operator to retry by continuing to type. The timeout is measured from the moment the RPC request is sent on the transport.

CLI degradation. The built-in `prompt` provider does not support real-time search. When an `autocomplete` field is dispatched to a CLI provider, the run engine issues a single `inputProvider/search` call with `query: ""` to retrieve all available options, then presents the result as a standard `select` list. Typing ahead is not supported in CLI mode.

Capability declaration. Providers that implement `inputProvider/search` MUST declare `search: true` in their capability advertisement (see matrix below). A provider that does not advertise `search: true` will have autocomplete fields degraded to the CLI path automatically.

14.5.6 Provider Capability Matrix

Not all providers support all interactive step types. The following table shows which step types each built-in provider supports:

Provider	Input Resolution	choice	decision	collector	getOptions	search
<code>prompt</code>	Yes	Yes	Yes	Yes	No	No
<code>env</code>	Yes	No	No	No	No	No
<code>file</code>	Yes	No	No	No	No	No
<code>workspace</code>	Yes	No	No	No	No	No

`search: false` for all built-in providers reflects CLI degradation: autocomplete fields degrade to a standard select when the active provider cannot perform live search. Rich UI providers (e.g., the VS Code extension's built-in provider) SHOULD declare `search: true` to enable real-time autocomplete rendering.

External providers (PagerDuty, ServiceNow, Slack, etc.) MAY implement any combination of these contracts. A provider that implements `provider/choice` MUST declare this in its capability advertisement during the handshake phase:

```
# Example provider capability declaration
{
  "jsonrpc": "2.0",
  "method": "provider/initialize",
  "params": {
    "capabilities": {
      "resolution": true,

```

```
    "choice": true,  
    "decision": false,  
    "collector": true,  
    "getOptions": true,  
    "search": true  
  }  
}  
}
```

If the run engine encounters a `choice`, `decision`, or `collector` step and the configured provider does not advertise the corresponding capability, the engine falls back to the built-in `prompt` provider.

14.6 Provider Composition

A `from:` binding may declare a priority-ordered chain of providers. The host attempts each provider in order and uses the first successful result.

```
inputs:  
  api_token:  
    from:  
      - env.ACME_API_TOKEN           # Try env var first  
      - vault.secret/acme/api-token # Fall back to Vault  
      - prompt                       # Final fallback: ask the operator  
    description: "API token for Acme service"
```

Chaining is evaluated left-to-right. A provider is considered to have failed if it returns an error, if the binding key is absent from the resolved map, or if the resolved value is an empty string (unless the input declares `allow-empty: true`).

14.7 Provider Lifecycle

Process model. Providers that use the `stdio-jsonrpc` or `mcp` transport are *persistent*: the host spawns each provider process once at the start of the pre-flight resolution phase and keeps it alive for the entire run. This avoids repeated startup latency for providers that need to authenticate or maintain session state (e.g., Vault token renewal, PagerDuty OAuth).

Providers using the `stdio` transport are *per-resolution*: the host spawns a new process for each resolution request.

Startup. Provider processes are started lazily on first use, following the same `ready-pattern` / timeout mechanism as `stdio-jsonrpc` tools (Section 6.3).

Shutdown. At run completion (success, failure, or cancellation) the host iterates over all running provider processes and sends the method declared in `transport.startup.shutdown-method` (default: `shutdown`). Providers are given a 5-second grace period to clean up before the host sends `SIGTERM`, then `SIGKILL` after a further 5 seconds.

Crash handling. If a provider process crashes during a resolution call, the host logs a structured `provider.crashed` trace event and applies the input's `fallback:` strategy for all bindings that were in-flight at the time of the crash.

Chapter 15

Observability and Diagnostics

Operational visibility is essential for production systems. This chapter specifies the observability model for gert: structured logging, metrics, distributed tracing, health probes, and diagnostics tooling. It is distinct from § 12 (Evidence, Tracing, and Resumption): **§12 specifies immutable audit records for compliance and resumption**; this chapter specifies **operational signals for operators** to answer questions like *is the system healthy?*, *what is slow?*, *where did this run fail?*, and *what is the resource utilization?*

The design follows the three-pillar observability model [13] adapted for workflow execution: metrics for aggregated performance, distributed tracing for execution flow, and structured logging for detailed context. All three are instrumented using industry standards (OpenTelemetry [6], W3C Trace Context [2], RFC 3339 timestamps [19]) to ensure interoperability with observability platforms.

15.1 Observability Model

Gert applies the three-pillar observability model to workflow execution. Each pillar serves a distinct purpose:

15.1.1 Metrics: Aggregated Performance Indicators

Purpose: Track aggregate performance, throughput, error rates, and resource utilization over time.

Use Cases:

- *Alerting:* Trigger alerts when error rates exceed thresholds or latency degrades

- *Capacity Planning*: Measure run throughput and active concurrency to inform scaling decisions
- *SLO Monitoring*: Track percentile latencies (p50, p95, p99) for step execution times

Metric Types:

- **Counter**: Monotonically increasing (total runs, total errors)
- **Gauge**: Point-in-time measurement (active runs, loaded extensions)
- **Histogram**: Distribution of values (run duration, tool invocation latency)

Exposition Format: Prometheus OpenMetrics format exposed via HTTP endpoint (GET /metrics) when running in server mode (`gert serve -http`).

15.1.2 Distributed Tracing: Execution Flow Visualization

Purpose: Capture the hierarchical execution tree with timing and outcome for every run, step, and tool invocation.

Use Cases:

- *Root-Cause Analysis*: Identify which step in a failed run caused the failure
- *Latency Diagnosis*: Find which tool invocations are slow
- *Cross-System Correlation*: Link gert spans to HTTP API call traces via W3C Trace Context propagation

Implementation: OpenTelemetry SDK with OTLP exporter [6]. Spans emitted for every run (root span), step (child spans), and tool invocation (grandchild spans).

Span Hierarchy:

```
gert.run (run_id=...)
+- gert.step (step_id=backup-db, step_type=cli)
| +- gert.tool.invoke (tool=ssh, exit_code=0)
|   L- gert.governance.check (policy=allow-ssh)
+- gert.step (step_id=deploy, step_type=tool)
|   L- gert.tool.invoke (tool=kubectl, exit_code=0)
L- gert.step (step_id=verify, step_type=manual)
    L- gert.input.resolve (provider=approval-gate)
```

Trace Backends: Jaeger [15], Zipkin [30], Datadog, New Relic, AWS X-Ray (all OTLP-compatible).

15.1.3 Structured Logging: Detailed Contextual Events

Purpose: Emit detailed, machine-parseable log lines with run and step context for debugging and incident investigation.

Use Cases:

- *Debugging:* Search logs by `run_id` to see all activity for a specific run
- *Error Investigation:* View full error messages and stack traces in context
- *Compliance Audit:* Query logs for all runs by a specific actor or runbook

Log Format: JSON, one object per line, emitted to `stderr` by default.

Required Fields:

- `timestamp` (RFC 3339 [19])
- `level` (debug, info, warn, error)
- `msg` (human-readable message)
- `run_id` (when in run context)
- `step_id` (when in step context)

Aggregation: Loki, Elasticsearch, Splunk, CloudWatch Logs (any system that can parse JSON logs).

15.2 OpenTelemetry Integration

Gert integrates OpenTelemetry as the standard tracing layer. All spans, metrics, and logs follow the OpenTelemetry specification [6] to ensure portability across observability backends.

15.2.1 Span Hierarchy and Attributes

OpenTelemetry spans form a hierarchical tree representing the execution structure. Each span level has required attributes for filtering and analysis.

Root Span: `gert.run`

The root span represents the entire run from start to completion.

Span Name: `gert.run`

Required Attributes:

```
runbook.id      = "deploy-app-v3"           // runbook identifier
runbook.version = "1.2.3"                   // version (if specified)
actor.id        = "alice@example.com"       // human or service principal
run.id          = "20260418T143022-a3f9c1"  // unique run ID
run.mode        = "real"                    // "real" | "replay" | "dry-run"
client.type     = "cli"                     // "cli" | "tui" | "web" | "mcp"
```

Span Status:

- OK if run outcome is `success`
- ERROR if run outcome is `failure`, `cancelled`, or `governance-blocked`

Child Span: `gert.step`

Each step execution produces a child span under `gert.run`.

Span Name: `gert.step`

Required Attributes:

```
step.id         = "backup-db"               // step identifier from YAML
step.type       = "cli"                     // "cli" | "manual" | "tool" | "invoke"
step.index      = 2                         // zero-based position in sequence
run.id          = "20260418T143022-a3f9c1"  // run correlation
step.name       = "Backup production database" // human-readable name
```

Optional Attributes:

```
step.outcome    = "success"                 // "success" | "failure" | "skipped"
step.retries    = 2                         // number of retry attempts
```

Span Events: Emit span events for key transitions:

- `governance.check.started` — governance evaluation started
- `governance.check.passed` — governance check passed
- `approval.requested` — human approval requested
- `approval.granted` — approval decision logged

Grandchild Span: gert.tool.invoke

Tool invocations create grandchild spans under `gert.step`.

Span Name: `gert.tool.invoke`

Required Attributes:

```
tool.name      = "kubectl"           // tool identifier
tool.transport = "exec"              // "exec" | "ssh" | "http"
tool.exit_code = 0                   // process exit code (exec only)
run.id         = "20260418T143022-a3f9c1" // run correlation
step.id        = "deploy"            // parent step
```

Optional Attributes:

```
tool.args      = ["apply", "-f", "deployment.yaml"] // redacted args
tool.duration_ms = 1234                             // invocation latency
```

Grandchild Span: gert.input.resolve

Input provider resolutions create grandchild spans.

Span Name: `gert.input.resolve`

Required Attributes:

```
input.name      = "db_password"      // input key from runbook
provider.type   = "vault"            // provider type
provider.address = "https://vault:8200" // endpoint (if applicable)
run.id          = "20260418T143022-a3f9c1"
```

Grandchild Span: gert.governance.check

Governance evaluations create grandchild spans.

Span Name: `gert.governance.check`

Required Attributes:

```
policy.name      = "allow-ssh"        // policy identifier
policy.outcome   = "allow"            // "allow" | "deny"
run.id           = "20260418T143022-a3f9c1"
step.id          = "backup-db"
```

15.2.2 Span Status and Error Handling

OpenTelemetry spans have three status codes:

- UNSET — default, no explicit status set
- OK — operation completed successfully
- ERROR — operation failed

Status Mapping:

- `gert.run` span:
 - OK if run outcome is `success`
 - ERROR if run outcome is `failure`, `cancelled`, `governance-blocked`
- `gert.step` span:
 - OK if step outcome is `success` or `skipped`
 - ERROR if step outcome is `failure`
- `gert.tool.invoke` span:
 - OK if `exit_code == 0`
 - ERROR if `exit_code != 0` or invocation crashes

When a span is marked ERROR, the runtime MUST record the error message as a span event or exception:

```
span.RecordError(err, trace.WithAttributes(  
    attribute.String("error.type", "tool_invocation_failed"),  
    attribute.String("error.message", err.Error()),  
))
```

15.2.3 Context Propagation

Gert propagates trace context across system boundaries using W3C Trace Context [2].

W3C Trace Context Headers

When a tool invokes an HTTP API, the runtime **MUST** inject W3C headers:

```
traceparent: 00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-01
tracestate: gert=run_id:20260418T143022-a3f9c1
```

The `traceparent` header contains:

- `version` (00)
- `trace-id` (128-bit, globally unique)
- `parent-id` (64-bit span ID)
- `trace-flags` (sampling decision)

The `tracestate` header carries vendor-specific metadata. Gert includes `run_id` as baggage for cross-system correlation.

Baggage

The `run_id` is propagated as OpenTelemetry baggage to all child spans and external systems:

```
| baggage.Set("run_id", runID)
```

This allows downstream systems to tag their own spans with the originating gert run ID, enabling end-to-end tracing across multi-system workflows.

15.2.4 OTLP Exporter Configuration

The runtime supports two trace export modes:

OTLP Export (Production)

Configure via environment variable or CLI flag:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://jaeger:4317
gert run deploy.yaml
```

Or:

```
gert run deploy.yaml --otel-endpoint=http://jaeger:4317
```

The runtime uses the OTLP/gRPC exporter by default. For HTTP, use:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://jaeger:4318
export OTEL_EXPORTER_OTLP_PROTOCOL=http/protobuf
```

Console Export (Development)

For local development without a trace backend:

```
gert run deploy.yaml --otel-console
```

This prints spans to `stderr` in a human-readable format.

15.2.5 Sampling Strategy

Gert uses **parent-based sampling** with a configurable default rate:

- If an incoming request already has a trace context (e.g., gert invoked via HTTP from another traced system), **inherit the parent's sampling decision**.
- If no parent context exists, sample at the configured rate (default: 100% = always sample).

Configuration:

```
export OTEL_TRACES_SAMPLER=parentbased_traceidratio
export OTEL_TRACES_SAMPLER_ARG=0.1 # sample 10% of traces
```

For production systems with high throughput, consider 0.01 (1%) or 0.001 (0.1%) to reduce backend load.

15.3 Metrics Catalog

Gert exposes the following metrics in Prometheus OpenMetrics format. All metrics are prefixed with `gert_`.

Metric	Type	Labels	Description
<code>gert_runs_total</code>	Counter	<code>outcome</code> , <code>runbook_id</code>	Total run completions by outcome (success , failure , cancelled)
<code>gert_run_duration_seconds</code>	Histogram	<code>outcome</code> , <code>runbook_id</code>	End-to-end run duration (start to completion)
<code>gert_active_runs</code>	Gauge	—	Currently executing runs

TABLE 15.1: Run-level metrics

15.3.1 Run Metrics

Histogram Buckets: `gert_run_duration_seconds` uses exponential buckets: 0.5, 1, 2, 5, 10, 30, 60, 120, 300, 600, 1800, 3600 (seconds).

15.3.2 Step Metrics

Metric	Type	Labels	Description
<code>gert_steps_total</code>	Counter	<code>outcome</code> , <code>step_type</code>	Total step completions by outcome and type
<code>gert_step_duration_seconds</code>	Histogram	<code>step_type</code>	Per-step duration by type (cli , manual , tool , invoke)

TABLE 15.2: Step-level metrics

Histogram Buckets: `gert_step_duration_seconds` uses buckets: 0.1, 0.5, 1, 2, 5, 10, 30, 60, 120, 300 (seconds).

15.3.3 Tool Invocation Metrics

Metric	Type	Labels	Description
<code>gert_tool_invocations_total</code>	Counter	<code>tool_name</code> , <code>outcome</code>	Tool invocation count by tool and outcome (success , failure)
<code>gert_tool_duration_seconds</code>	Histogram	<code>tool_name</code>	Tool invocation latency

TABLE 15.3: Tool invocation metrics

Histogram Buckets: `gert_tool_duration_seconds` uses buckets: 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10, 30 (seconds).

15.3.4 Approval Metrics

Metric	Type	Labels	Description
<code>gert_approval_wait_seconds</code>	Histogram	<code>step_id</code>	Time from approval request to decision (approved or rejected)
<code>gert_approvals_total</code>	Counter	<code>decision, step_id</code>	Total approval decisions (approved, rejected)

TABLE 15.4: Approval metrics

Histogram Buckets: `gert_approval_wait_seconds` uses buckets: 10, 30, 60, 300, 600, 1800, 3600, 7200, 14400, 28800, 86400 (seconds, up to 24 hours).

15.3.5 Extension Metrics

Metric	Type	Labels	Description
<code>gert_extension_crashes_total</code>	Counter	<code>extension_id</code>	Extension process crashes
<code>gert_extension_loaded</code>	Gauge	<code>extension_id</code>	1 if extension is loaded, 0 otherwise

TABLE 15.5: Extension metrics

15.3.6 Metrics Endpoint

When running in server mode, metrics are exposed at:

GET `http://localhost:8080/metrics`

Configure the server:

```
gert serve --http --addr=:8080
```

The response is in Prometheus text exposition format. Example:

```
# TYPE gert_runs_total counter
gert_runs_total{outcome="success",runbook_id="deploy-app-v3"} 42
gert_runs_total{outcome="failure",runbook_id="deploy-app-v3"} 3

# TYPE gert_run_duration_seconds histogram
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="1"} 5
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="5"} 18
gert_run_duration_seconds_bucket{outcome="success",runbook_id="deploy-app-v3",le="+Inf"} 42
gert_run_duration_seconds_sum{outcome="success",runbook_id="deploy-app-v3"} 187.4
gert_run_duration_seconds_count{outcome="success",runbook_id="deploy-app-v3"} 42

# TYPE gert_active_runs gauge
gert_active_runs 3
```

15.3.7 Push-Based Metrics (OTLP)

For environments where scraping is not feasible (e.g., ephemeral CI jobs), use OTLP metrics export:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://collector:4317
export OTEL_METRICS_EXPORTER=otlp
gert run deploy.yaml
```

The runtime pushes metrics to the OTLP endpoint at configurable intervals (default: 60 seconds).

15.4 Structured Logging

Gert emits structured JSON logs to `stderr`. Every log line is a complete JSON object.

15.4.1 Log Line Format

```
{
  "timestamp": "2026-04-18T14:30:22.123456Z",
  "level": "info",
  "msg": "step execution started",
  "run_id": "20260418T143022-a3f9c1",
```

```
"step_id": "backup-db",  
"step_type": "cli",  
"actor": "alice@example.com"  
}
```

15.4.2 Required Fields

- **timestamp** — RFC 3339 format [19] with microsecond precision, UTC timezone
- **level** — one of: `debug`, `info`, `warn`, `error`
- **msg** — human-readable message describing the event

15.4.3 Contextual Fields (When Applicable)

- **run_id** — run identifier (present in all logs within run context)
- **step_id** — step identifier (present in all logs within step context)
- **step_type** — step type (`cli`, `manual`, `tool`, `invoke`)
- **actor** — user or service principal executing the run
- **tool_name** — tool identifier (for tool invocation logs)
- **error** — error message (for `error` level logs)
- **exit_code** — process exit code (for CLI step completions)

15.4.4 Log Levels

- **debug** — Detailed diagnostic information (e.g., input resolution, policy evaluation details)
- **info** — Operational milestones (e.g., step started, run completed)
- **warn** — Recoverable issues (e.g., retry attempt, fallback to default)
- **error** — Failures requiring attention (e.g., step failure, governance denial)

Configure via environment variable or CLI flag:

```
export GERT_LOG_LEVEL=debug
gert run deploy.yaml
```

Or:

```
gert run deploy.yaml --log-level=debug
```

Default: info.

15.4.5 Sensitive Data Handling

Log lines MUST NOT contain sensitive input values. The runtime applies the same redaction rules as the trace file (see § 12.4).

Safe:

```
{"msg": "resolved input 'db_password' from provider 'vault'"}
```

Unsafe:

```
{"msg": "resolved input 'db_password' = 'hunter2'"}
```

When logging errors that may contain sensitive data, scrub or omit the error message:

```
{"msg": "input resolution failed", "input_name": "db_password",  
  "provider": "vault", "error": "connection timeout"}
```

15.4.6 Log Output Configuration

By default, logs are written to `stderr`. To write to a file:

```
gert run deploy.yaml --log-output=file:/var/log/gert/gert.log
```

This writes logs to the file *in addition to* `stderr`.

For syslog output:

```
gert run deploy.yaml --log-output=syslog://localhost:514
```

15.4.7 Log Aggregation and Correlation

Every log line in run context includes `run_id`, enabling log aggregation by run:

Loki Query:

```
{job="gert"} | json | run_id="20260418T143022-a3f9c1"
```

Elasticsearch Query:

```
GET /gert-logs-*/_search
{
  "query": {
    "term": {"run_id": "20260418T143022-a3f9c1"}
  }
}
```

This returns all log lines for the specified run, ordered by timestamp.

15.5 Health and Readiness Endpoints

When running in server mode (`gert serve -http`), the runtime exposes HTTP endpoints for health probes. These are designed for Kubernetes liveness and readiness probes.

15.5.1 Liveness Endpoint

GET /health

Purpose: Indicates whether the gert process is alive.

Response (200 OK):

```
{
  "status": "ok",
  "version": ".0",
  "uptime_seconds": 3600
}
```

Use Case: Kubernetes liveness probe to restart crashed pods.

15.5.2 Readiness Endpoint

GET /ready

Purpose: Indicates whether the runtime is ready to accept new runs.

Response (200 OK):

```
{
  "status": "ready",
  "extensions_loaded": 3,
  "extensions_failed": 0
}
```

Response (503 Service Unavailable):

```
{
  "status": "not_ready",
  "reason": "extension 'aws-tools' failed to load",
  "extensions_loaded": 2,
  "extensions_failed": 1
}
```

Use Case: Kubernetes readiness probe to delay traffic until extensions are loaded.

15.5.3 Kubernetes Probe Configuration

```
apiVersion: v1
kind: Pod
metadata:
  name: gert-server
spec:
  containers:
  - name: gert
    image: gert:.0
    command: ["gert", "serve", "--http", "--addr=:8080"]
    livenessProbe:
      httpGet:
        path: /health
        port: 8080
      initialDelaySeconds: 5
```

```
    periodSeconds: 10
  readinessProbe:
    httpGet:
      path: /ready
      port: 8080
    initialDelaySeconds: 10
    periodSeconds: 5
```

15.6 Diagnostics Commands

Gert provides CLI commands for self-diagnosis and trace inspection.

15.6.1 gert diagnose

Purpose: Run pre-flight checks to verify that the runtime environment is correctly configured.

Usage:

```
gert diagnose [--runbook=<path>]
```

Checks Performed:

1. **Runbook Schema Validation:** Parse and validate the runbook YAML against JSON Schema
2. **Tool Availability:** Verify that all declared tools (`.tool.yaml`) are executable and findable in `PATH` or `.runbook/tools/`
3. **Input Provider Connectivity:** Test that all input providers (`.provider.yaml`) can be reached (e.g., Vault is reachable, API keys are valid)
4. **Extension Loading:** Attempt to load all declared extensions and report any load failures
5. **Governance Policy Syntax:** Validate OPA policies (if policy-as-code is configured)

Output (Success):

- ✓ Runbook schema validation passed
- ✓ Tool 'kubectl' found at /usr/local/bin/kubectl
- ✓ Tool 'aws' found at /usr/local/bin/aws
- ✓ Input provider 'vault' reachable at https://vault:8200
- ✓ Extension 'aws-tools' loaded successfully
- ✓ Extension 'k8s-tools' loaded successfully

Diagnostics: PASS

Output (Failure):

- ✓ Runbook schema validation passed
- × Tool 'kubectl' not found in PATH or .runbook/tools/
- ✓ Tool 'aws' found at /usr/local/bin/aws
- × Input provider 'vault' unreachable: connection refused
- ✓ Extension 'aws-tools' loaded successfully
- × Extension 'k8s-tools' failed to load: missing dependency

Diagnostics: FAIL (3 errors)

Exit Code: 0 if all checks pass, 1 if any check fails.

15.6.2 gert trace show

Purpose: Pretty-print a completed run's trace from the JSONL file.

Usage:

```
gert trace show <run_id>
```

Output:

```
Run: 20260418T143022-a3f9c1
Runbook: deploy-app-v3
Actor: alice@example.com
Started: 2026-04-18T14:30:22Z
Completed: 2026-04-18T14:33:15Z
Duration: 2m53s
Outcome: success
```


Steps:

```
[0] backup-db (cli) - 12.3s - success
  L- tool: ssh - 12.1s - exit_code=0
[1] deploy (tool) - 45.2s - success
  L- tool: kubectl - 44.8s - exit_code=0
[2] verify (manual) - 2m10s - success
  L- approval: alice@example.com - granted
```

15.6.3 gert trace export

Purpose: Convert a JSONL trace to OpenTelemetry OTLP format for import into Jaeger or Zipkin.

Usage:

```
gert trace export <run_id> --format=otlp --output=trace.json
```

Supported Formats:

- **otlp** — OTLP JSON format
- **jaeger** — Jaeger JSON format
- **zipkin** — Zipkin JSON format

Use Case: Retrospectively analyze a run in a trace visualization tool when the runtime was not configured to export traces at execution time.

15.6.4 gert run list

Purpose: List recent runs with status.

Usage:

```
gert run list [--limit=N] [--json]
```

Output (Human-Readable):

RUN_ID	RUNBOOK	ACTOR	OUTCOME	DURATION
20260418T143022-a3f9c1	deploy-app-v3	alice@example.com	success	2m53s
20260418T140015-f3a7b2	rollback-prod	bob@example.com	failure	45s
20260418T135512-c9e8d1	backup-db	cron@system	success	3m12s

Output (JSON, for CI integration):

```
[
  {
    "run_id": "20260418T143022-a3f9c1",
    "runbook_id": "deploy-app-v3",
    "actor": "alice@example.com",
    "outcome": "success",
    "duration_ms": 173000,
    "started_at": "2026-04-18T14:30:22Z",
    "completed_at": "2026-04-18T14:33:15Z"
  },
  ...
]
```

Use Case: CI pipelines can parse JSON output to determine run status.

15.7 Integration Recipes

This section provides practical integration patterns for common observability platforms.

15.7.1 Prometheus + Grafana

Step 1: Start gert server

```
gert serve --http --addr=:8080
```

Step 2: Configure Prometheus scrape target

Add to `prometheus.yml`:

```
scrape_configs:
- job_name: 'gert'
  static_configs:
    - targets: ['localhost:8080']
  metrics_path: '/metrics'
  scrape_interval: 15s
```

Step 3: Import Grafana dashboard

Gert provides a reference Grafana dashboard JSON at `.runbook/observability/grafana-dashboard.json`. Import it into Grafana to visualize:

- Run throughput (runs/minute)
- Success vs. failure rate
- p50, p95, p99 run duration
- Active runs (gauge)
- Step execution time by type
- Tool invocation latency

15.7.2 Jaeger (Distributed Tracing)

Step 1: Start Jaeger (all-in-one)

```
docker run -d --name jaeger \  
-p 16686:16686 \  
-p 4317:4317 \  
jaegertracing/all-in-one:latest
```

Step 2: Configure gert to export traces

```
export OTEL_EXPORTER_OTLP_ENDPOINT=http://localhost:4317  
gert run deploy.yaml
```

Step 3: View traces in Jaeger UI

Open <http://localhost:16686>, select service `gert`, and search for traces.

15.7.3 Loki + Grafana (Log Aggregation)

Step 1: Configure Promtail to scrape gert logs

Add to `promtail-config.yaml`:

```
scrape_configs:  
- job_name: gert  
  static_configs:  
    - targets:  
      - localhost  
    labels:  
      job: gert
```

```
    __path__: /var/log/gert/*.log
  pipeline_stages:
    - json:
        expressions:
          timestamp: timestamp
          level: level
          msg: msg
          run_id: run_id
    - labels:
        level:
        run_id:
```

Step 2: Start gert with file logging

```
gert run deploy.yaml --log-output=file:/var/log/gert/gert.log
```

Step 3: Query logs in Grafana Explore

```
{job="gert"} | json | run_id="20260418T143022-a3f9c1"
```

15.7.4 Datadog / New Relic (OTLP-Compatible SaaS)

Both platforms support OpenTelemetry OTLP ingestion.

Datadog:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=https://api.datadoghq.com
export DD_API_KEY=<your-api-key>
gert run deploy.yaml
```

New Relic:

```
export OTEL_EXPORTER_OTLP_ENDPOINT=https://otlp.nr-data.net:4317
export NEW_RELIC_API_KEY=<your-api-key>
gert run deploy.yaml
```

Consult platform documentation for exact endpoint URLs and authentication headers.

15.8 Observability for CI/CD Pipelines

Gert provides machine-readable output for CI/CD integration.

15.8.1 JSON Output Mode

```
gert run deploy.yaml --output=json
```

On completion, emits a JSON summary to `stdout`:

```
{
  "run_id": "20260418T143022-a3f9c1",
  "outcome": "success",
  "step_count": 3,
  "failed_step": null,
  "duration_ms": 173000,
  "started_at": "2026-04-18T14:30:22Z",
  "completed_at": "2026-04-18T14:33:15Z"
}
```

Exit Code:

- 0 if outcome == "success"
- 1 if outcome == "failure"
- 2 if outcome == "governance-blocked"
- 3 if outcome == "cancelled"

15.8.2 GitHub Actions Example

```
- name: Run gert runbook
  id: gert_run
  run: |
    OUTPUT=$(gert run deploy.yaml --output=json)
    echo "$OUTPUT" | jq .
    RUN_ID=$(echo "$OUTPUT" | jq -r .run_id)
    OUTCOME=$(echo "$OUTPUT" | jq -r .outcome)
    echo "run_id=$RUN_ID" >> $GITHUB_OUTPUT
    echo "outcome=$OUTCOME" >> $GITHUB_OUTPUT

- name: Upload trace artifact
  if: always()
```

```
uses: actions/upload-artifact@v3
with:
  name: gert-trace
  path: .runbook/runs/${{ steps.gert_run.outputs.run_id }}/trace.jsonl
```

15.8.3 GitLab CI Example

```
deploy:
  script:
    - gert run deploy.yaml --output=json | tee run-output.json
    - RUN_ID=$(jq -r .run_id run-output.json)
    - echo "Run ID: $RUN_ID"
  artifacts:
    when: always
    paths:
      - .runbook/runs/
  reports:
    junit: .runbook/runs/*/trace.jsonl
```

15.9 Summary

This chapter specifies a production-grade observability model for gert grounded in industry standards:

- **Three-pillar model:** Metrics (Prometheus), distributed tracing (OpenTelemetry), structured logging (JSON)
- **Span hierarchy:** `gert.run` → `gert.step` → `gert.tool.invoke` / `gert.governance.check` / `gert.input.resolve`
- **Context propagation:** W3C Trace Context headers and OpenTelemetry baggage for cross-system correlation
- **Metrics catalog:** 9 metrics covering runs, steps, tools, approvals, and extensions
- **Health endpoints:** `/health` (liveness), `/ready` (readiness), `/metrics` (Prometheus)
- **Diagnostics commands:** `gert diagnose`, `gert trace show`, `gert trace export`, `gert run list`

- **Integration recipes:** Prometheus, Grafana, Jaeger, Loki, Datadog, New Relic
- **CI/CD support:** JSON output mode with exit codes for pipeline integration

Operators gain visibility into run health, performance bottlenecks, and failure root causes. The design ensures portability across observability platforms by adhering to open standards [2, 6, 13].

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